

GLOBAL CREDIT TRADER

IG Private Placements—No Longer Niche

- The wave of AI-related debt issuance has impacted multiple aspects of the corporate credit markets, and IG private placements are no exception to this trend. The IG non-financial market has seen over \$270 billion of private placement supply year to date, which is the entire supply seen last year, and puts 2026 on track for a new record.
- While IG private placements have traditionally been dominated by Utilities, Oil & Gas, and Autos, the last 12-months of issuance has been led by AI-related sectors. For example, in 2024 only 3% of private placement issuance was AI-related; so far this year, that share has grown to 42%.
- Of the 20 largest USD IG bond tranches issued in the past year, all have been AI-related and the top five tranches have all been private placements. The average private placement new issue deal has grown to \$1.5 billion this year from \$1.1 billion just two years ago as large AI deals have pulled up the average.
- The issuance of “144A for life” USD IG private placements without registration rights has grown meaningfully over the last couple of years, driven in part by AI infrastructure-focused special purpose vehicles (SPVs). We estimate approximately \$2.4 trillion of USD IG bonds are currently index-ineligible due to the absence of registration rights, representing 20% of the broader USD IG market, though AI-related names still represent a small fraction of this index-ineligible private placement universe at just over 4%.
- With the TMT cohort, which includes most AI-related issuers, we find that the private placement excess spread premium (over registered bonds) has moved sharply wider over the last four months.

Amanda Lynam, CPA
+1(212)934-1895 |
amanda.lynam@gs.com
Goldman Sachs & Co. LLC

Spencer Rogers, CFA
+1(801)884-1104 |
spencer.rogers@gs.com
Goldman Sachs & Co. LLC

Sara Grut
+44(20)7774-8622 | sara.grut@gs.com
Goldman Sachs International

Shamshad Ali
+1(212)902-6712 |
shamshad.ali@gs.com
Goldman Sachs & Co. LLC

IG Private Placements—No Longer Niche

Given the size and liquidity of the USD Investment Grade (IG) credit market, most bonds have traditionally been issued as registered bonds by large, repeat borrowers with established debt capital structures. However, a smaller contingent of IG bonds has historically been issued as private placements, typically by foreign issuers utilizing the Regulation S (Reg S) exemption.

Over the past year, this dynamic has shifted. We have observed an acceleration in large-scale IG private placements issued by US-based firms, primarily aimed at funding the large capital expenditures required for AI infrastructure and data centers. The IG non-financial market has seen over \$270 billion of private placement supply year to date, which is the entire supply seen last year, and puts 2026 on track for a new record. For context, in 2024 only 3% of private placement issuance was AI-related; so far this year, that share has grown to 42%. Of the 20 largest USD IG bond tranches issued in the past year, all have been AI-related and the top five tranches have all been private placements. The average private placement new issue deal this year has grown to \$1.5 billion from \$1.1 billion just two years ago as large AI deals have pulled up the average.

The acceleration in large-sized private placement issuance has prompted a range of questions from market participants, which we address below.

1. Have private placements become more common in the IG market?
2. Under what conditions are private placements eligible for index inclusion?
3. How do IG private placements compare to their registered counterparts from a liquidity standpoint?
4. Do IG private placements offer a spread premium over their registered counterparts?

A Brief Refresher on Private Placements

Before diving into these questions, a [brief refresher on private placements](#) is warranted. There are two primary types of private placements created under the Securities Act of 1933: Rule 144A and Regulation S.

- **Rule 144A:** Allows issuers to bypass the SEC registration process by selling bonds exclusively to Qualified Institutional Buyers (QIBs) in the US market. This offers expediency and lower syndication costs for borrowers. Since mid-2014, 144A bond trades have been disseminated through TRACE, bringing additional transparency to the market.
- **Regulation S (Reg S):** Commonly utilized by corporations seeking to tap foreign investor capital. Reg S permits issuers to bypass SEC registration by selling bonds overseas in offshore transactions to foreign QIBs, granting international investors greater access to the USD corporate bond market.

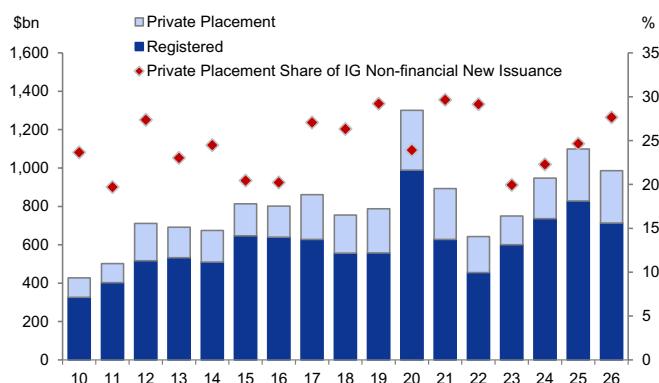
In the current market evolution, most IG private placements create two distinct ISINs—one for the 144A market and one for Reg S—and the bonds are sold simultaneously to both investor bases. Given this dynamic, we will refer to these bonds generically as “private placements” throughout the remainder of this note, without distinguishing between the two.

Have private placements become more common in the IG market?

Over the past year, several large AI-related and data center financing deals have been issued in the IG market as private placements. While these transactions have been large and captured the focus of market participants, Exhibit 1 shows that the share of primary market supply issued via private placement has trended upward only mildly since the end of the Global Financial Crisis (GFC)—although the move higher in recent years is notable. Similarly, Exhibit 2 illustrates that the overall share of the IG non-financial secondary market composed of private placements has remained roughly flat, between 15% to 20%, for nearly 15 years.

Exhibit 1: The share of IG new issuance that is private placement has increased over the past few years

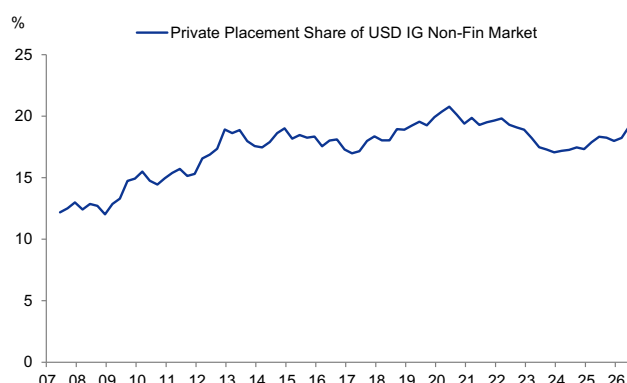
Notional amount and share (RHS) of USD IG non-financial private placement issuance



Source: Dealogic, Goldman Sachs Global investment Research

Exhibit 2: IG private placements have hovered between 15-20% of the secondary market since 2010

Notional share of the USD IG non-financial universe that is private placement irrespective of index eligibility

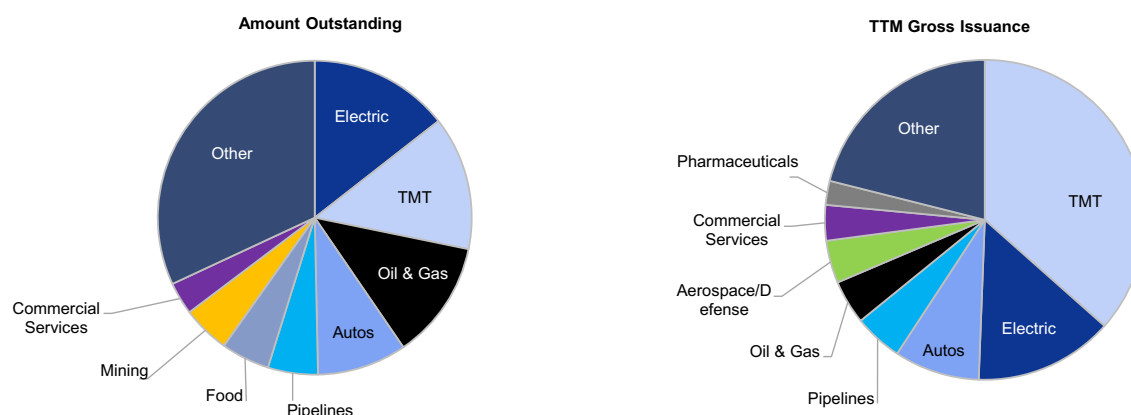


Source: Bloomberg, Goldman Sachs Global investment Research

While the *overall share* has stayed relatively the same, the *underlying composition* of this private placement issuance has shifted. Exhibit 3 shows the sector distribution of the current outstanding universe of USD IG non-financial private placement bonds vs. the last 12 months of USD IG private placement supply. Whereas traditionally IG private placements have been dominated by Utilities, Oil & Gas, and Autos, the last 12-months of issuance has been led by AI-related sectors.

Exhibit 3: A larger share of recent USD IG private placement issuance has come from AI-related sectors

Notional sector distribution of the outstanding universe of IG private placements vs. the sector distribution of the trailing 12-month supply

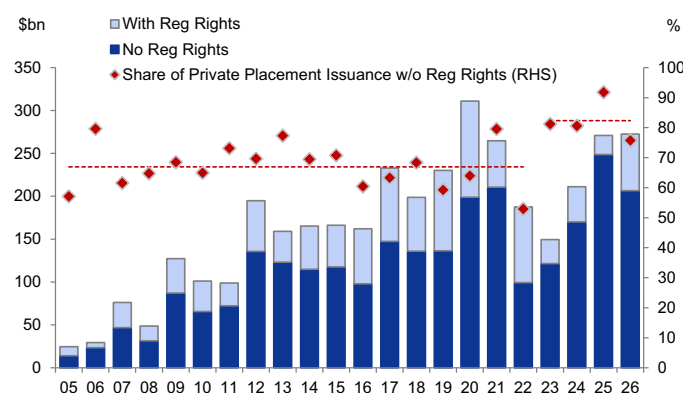


Source: Dealogic, Bloomberg, Goldman Sachs Global Investment Research

The other meaningful structural shift has been the proportion of these private placement deals issued *with* registration rights versus those remaining “144A-for-life.” As shown in Exhibit 4, the issuance of USD IG private placements without registration rights has grown meaningfully over the last couple of years, driven in part by AI infrastructure-focused special purpose vehicles (SPVs).

Exhibit 4: The share of index-ineligible private placement issuance has shifted higher in recent years, though 2026 has generated the lowest share since 2022

USD IG private placement issuance with and without registration rights



Source: Dealogic, Goldman Sachs Global investment Research

Under what conditions are private placements eligible for index inclusion?

There is a distinct bifurcation in how private placements are treated for index inclusion across the USD High Yield (HY) and IG markets. In the HY market, all 144A and Reg S private placements are generally eligible for index inclusion. Conversely, major USD IG indices require private placements to possess *registration rights* to be eligible.

Registration rights allow the issuer to eventually exchange the privately placed bonds for SEC-registered bonds. Consequently, per the current index inclusion rules, the recent data center deals issued as “144A-for-life” will remain off-index. We estimate approximately \$2.4 trillion of USD IG bonds are currently index-ineligible due to the absence of registration rights, representing 20% of the broader USD IG market. We further estimate that AI-related deals still represent a small fraction of this index-ineligible private placement universe at just over 4%.

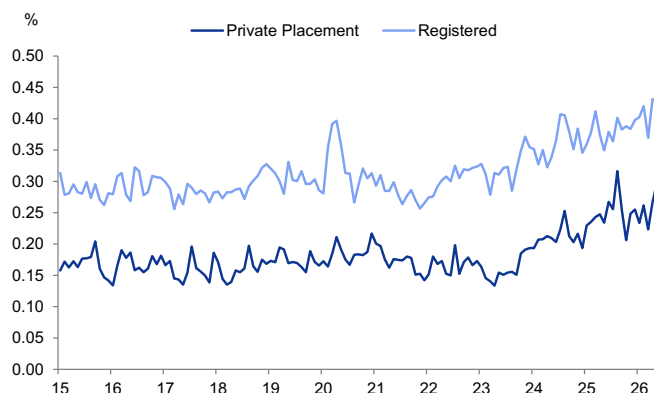
How do IG private placements compare to their registered counterparts from a liquidity standpoint?

Because the subset of eligible buyers (QIBs) for private placements is inherently smaller than the universe of investors capable of purchasing registered bonds, one would expect lower liquidity due to a more constrained secondary market. Exhibit 5 confirms this assumption: turnover for USD IG private placements runs at roughly half the level of registered bonds.

However, it is worth noting that trading volumes for both cohorts have trended upward over the last few years. Furthermore, our prior research indicates no material difference in bid-ask spreads between registered and 144A bonds, suggesting that, while turnover is lower, execution costs remain highly comparable.

Exhibit 5: Turnover is higher for registered vs. private placement bonds

Daily average trading volume as a share of notional outstanding



Source: TRACE, Bloomberg, Goldman Sachs Global investment Research

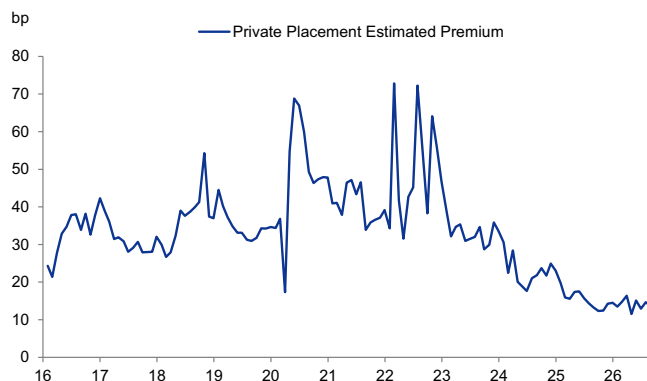
Do IG private placements offer a (wider) spread premium over their registered counterparts?

From first principles, a bond with a smaller investor base and lower trading velocity should trade at a premium (i.e., a wider spread) to a more available, freely traded bond, all else being equal. To quantify this private placement premium, we built a regression model controlling for rating, seniority, sector, and duration.

Historically, private placement bonds offered roughly 30bp of excess spread over their registered counterparts. However, as shown in Exhibit 6, this premium has compressed over the last several years and currently averages closer to 15bp. If we isolate the TMT cohort, which includes most AI-related issuers, we find that the private placement premium has corrected sharply wider over the last four months—after several years of trending tighter and briefly dipping into negative territory (Exhibit 7). Notwithstanding the localized widening of the premium for AI-related issuers, the pickup in trading volumes and compression of the overall private placement premium indicate to us a growing institutional comfort and demand for this asset class.

Exhibit 6: The premium demanded by IG private placement investors on an aggregate basis has been trending tighter

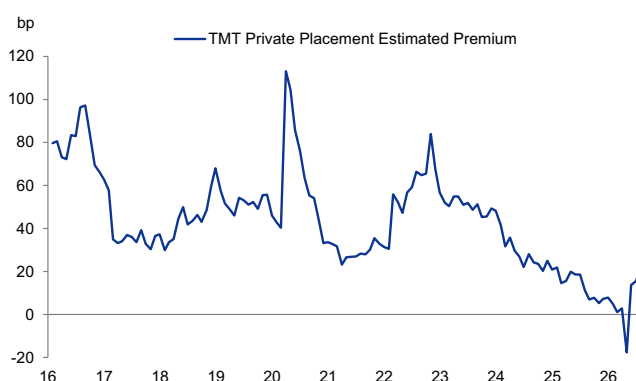
Model-estimated private placement spread premium adjusting for rating, sector, seniority, and duration



Source: iBoxx, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 7: In contrast to the broader market, the private placement premium for TMT bonds has shifted wider alongside the acceleration in supply

Model-estimated TMT private placement spread premium adjusting for rating, seniority, and duration



Source: iBoxx, Bloomberg, Goldman Sachs Global Investment Research

In the primary market, we observe a negligible difference in new issue concessions between registered and privately placed bonds. Over the trailing 12-months, new issue concessions have averaged 6bp for USD IG private placements versus 5bp for registered bonds.

The authors would like to thank Patricia Pacheco for her contribution to this report.

Forecast dashboard: Spreads, returns, issuance, defaults, and ratings migrations

Exhibit 8: USD and EUR spread forecasts

Market	Updated as of Aug. 19, 2026				
	Current	2026Q3	2026Q4	2027Q1	2027Q2
USD spreads (bp)					
USD IG	80	85	82	80	78
USD HY	270	305	295	290	280
EUR spreads (bp)					
EUR IG	77	91	88	86	84
EUR HY	280	320	315	310	305

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 10: Issuance forecasts

Market	Updated as of Aug. 19, 2026	
	2026 YTD (bn)	2026 Full-year forecast (bn)
USD market		
Gross issuance		
IG	\$1,627	\$2,100
HY	\$223	\$400
Leveraged loans	\$269	--
Net issuance		
IG	\$725	\$850
HY	\$46	\$160
EUR market		
Gross issuance		
IG	€ 551	€ 850
HY	€ 71	€ 125
Leveraged loans	€ 57	--
Net issuance		
IG	€ 142	€ 265
HY	€ 14	€ 60

Source: Dealogic, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 9: USD and EUR excess returns forecast

Market	Updated as of Aug. 19, 2026		
	Current duration	2026 YTD (%)	2026 Full-year forecast (%)
USD excess returns			
USD IG	6.8	0.4%	0.5%
USD HY	3.2	1.7%	1.3%
EUR excess returns			
EUR IG	4.5	0.7%	0.5%
EUR HY	3.1	1.7%	1.3%

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 11: Ratings migration and default forecasts

Default rates are issuer-weighted and 12-month trailing

Market	Updated as of Aug. 19, 2026	
	2026 YTD	2026 Full-year forecast
USD market		
Rising stars (bn)	\$75	\$75
Fallen angels (bn)	\$35	\$60
HY defaults (12-month trailing)	3.7%	4.0%
Leveraged loan defaults (12-month trailing)	4.5%	4.75%
EUR market		
Rising stars (bn)	€ 18	€ 20
Fallen angels (bn)	€ 5	€ 30
HY defaults (12-month trailing)	4.9%	5.0%
Leveraged loan defaults (12-month trailing)	3.0%	4.0%

Source: Bloomberg, Moody's, Goldman Sachs Global Investment Research

Forecast dashboard: Relative value views

Exhibit 12: Relative value views

Relative value views		
Corporate Credit		
Regional	Overweight USD vs. EUR	We have a slight preference for USD credit vs. its EUR peer, owing to a somewhat more challenging growth-inflation-monetary policy mix in the Euro Area. That said, the lines are blurring between the regional classifications, given the acceleration in cross-border issuance.
	Overweight EUR IG CLO tranches vs. USD IG CLO tranches	EUR IG tranches offer carry pickup relative to USD IG CLO tranches. While the policy and macroeconomic picture is less supportive in Europe, subordination should protect tranches up in the capital structure from a continued idiosyncratic default picture.
USD market	Neutral IG vs. HY	While we previously had a slight preference for IG vs. HY, we now view the case for an outright overweight in IG as less obvious based on valuations, rate sensitivity and supply related technicals.
	Overweight BBBs vs. higher-rated cohorts	We prefer owning BBBs over the higher-rated segments of USD IG (As and AAs), as a way to capture additional spread. We also see scope for balance sheet deterioration at the highest end of the rating spectrum in IG.
	Overweight Bs, Neutral BBs, Underweight CCCs	We expect BBs will continue to encounter a challenging supply technical due to AI-related supply. As a result, we favor the B rated cohort. The CCC universe is highly idiosyncratic and requires careful credit selection, in our view.
	Overweight HY bonds vs. loans	The substantial software sector exposure in the loan market leaves us favoring the HY market, at least until we receive more clarity on the refinancing of the 2028 loan maturity walls. The lower-rated distribution of loans and our economists' base case for no rate hikes in 2026 are also contributing factors.
	Overweight CLO mezz tranches vs. leveraged loans Neutral CLO vs. USD IG Neutral CLO vs. CMOs	For floating rate investors, we prefer CLO mezzanine tranches vs. leveraged loans. Historical data points to low impairment rates for BB and BBB CLO tranches, and current valuations offer attractive carry and a cushion against moderate to elevated default outcomes. Elsewhere, we are neutral between senior CLO tranches and USD IG; we would use spread widening to add exposure in CLOs. We are also neutral between CLO AAA and CMO floaters.
	Prefer Agency MBS vs. IG	Agency MBS offers an attractive diversification benefit vs. a portfolio of IG corporate bonds.
	Overweight Hybrids vs. BBBs	We view the additional spread compensation of hybrids vs. BBBs as moderately attractive in the context of the tight spread environment. We also expect the hybrid spread premium to compress vs. BBBs, as the hybrid market grows further.
	Neutral IG vs. HY	While we previously had a preference for IG vs. HY, we now view the case for an outright overweight in IG as less obvious based on valuations, rate sensitivity and supply related technicals.
EUR market	Overweight BBBs vs. higher-rated cohorts	We recently shifted our view to favor BBBs over both AAs and As in the EUR market, owing to an improved macroeconomic backdrop which now favors a broad-based, down-in-quality stance within IG.
	Overweight HY bonds vs. loans	Similar to the US, we prefer HY bonds given the loan market's weighting toward software (and the potential negative sentiment overhang). The reaction function of the ECB is increasing borrowing costs for floating rate issuers, given the central bank's focus on inflationary pressures.
	Overweight BBs, Neutral Bs, Underweight CCCs	At current valuations and given our expectation for heightened dispersion, we prefer moving up in quality within EUR HY.

Source: Goldman Sachs Global Investment Research

Earnings calendar

Exhibit 13: USD IG earnings report dates

Upcoming earnings report dates for the Bloomberg USD IG Index (LUACTRUU); minimum \$1 billion notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)
8/25/2026			8/26/2026			8/27/2026			8/28/2026		
BNS	Bank of Nova Scotia/The	16.5	CRM	Salesforce Inc	33.5	RY	Royal Bank of Canada	29.2			
BMO	Bank of Montreal	14.9	NVDA	NVIDIA Corp	32.5	TD	Toronto-Dominion Bank/The	24.0			
WDSAU	Woodside Energy Group Ltd	5.5	SNPS	Synopsys Inc	9.0	CM	Canadian Imperial Bank of Con	13.5			
HEI	HEICO Corp	2.4	SJM	J M Smucker Co/The	6.1	DG	Dollar General Corp	4.5			
DKS	Dick's Sporting Goods Inc	1.9	NACN	National Bank of Canada	4.3	DLTR	Dollar Tree Inc	2.5			
			A	Agilent Technologies Inc	3.7	HRL	Hormel Foods Corp	2.4			
			PRUFIN	Prudential PLC	1.4	WDAY	Workday Inc	2.0			
9/31/2026			9/1/2026			9/2/2026			9/3/2026		
			MDT	Medtronic PLC	6.1	AVGO	Broadcom Inc	54.0	CPB	Campbell's Company/The	6.0
						VMW	Broadcom Inc	3.0			
						NTAP	NetApp Inc	2.0			
						BFB	Brown-Forman Corp	1.5			

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 14: USD HY earnings report dates

Upcoming earnings report dates for the Bloomberg USD HY index (LF98TRUU); minimum \$500 million notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)	Ticker	Name	Notional (\$bn)
8/21/2026			8/24/2026			8/25/2026			8/26/2026		
GALCLD	Bohai Leasing Co Ltd	1.0	PLSAU	PLS Group Ltd	0.6				BBWI	Bath & Body Works Inc	3.4
									DY	Dycom Industries Inc	0.5
9/27/2026			8/28/2026			8/31/2026			9/1/2026		
MINAU	Mineral Resources Ltd	3.4				MPEL	Melco International Developmei	3.3			
GAP	Gap Inc/The	1.5				STCITY	Melco International Developmei	1.4			
HQY	HealthEquity Inc	0.6				SAIC	Science Applications Internati	0.9			
ESTC	Elastic NV	0.6									

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 15: EUR IG earnings report dates

Upcoming earnings report dates for the Bloomberg EUR IG Index (LECP TREU); minimum €1 billion notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)
8/21/2026			8/24/2026			8/25/2026			8/26/2026		
TVLRO	Banca Transilvania SA	1.5				BMO	Bank of Montreal	3.3	ARRFP	Eiffage SA	6.4
						BNS	Bank of Nova Scotia/The	2.5	ARNDTN	Aroundtown SA	4.1
						SCGAU	Scentre Group	1.5	SYDBDC	AL Sydbank	2.3
						MUNHYP	Muenchener Hypothekenbank €	1.0	GYCGR	Aroundtown SA	2.0
									WOWAU	Woolworths Group Ltd	1.1
									NACN	National Bank of Canada	1.0
8/27/2026			8/28/2026			8/31/2026			9/1/2026		
RIFP	Pernod Ricard SA	9.0	BERTEL	Bertelsmann SE & Co KGaA	3.8				MDT	Medtronic PLC	16.5
TD	Toronto-Dominion Bank/The	8.6							SLHNVX	Swiss Life Holding AG	2.8
RY	Royal Bank of Canada	6.8							ATDBCN	Alimentation Couche-Tard Inc	2.1
CM	Canadian Imperial Bank of Con	2.5							AEDBB	Aedifica SA	1.5
AGSBB	Ageas SA/NV	1.5									
WESAU	Wesfarmers Ltd	1.2									
ARLA	Arla Foods ambia	1.0									

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 16: EUR HY earnings report dates

Upcoming earnings report dates for the Bloomberg Pan-Euro HY index (LP01TREU); minimum €100 million notional value and including up to the top 20 capital structures in the index reporting that day

Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)	Ticker	Name	Notional (€bn)
8/26/2026			8/27/2026			8/28/2026			8/31/2026		
ARNDTN	Aroundtown SA	2.1				INTRUM	Intrum AB	2.8			
GYCGR	Aroundtown SA	1.2									
9/1/2026			9/2/2026			9/3/2026			9/4/2026		

Source: Bloomberg, Goldman Sachs Global Investment Research

Performance dashboard: Cash and synthetic markets

Exhibit 17: Performance of USD and EUR markets

Market	Current*		Total return			Spread change (YTD,bp)	Yield change (YTD, %)
	Spread	Yield	MTD	QTD	YTD		
USD Corporate bonds							
IG	80	5.4	0.75	-0.93	-0.09	2	0.58
AA	60	5.2	0.73	-1.43	-0.93	11	0.61
A	68	5.3	0.70	-0.78	0.09	3	0.52
BBB	100	5.6	0.74	-0.75	0.49	-1	0.50
HY	270	7.2	0.84	0.59	2.56	4	0.71
BB	161	6.3	0.92	0.54	2.39	-8	0.41
B	291	7.8	0.87	0.86	3.33	-10	0.40
CCC	1030	14.8	0.51	-0.28	-0.14	145	1.81
USD Leveraged Loans							
Leveraged loan index	423	8.1	0.59	1.39	2.72	28	0.2
BB	248	6.2	0.46	1.11	3.68	-7	-0.2
B	419	8.0	0.74	1.70	2.78	27	0.2
CCC	2006	25.3	-0.04	-0.13	-3.56	442	4.6
USD Securitized products							
Agency MBS	24	5.2	1.17	-0.72	0.72	2	0.5
ABS (Autos)	55	4.6	0.35	0.63	2.11	-11	0.4
ABS (Credit cards)	28	4.4	0.39	0.54	1.75	-5	0.5
ABS (Student loans)	96	4.8	0.27	0.78	3.07	123	-0.1
USD Macro indices							
CDX IG	51	--	0.11	0.08	0.55	1	--
CDX HY	303	--	0.42	0.36	1.95	-13	--
EUR Corporate bonds							
IG	89	4.0	-0.01	-1.01	0.35	0	0.5
AA	67	3.9	-0.13	-1.43	-0.15	5	0.5
A	82	3.9	-0.03	-1.05	0.30	-2	0.5
BBB	102	4.1	0.04	-0.89	0.50	0	0.5
HY	255	6.0	0.36	0.02	1.92	-15	0.4
AT1	245	5.4	0.42	0.14	3.53	-41	0.1
EUR Leveraged loans							
Leveraged loan index	454	7.2	0.41	0.64	2.52	4	0.6
BB	268	5.2	0.36	0.71	3.50	-84	-0.4
B	420	6.8	0.44	0.76	2.51	5	0.6
EUR Macro indices							
iTraxx Main	52	--	0.13	0.07	0.6	1	--
iTraxx Xover	250	--	0.65	0.12	3.4	6	--
iTraxx SenFin	54	--	0.12	0.10	0.8	-1	--
iTraxx SubFin	87	--	0.17	0.25	1.7	-6	--

*Data as of previous day close.

For USD IG and HY we use the Bloomberg indices with history back to Jun-89 and Jan-94, respectively. Corp rating level, Agency MBS and ABS we use ICE-BAML indices with history back to Dec-96. EUR IG we use the iBoxx QW5A index with history back to Jan-99. EUR HY we use the ICE-BAML index with history back to Dec-97, and USD and EUR leveraged loans we use the Morningstar LSTA LLI and ELLI indices with data going back to Jan-97 and Jan-02, respectively.

Source: Bloomberg, iBoxx, ICE-BAML, Pitchbook LCD, Goldman Sachs Global Investment Research

Performance dashboard: Sectors

Exhibit 18: Sector performance: USD IG

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	2%	180.2	80		5.5	7.2	BAA1/BAA2	1	2	0	-9	0.1	0.8	-0.6	0.7
Chemicals	1%	81.8	84		5.5	7.5	BAA1/BAA2	1	2	-1	-14	0.2	0.9	-0.5	1.2
Metals & Mining	1%	72.4	73		5.4	7.4	A3/BAA1	2	3	3	-3	0.0	0.8	-0.9	0.2
Paper	0%	26.1	86		5.4	5.9	BAA2/BAA3	-2	-1	-2	-9	0.1	0.8	-0.2	0.7
Capital Goods	6%	417.2	61		5.2	6.6	A3/BAA1	0	3	7	-3	0.1	0.7	-0.8	0.2
Aerospace & Defense	2%	149.7	68		5.4	8.0	BAA1/BAA2	1	3	6	-5	0.1	0.8	-1.3	-0.1
Building Materials	1%	46.8	76		5.3	6.5	BAA1/BAA2	0	2	5	0	0.1	0.7	-0.8	0.1
Diversified Manufacturing	2%	117.3	61		5.2	6.3	A3/BAA1	1	3	8	-1	0.1	0.7	-0.8	0.1
Construction Machinery	1%	57.9	38		4.8	4.5	A1/A2	0	6	12	-7	0.1	0.6	-0.2	0.6
Packaging	0%	14.3	69		5.1	4.3	BAA2/BAA3	1	3	3	-12	0.0	0.5	0.0	1.0
Environmental	0%	31.2	47		5.0	5.8	A3/A3	0	5	9	-5	0.0	0.6	-0.5	0.2
Communications	8%	578.2	113		5.9	8.3	A3/BAA1	0	1	13	19	0.1	1.1	-2.0	-1.9
Cable & Satellite	2%	130.1	162		6.4	8.7	BAA1/BAA2	-1	-3	10	24	0.2	1.6	-2.2	-2.2
Media & Entertainment	2%	129.1	102		5.8	8.4	A1/A2	3	4	20	20	-0.2	0.8	-2.5	-3.7
Wireless	2%	129.6	87		5.5	7.0	BAA1/BAA2	0	0	10	8	0.1	0.8	-1.3	-0.5
Wirelines	2%	189.3	104		5.8	8.8	BAA1/BAA2	0	1	13	17	0.2	1.1	-2.1	-1.6
Consumer Cyclical	7%	558.8	72		5.3	6.2	A2/A3	1	2	9	2	0.0	0.7	-0.7	0.0
Automotive	2%	167.4	81		5.2	4.2	A3/BAA1	0	1	1	-9	0.0	0.6	0.2	1.3
Leisure	0%	6.0	100		5.5	6.2	BAA2/BAA2	2	3	11	-1	-0.1	0.6	-0.9	-0.1
Gaming	0%	24.2	111		5.5	4.7	BAA2/BAA3	2	4	7	0	-0.1	0.5	-0.3	0.8
Home Construction	0%	9.6	82		5.3	5.0	BAA1/BAA2	0	2	5	-2	0.0	0.6	-0.3	0.7
Lodging	0%	20.8	73		5.1	4.8	BAA2/BAA3	0	5	12	-2	0.0	0.5	-0.2	0.6
Retailers	4%	271.1	63		5.3	7.6	A1/A2	1	3	18	19	0.1	0.8	-1.4	-1.0
Restaurants	0%	31.9	61		5.3	7.5	BAA1/BAA2	1	3	8	-3	0.1	0.7	-1.0	0.1
Consumer Cyc Services	0%	27.8	75		5.3	6.1	A3/BAA1	0	1	3	5	0.0	0.8	-0.5	0.1
Consumer Non-Cyclical	15%	1126.4	66		5.3	7.6	A2/A3	0	4	7	-2	0.1	0.7	-1.1	-0.2
Consumer Products	1%	58.3	48		5.0	5.9	A2/A3	0	7	8	-5	0.0	0.7	-0.4	0.4
Food & Beverage	3%	230.8	71		5.4	7.5	A3/BAA1	1	3	5	2	0.1	0.7	-1.0	-0.2
Healthcare	4%	325.8	72		5.4	7.6	A3/BAA1	0	3	4	-5	0.1	0.8	-1.0	0.0
Pharmaceuticals	5%	416.0	60		5.3	8.1	A2/A3	0	6	10	1	0.2	0.7	-1.4	-0.5
Supermarkets	0%	12.6	90		5.8	9.7	BAA1/BAA2	0	5	7	5	0.2	0.9	-1.6	-0.8
Tobacco	1%	82.9	74		5.3	6.6	A3/BAA1	1	7	8	-4	0.0	0.6	-0.8	0.2
Energy	7%	524.4	79		5.5	7.5	A3/BAA1	0	0	0	-12	0.1	0.9	-0.8	0.9
Independent Energy	1%	91.2	79		5.5	7.3	BAA1/BAA2	0	-1	-2	-17	0.1	1.0	-0.6	1.7
Integrated Energy	1%	100.7	55		5.2	8.1	AA3/A1	1	1	0	-8	0.1	0.9	-0.9	0.0
Oil Field Services	0%	24.3	80		5.5	7.5	A3/BAA1	1	-1	-2	-17	0.1	1.0	-0.6	1.0
Refining	0%	30.2	80		5.4	6.9	BAA1/BAA2	0	-3	-6	-24	0.2	1.0	-0.3	1.9
Midstream	4%	278.0	87		5.6	7.4	BAA1/BAA2	0	0	2	-10	0.1	0.9	-0.8	0.8
Technology	11%	807.6	89		5.5	7.0	A2/A3	1	0	13	17	-0.1	0.8	-1.5	-1.0
Transportation	2%	141.5	68		5.5	8.9	A3/BAA1	1	2	5	-6	0.2	0.9	-1.5	-0.1
Airlines	0%	17.0	84		5.2	3.8	A2/A3	3	7	12	-6	0.0	0.4	0.0	1.3
Railroads	1%	87.1	65		5.6	10.7	A3/BAA1	0	2	3	-4	0.2	1.0	-1.9	-0.7
Transportation Services	0%	37.4	68		5.3	7.1	A3/BAA1	0	1	6	-9	0.1	0.8	-1.2	0.6
Other Industrial	1%	41.2	64		5.5	9.9	AA3/A1	1	5	9	2	0.2	0.9	-1.6	-0.3
Utility	10%	771.3	84		5.5	8.0	A3/BAA1	1	3	7	-1	0.1	0.8	-1.2	-0.1
Electric	9%	695.6	83		5.5	8.0	A3/BAA1	1	3	7	-1	0.1	0.8	-1.3	-0.1
Natural Gas	1%	59.6	92		5.6	7.6	A3/BAA1	2	5	6	-4	0.1	0.7	-1.1	0.2
Other Utility	0%	16.2	71		5.4	8.0	BAA1/BAA2	0	4	7	2	0.1	0.7	-1.2	-0.5
Financial Institutions	32%	2428.1	80		5.3	5.3	A2/A3	1	2	6	3	0.0	0.6	-0.4	0.4
Banking	22%	1657.6	75		5.2	4.9	A2/A3	1	1	8	5	0.0	0.6	-0.2	0.4
Brokerage Asset Managers Exchanges	2%	127.9	80		5.3	6.0	A3/BAA1	1	1	3	9	0.0	0.7	-0.6	-0.1
Finance Companies	1%	110.8	141		5.7	3.5	BAA2/BAA3	2	-2	-3	-1	0.0	0.7	0.5	1.5
Insurance	4%	329.4	88		5.6	8.1	A3/BAA1	2	5	7	-1	0.0	0.6	-1.4	-0.4
REITS	3%	194.7	73		5.2	5.3	A3/BAA1	1	5	8	-4	0.0	0.6	-0.3	0.6
Other Financial	0%	7.6	103		5.5	5.0	A3/BAA1	1	0	4	2	0.0	0.8	0.2	1.6
US Corporate IG		7574.8	80		5.4	6.8	A3/BAA1	1	2	7	3	0.0	0.8	-0.9	-0.1

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 19: Sector performance: USD HY

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	6%	92.5	252		7.1	3.2	BA3/B1	2	-2	-4	-13	-0.1	0.7	0.7	3.4
Chemicals	3%	47.7	227		6.8	3.4	BA3/B1	1	-2	-6	-30	0.0	1.0	0.8	4.8
Metals & Mining	3%	38.1	196		6.5	3.1	BA3/B1	4	-5	-3	-6	-0.1	0.9	0.9	2.7
Paper	0%	6.8	732		11.8	2.4	B1/B2	3	5	4	46	-1.2	-2.2	-0.9	-2.5
Capital Goods	10%	155.8	214		6.7	3.0	BA3/B1	2	-8	3	0	-0.1	1.0	0.4	1.2
Aerospace & Defense	2%	30.9	96		5.6	2.8	BA3/B1	2	-8	-6	-4	0.0	0.8	0.7	2.4
Building Materials	3%	41.0	341		8.0	3.2	BA3/B1	3	-5	5	19	-0.3	1.0	0.0	0.0
Diversified Manufacturing	2%	23.9	155		6.1	2.8	BA2/BA3	-3	-14	1	48	0.0	1.0	0.8	2.2
Construction Machinery	2%	23.4	103		5.6	3.0	BA3/B1	5	-22	-22	-21	0.0	1.1	0.8	2.9
Packaging	2%	27.7	321		7.8	3.1	B1/B2	1	-5	7	-23	-0.1	1.1	0.4	-0.5
Environmental	1%	8.9	158		6.1	3.2	BA3/B1	-1	-6	15	7	0.2	1.0	-0.1	1.7
Communications	14%	204.2	449		9.0	3.7	B1/B2	1	-2	5	29	0.0	1.0	0.0	1.5
Cable & Satellite	5%	68.8	660		11.0	3.0	B1/B2	1	0	14	65	-0.1	0.9	-0.5	-0.6
Media & Entertainment	5%	81.1	315		7.8	4.1	BA3/B1	0	-6	-4	-6	-0.1	1.4	1.1	3.0
Wireless	1%	17.2	314		7.7	4.2	B1/B2	2	-2	8	25	0.1	0.9	-0.1	0.5
Wirelines	2%	37.1	413		8.8	4.0	B1/B2	2	0	6	32	0.0	0.5	-1.2	2.9
Consumer Cyclical	18%	263.9	233		6.8	3.0	BA3/B1	2	-3	-1	-8	0.0	0.7	0.8	3.0
Automotive	3%	47.4	265		7.1	2.9	BA3/B1	2	-1	8	-4	0.0	0.8	0.5	2.6
Leisure	1%	21.0	228		6.7	3.0	B2/B3	4	-5	6	27	-0.1	0.8	0.3	2.2
Gaming	3%	45.1	266		7.2	2.8	BA3/B1	2	-1	5	10	-0.1	0.6	0.5	2.5
Home Construction	1%	18.5	157		6.0	2.6	BA3/B1	-1	-10	-8	-22	-0.1	0.9	1.0	3.7
Lodging	1%	16.5	116		5.7	3.7	BA2/BA3	2	-4	3	-13	0.0	0.8	0.4	2.1
Retailers	4%	60.5	215		6.6	3.3	BA3/B1	1	-2	-10	-34	0.0	0.7	1.0	3.9
Restaurants	1%	11.4	148		6.0	3.9	BA3/B1	2	-2	-3	-9	0.0	0.6	0.6	2.6
Consumer Cyc Services	3%	43.7	288		7.4	2.6	B1/B2	2	-3	-7	4	0.0	0.8	1.3	3.2
Consumer Non-Cyclical	11%	168.0	260		7.1	3.2	B1/B2	2	-3	3	6	0.1	0.9	0.7	3.0
Consumer Products	2%	27.5	308		7.6	3.9	BA3/B1	5	-3	-3	-6	-0.2	1.3	0.8	1.1
Food & Beverage	2%	29.2	182		6.4	3.4	B1/B2	-3	1	5	7	0.1	0.7	0.4	2.3
Healthcare	5%	75.5	249		6.9	2.8	B1/B2	1	-4	10	20	0.2	0.7	0.5	3.2
Pharmaceuticals	2%	26.0	358		8.2	2.9	B2/B3	0	-3	-8	-18	0.1	0.9	1.8	6.0
Supermarkets	1%	9.6	183		6.3	3.9	BA2/BA3	9	1	17	165	0.2	1.0	0.1	0.7
Tobacco	0%	0.3	198		6.7	3.1	BA3/BA3	-3	-10	2	16	0.4	1.3	0.5	1.1
Energy	11%	166.0	138		6.0	3.2	BA2/BA3	-7	-15	-21	-49	0.1	1.0	1.3	5.5
Independent Energy	3%	45.9	115		5.8	2.8	BA3/B1	-25	-35	-42	-54	0.2	1.2	1.9	5.9
Oil Field Services	2%	27.7	176		6.5	3.1	BA3/B1	-1	-10	-10	-42	0.1	1.1	1.3	5.4
Refining	0%	5.4	144		6.1	2.1	BA3/B1	1	257	-41	-60	0.1	0.9	2.8	8.8
Midstream	6%	87.1	137		6.0	3.4	BA2/BA3	0	-9	-9	-49	0.1	0.9	0.9	5.2
Technology	10%	147.4	369		8.2	3.2	BA3/B1	3	-1	10	17	-0.4	0.7	0.1	1.3
Transportation	2%	33.7	416		8.7	2.8	BA3/B1	4	-4	3	15	-0.5	1.0	0.0	-0.2
Airlines	1%	13.8	383		8.2	2.7	BA3/B1	9	0	17	38	-1.0	0.5	-0.6	0.8
Railroads	0%	1.8	883		13.5	2.5	B1/B2	-3	-9	-2	-8	0.1	-0.3	0.0	-6.9
Transportation Services	1%	18.0	396		8.6	2.9	BA3/B1	3	-5	-6	13	-0.2	1.6	0.4	-0.2
Other Industrial	1%	15.4	228		5.5	3.1	BA3/B1	2	-2	-4	-12	0.0	0.7	1.3	2.4
Utility	3%	44.0	169		6.3	4.0	BA2/BA3	8	1	5	15	-0.1	0.6	0.1	2.8
Financial Institutions	13%	197.7	234		6.9	2.9	BA3/B1	3	-2	-4	5	-0.1	0.7	0.9	2.0
Banking	0%	3.0	176		6.2	4.2	BA1/BA2	0	-2	-3	-19	0.1	0.8	0.4	1.3
Brokerage Asset Managers Exchanges	1%	18.9	131		6.4	2.0	BA3/B1	3	-24	-27	-31	0.0	1.6	1.8	3.9
Finance Companies	5%	68.1	242		7.0	3.0	BA2/BA3	2	-5	-2	16	-0.1	1.0	1.0	1.6
Insurance	4%	59.6	239		6.9	3.2	BA3/B1	4	8	-2	25	-0.2	0.1	0.7	0.8
REITS	2%	30.2	237		6.8	2.6	B1/B2	4	-3	-2	-13	-0.1	0.5	0.6	3.2
Other Financial	1%	17.8	305		7.6	2.9	BA3/B1	1	2	-5	-4	0.0	0.5	1.1	3.7
US Corp High Yield		1488.7	270		7.2	3.2	BA3/B1	2	-3	0	1	-0.1	0.8	0.6	2.6

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 20: Sector performance: EUR IG

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumer Goods	12.5	€ 395.4	84.8		3.9	4.5	A	0	-2	-3	0	-0.35	0.03	-0.91	0.28
Automobiles & Parts	5.2	€ 165.4	104.8		4.1	3.7	A	0	-4	-4	3	-0.26	0.15	-0.54	0.42
Beverages	2.9	€ 90.4	79.6		4.0	5.8	A	0	-1	-2	-1	-0.48	-0.12	-1.37	0.17
Food Producers	1.9	€ 60.9	67.2		3.8	5.0	A	0	-1	-3	-3	-0.39	-0.04	-1.15	0.13
Household Goods	0.2	€ 6.2	49.0		3.7	5.8	AA	0	0	-2	-2	-0.51	-0.16	-1.46	-0.07
Personal Goods	1.7	€ 53.1	63.8		3.7	4.2	A	0	-1	-2	1	-0.32	-0.01	-0.92	0.11
Tobacco	0.6	€ 19.4	103.7		4.1	4.8	BBB	0	-2	-3	-8	-0.38	0.02	-1.00	0.65
Consumer Services	4.6	€ 146.4	86.8		4.0	5.2	A	0	-2	0	4	-0.43	0.01	-1.28	0.04
Food & Drug Retailers	0.6	€ 20.2	85.4		3.9	4.4	BBB	0	0	1	0	-0.36	-0.03	-0.99	0.36
General Retailers	0.6	€ 19.7	95.1		4.3	6.4	AA	0	-1	3	16	-0.56	-0.14	-2.06	-0.66
Media	1.1	€ 34.3	87.9		3.9	4.3	BBB	0	-2	0	4	-0.33	0.07	-0.97	0.32
Travel & Leisure	2.3	€ 72.2	84.0		4.0	5.5	A	0	-3	-2	3	-0.46	0.04	-1.29	0.01
Energy	3.1	€ 98.9	92.1		4.0	4.8	A	-1	-3	-4	-6	-0.32	0.08	-0.96	0.67
Oil & Gas Producers	2.8	€ 88.3	92.1		4.0	4.8	A	-1	-4	-5	-7	-0.30	0.09	-0.93	0.68
Oil Equipment / Services & Distribution	0.3	€ 8.2	93.5		4.1	5.4	A	0	-2	1	17	-0.47	-0.04	-1.27	0.51
Renewable Energy	0.1	€ 2.4	88.5		3.9	4.9	BBB	0	-5	-10	-13	-0.36	0.15	-0.70	0.87
Industrials	9.5	€ 300.0	78.4		3.8	4.5	A	0	-2	-2	-2	-0.35	0.01	-0.90	0.34
Aerospace & Defense	0.5	€ 15.7	66.1		3.8	4.4	A	-1	-2	-6	2	-0.32	0.01	-0.85	0.05
Construction & Materials	1.5	€ 47.8	82.8		3.9	4.3	BBB	1	0	0	-1	-0.36	-0.02	-0.93	0.34
Electronic & Electrical Equipment	0.9	€ 28.6	69.7		3.8	4.8	A	0	-2	-2	0	-0.40	-0.03	-1.13	0.08
General Industrials	1.2	€ 37.3	66.6		3.8	5.3	A	0	-1	-4	-8	-0.44	-0.05	-1.19	0.21
Industrial Engineering	1.1	€ 34.9	72.1		3.7	3.3	A	-1	-3	-5	-5	-0.21	0.09	-0.47	0.69
Industrial Transportation	3.3	€ 105.3	82.6		3.9	4.6	BBB	0	-1	-2	-2	-0.36	0.01	-0.96	0.25
Support Services	1.0	€ 30.4	95.8		4.0	4.1	BBB	0	-4	0	1	-0.30	0.11	-0.61	0.76
Health Care	5.5	€ 172.1	76.5		4.0	5.7	A	0	-1	-1	-3	-0.46	-0.09	-1.40	0.13
Health Care Equipment & Services	2.4	€ 75.1	83.1		4.0	5.6	A	0	-1	1	0	-0.44	-0.10	-1.40	0.07
Pharmaceuticals & Biotechnology	3.1	€ 97.0	71.6		3.9	5.8	A	0	-1	-2	-5	-0.47	-0.09	-1.40	0.18
Basic Materials	2.6	€ 82.7	77.6		3.8	4.8	A	-1	-3	-5	-9	-0.33	0.07	-0.92	0.58
Chemicals	2.0	€ 62.5	77.5		3.9	4.9	A	0	-3	-4	-7	-0.38	0.05	-0.99	0.49
Forestry & Paper	0.2	€ 5.6	58.2		3.6	3.9	BBB	-13	-14	-19	-28	0.19	0.51	-0.16	1.42
Industrial Metals	0.1	€ 3.6	83.8		3.9	4.9	BBB	0	-1	-2	-9	-0.41	-0.02	-1.09	0.58
Mining	0.3	€ 11.0	85.0		3.9	4.0	BBB	0	-1	-2	-12	-0.31	0.02	-0.84	0.66
Technology	2.8	€ 88.1	95.3		4.2	5.5	A	1	-2	0	9	-0.52	-0.04	-1.60	-0.55
Software & Computer Services	2.4	€ 75.5	98.6		4.2	5.9	A	1	-2	0	9	-0.56	-0.05	-1.74	-0.68
Technology Hardware & Equipment	0.4	€ 12.6	63.7		3.6	3.6	A	0	-1	1	4	-0.26	0.04	-0.75	0.21
Telecommunications	5.5	€ 171.9	92.9		4.1	5.4	BBB	0	-3	1	1	-0.45	0.06	-1.39	0.07
Fixed Line Telecommunications	4.4	€ 138.7	92.1		4.1	5.5	BBB	0	-4	2	1	-0.45	0.07	-1.41	0.05
Mobile Telecommunications	1.1	€ 33.2	96.2		4.1	5.2	BBB	1	-3	0	0	-0.46	0.02	-1.34	0.18
Financials	42.2	€ 1,331.6	93.9		3.9	4.0	A	2	0	1	-1	-0.39	-0.02	-0.85	0.49
Banks Sen	26.5	€ 834.7	82.8		3.8	3.8	A	3	0	0	-1	-0.38	-0.03	-0.79	0.48
Banks Sub	4.1	€ 128.9	121.5		4.2	3.5	BBB	2	-2	-3	-5	-0.30	0.10	-0.55	0.93
Financial Services Sen	2.8	€ 89.0	80.2		3.9	4.4	A	0	0	-1	-3	-0.36	-0.02	-0.96	0.50
Financial Services Sub	0.0	€ 1.0	104.6		3.8	1.9	A	-8	-2	-8	-34	0.06	0.14	0.08	1.82
Insurance Sen	1.4	€ 43.1	79.2		3.8	4.2	A	3	1	2	4	-0.44	-0.11	-1.04	0.09
Insurance Sub	2.6	€ 83.4	134.5		4.4	5.1	A	3	-1	3	1	-0.53	0.00	-1.21	0.47
Real Estate Sen	4.8	€ 150.9	112.2		4.2	4.8	BBB	0	2	3	3	-0.39	-0.07	-1.13	0.32
Real Estate Sub	0.0	€ 0.5	173.5		4.7	5.5	A	-1	0	-11	0	-0.37	-0.09	-0.72	0.00
Utilities	11.6	€ 367.2	93.1		4.1	5.2	BBB	0	-1	-2	-1	-0.41	-0.04	-1.20	0.32
Electricity	6.9	€ 219.2	95.2		4.1	5.3	BBB	0	-2	-2	-2	-0.41	-0.03	-1.21	0.35
Gas / Water & Multiutilities	4.7	€ 148.0	90.0		4.0	5.2	BBB	0	-1	-1	-1	-0.42	-0.05	-1.20	0.26
Index	100.0	€ 3,154.3	89.1		4.0	4.6	A	1	-1	-1	0	-0.39	-0.01	-1.01	0.35

Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 21: Sector performance: EUR HY

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumers	26.8	€ 123.9	275		5.7	2.6	BB	-8	-10	-15	0	-0.03	0.34	0.23	2.21
Consumer Goods	14.8	€ 68.3	248		5.4	2.8	BB	-3	-6	-23	11	-0.04	0.42	0.46	2.32
Consumer Services	12.0	€ 55.6	313		6.1	2.4	BB	-16	-17	-3	-5	-0.02	0.23	-0.04	2.09
Energy	9.9	€ 45.7	238		5.3	3.0	BB	-8	-13	-18	-20	0.08	0.51	0.32	1.63
Energy	2.3	€ 10.5	213		5.0	2.6	BB	-13	-22	-31	-89	0.20	0.69	0.70	4.14
Utilities	7.6	€ 35.2	244		5.4	3.1	BB	-7	-11	-15	-2	0.04	0.46	0.21	0.92
Industrials	31.5	€ 145.6	296		5.9	2.7	BB	-6	-16	-31	-48	0.04	0.57	0.54	3.19
Basic Materials	6.2	€ 28.8	385		6.8	2.7	B	-9	-25	-33	-179	0.15	0.81	0.65	7.26
Construction & Materials	1.4	€ 6.6	545		8.4	2.9	BB	-6	46	16	-63	0.27	0.36	1.27	2.34
Health Care	8.7	€ 40.4	272		5.7	2.5	BB	3	-10	-40	5	-0.12	0.41	0.30	2.40
Industrials	15.1	€ 69.8	250		5.4	2.8	BB	-8	-20	-27	-37	0.08	0.58	0.56	2.27
TMT	19.7	€ 91.2	342		6.4	3.4	BB	-1	-6	12	56	-0.17	0.21	-0.51	0.71
Technology	4.6	€ 21.2	442		7.4	3.6	B	-1	-16	11	64	-0.22	0.75	0.59	0.29
Telecommunications	15.1	€ 69.9	309		6.1	3.4	BB	-2	-4	9	37	-0.15	0.05	-0.83	0.70
Financials	12.2	€ 56.5	347		6.4	3.1	BB	-3	-10	-17	20	-0.05	0.38	0.34	1.08
Core Financials	4.8	€ 22.0	188		4.8	3.1	BB	-2	-4	-7	9	-0.11	0.23	-0.11	1.11
Financial Services	1.6	€ 7.5	810		11.0	2.3	B	-6	-114	-157	209	0.00	0.29	2.32	-2.07
Real Estate	5.8	€ 27.0	380		6.7	3.4	BB	-5	-9	-3	2	-0.01	0.52	0.19	2.02
Index	100.0	€ 462.8	303		6.0	2.9	BB	-5	-11	-13	0	-0.03	0.41	0.20	2.03

0 50 100

Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: USD Credit ETFs

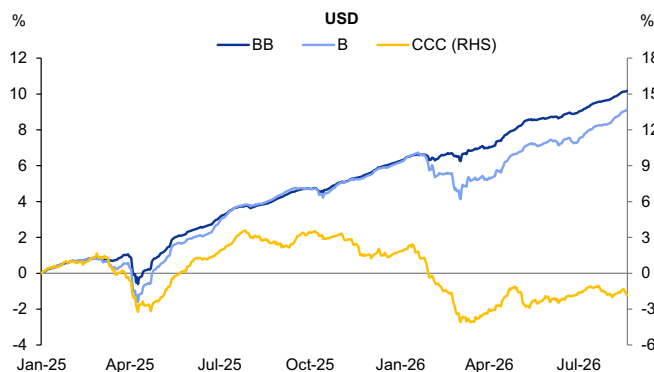
Exhibit 22: USD ETF performance

Sector	ETF Ticker	AUM	Weekly Flow (\$Mn.)	YTD cum flow (% of AUM)	Shares Outstanding (Mn.)	Short Interest (%)	Price	% NAV Premium	YTD Total Return	QTD Total Return	MTD Total Return	Benchmark Index
Investment Grade	LQD	\$32.98B	-324	13.4%	310	15.38	106.57	0.01%	-0.7	-1.5	0.7	iBoxx USD Liquid Investment Grade Index
	VCSH	\$45.16B	-21	14.0%	574	0.94	78.69	0.00%	1.3	0.3	0.5	Bloomberg Barclays US Corporate 1-5 Yrs Total Return Index Value Unhedged
	VCIT	\$68.36B	422	17.9%	839	1.97	81.42	-0.01%	0.0	-0.7	0.6	Bloomberg Barclays US Credit Corporate 5-10 Yrs Total Return Index Value Unhedged
	FLOT	\$10.6B	25	17.8%	208	--	50.97	0.05%	2.6	0.5	0.2	Bloomberg Barclays US FRN < 5 yr Total Return Index Value Unhedged
	IGSB	\$23.42B	23	9.6%	449	0.17	52.20	0.02%	1.4	0.4	0.4	ICE BofAML 1-5 Year US Corporate Bond Index
	IGIB	\$18.74B	8	11.3%	358	0.27	52.39	0.01%	0.0	-0.6	0.6	Bloomberg Barclays U.S. Interim
	SPSB	\$10.53B	-219	26.0%	351	3.60	29.96	0.01%	1.7	0.5	0.4	Bloomberg Barclays US Corporate 1-3 Yrs Total Return Index Value Unhedged
	SPIB	\$11.66B	225	8.4%	351	3.36	33.18	-0.03%	0.7	-0.1	0.5	Bloomberg Barclays Intermediate Corporate Total Return Index Value Unhedged
	VCIT	\$8.85B	-94	21.9%	122	14.27	72.38	0.00%	-1.5	-2.9	1.0	Bloomberg Barclays Long U.S. Corporate Total Return Index Value Unhedged
High Yield	HYG	\$17.58B	-191	-10.7%	221	42.36	79.71	0.13%	2.3	0.6	0.8	iBoxx USD Liquid High Yield Index
	JNK	\$6.66B	-77	-11.2%	69	10.49	95.96	0.17%	2.6	0.7	0.8	Bloomberg Barclays VLI: High Yield Total Return Index Value Unhedged USD
	SJNK	\$4.79B	15	-6.8%	193	1.19	24.89	0.21%	2.3	0.6	0.7	Bloomberg Barclays US High Yield 350M Cash 0-5yr 2% Cap Total Return Index
	SHYG	\$7.67B	63	-2.4%	182	0.16	42.25	0.19%	2.6	0.8	0.8	Markit iBoxx USD Liquid High Yield 0-5 Total Return Index
	HYS	\$1.78B	5	19.0%	19	0.09	92.95	0.08%	2.3	0.6	0.8	ICE BofAML 0-5 Year US High Yield Constrained Index
	HYLS	\$1.64B	0	-4.9%	40	0.12	40.86	0.27%	1.5	1.1	0.9	ICE BofAML US High Yield Constrained Index
	ANGL	\$3.17B	42	2.0%	109	0.52	29.07	0.08%	2.9	0.6	1.1	ICE BofAML US Fallen Angel High Yield Index
	BKLN	\$6.74B	-93	-0.2%	327	18.89	20.57	-0.20%	1.5	1.5	0.9	S&P/LSTA U.S. Leveraged Loan 100 Index
	SRLN	\$5.46B	0	-15.2%	135	9.14	40.57	-0.11%	2.4	1.9	1.0	Markit iBoxx USD Liquid Leverage Loans Total Return Index
Loans	FTSL	\$2.35B	4	1.9%	52	0.29	45.04	0.16%	1.8	1.2	0.5	S&P/LSTA US Leveraged Loan 100 Market Value Index
	EMB	\$14.95B	199	-8.1%	158	9.80	95.11	0.30%	1.8	-0.5	0.9	J.P. Morgan EMBI Global Core Index
	EMLC	\$4.81B	15	15.4%	187	0.59	25.81	0.37%	3.7	2.1	1.9	J.P. Morgan Government Bond Index Emerging Markets Global Core Index
EM	JAAA	\$30.02B	-78	23.4%	593	11.80	50.63	-0.02%	2.9	0.7	0.3	
	PAAA	\$11.77B	171	89.6%	229	--	51.43	-0.01%	3.0	0.7	0.3	
	JBBB	\$1.46B	9	23.0%	31	--	47.52	0.32%	3.1	0.8	0.5	
	CLOI	\$1.53B	0	15.6%	29	9.23	52.90	-0.03%	3.1	0.7	0.2	
	CLOA	\$2.27B	0	64.8%	44	--	51.88	-0.08%	3.0	0.7	0.2	
CLOs												

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

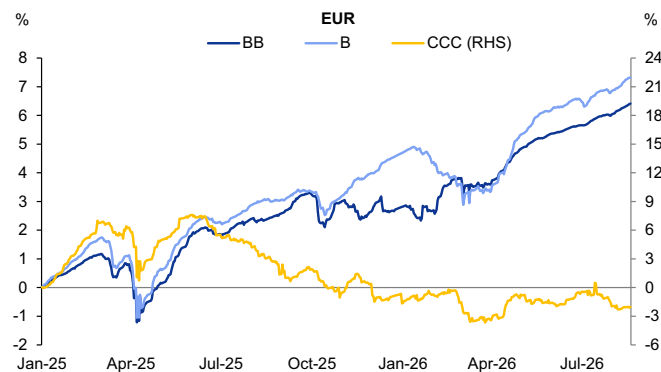
Performance dashboard: Leveraged Loans

Exhibit 23: Cumulative total return by rating for USD Leveraged Loans



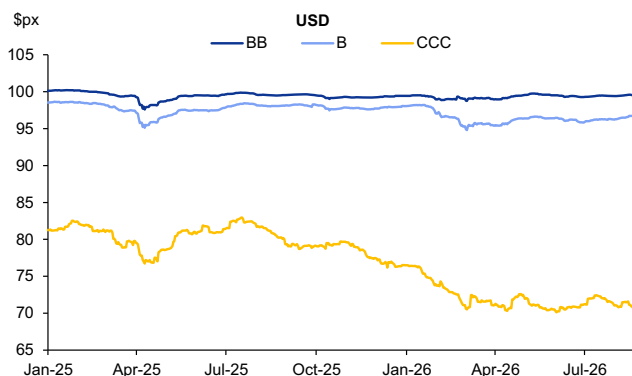
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 24: Cumulative total return by rating for EUR Leveraged Loans



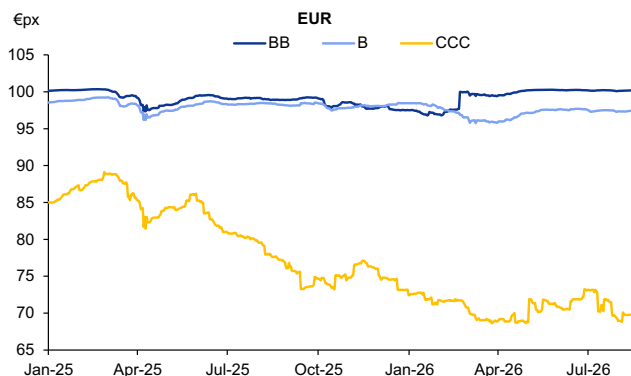
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 25: Bid price for USD Leveraged Loans by rating



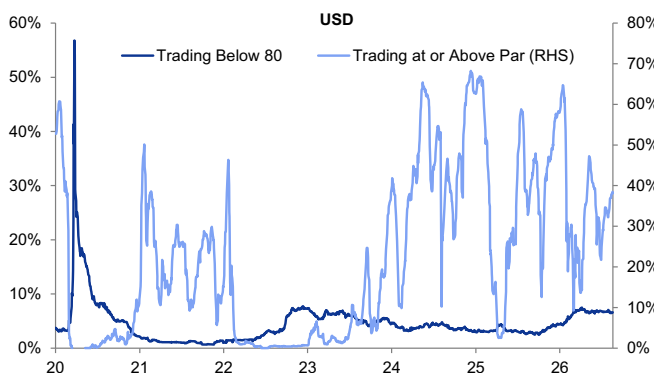
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 26: Bid price for EUR Leveraged Loans by rating



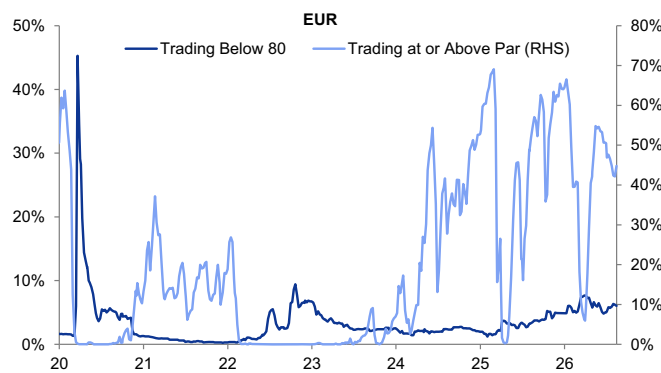
Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 27: Price breakdown for USD Leveraged Loans



Source: PitchBook, Goldman Sachs Global Investment Research

Exhibit 28: Price breakdown for EUR Leveraged Loans

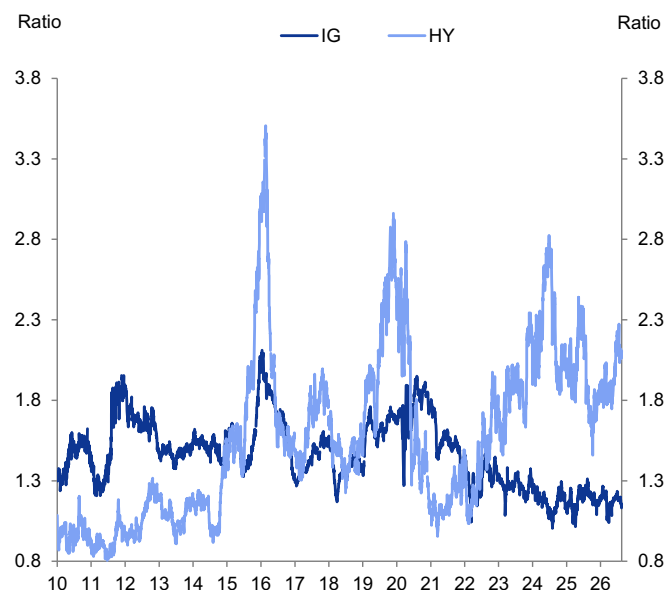


Source: PitchBook, Goldman Sachs Global Investment Research

Performance dashboard: Bond-level dispersion

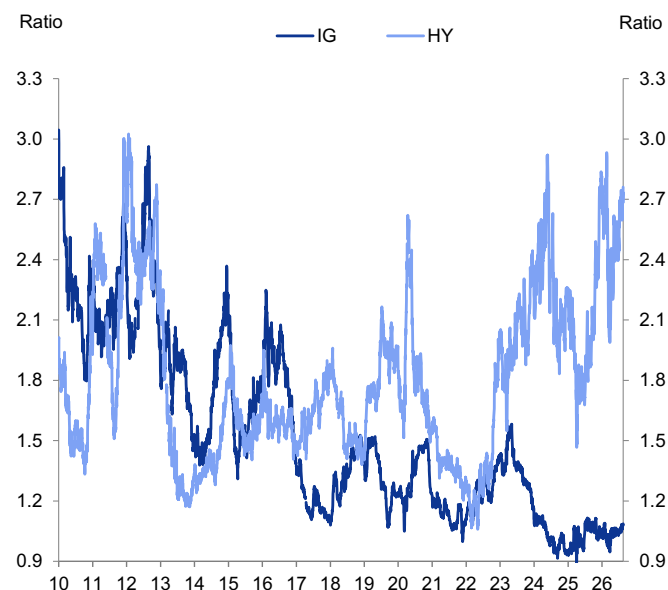
Note: Spread dispersion is the inter-decile range divided by the median across the bond index constituents and return dispersion is the inter-decile range of 10-day returns divided by the median across the bond index constituents.

Exhibit 29: USD spread dispersion



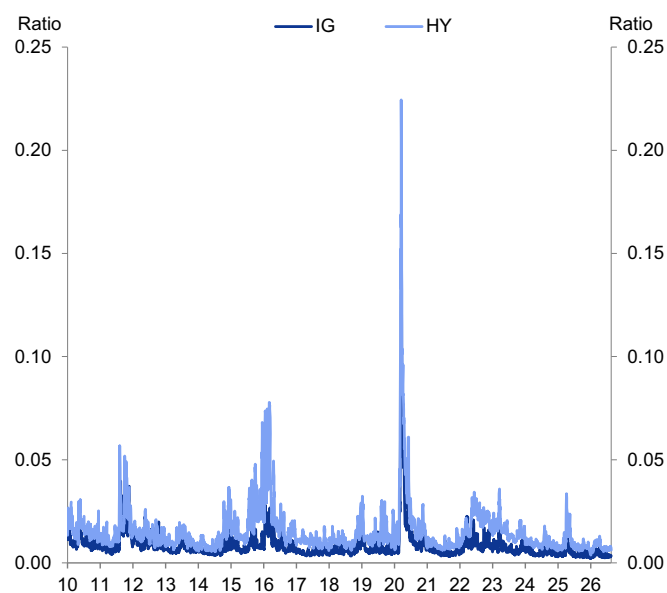
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 30: EUR spread dispersion



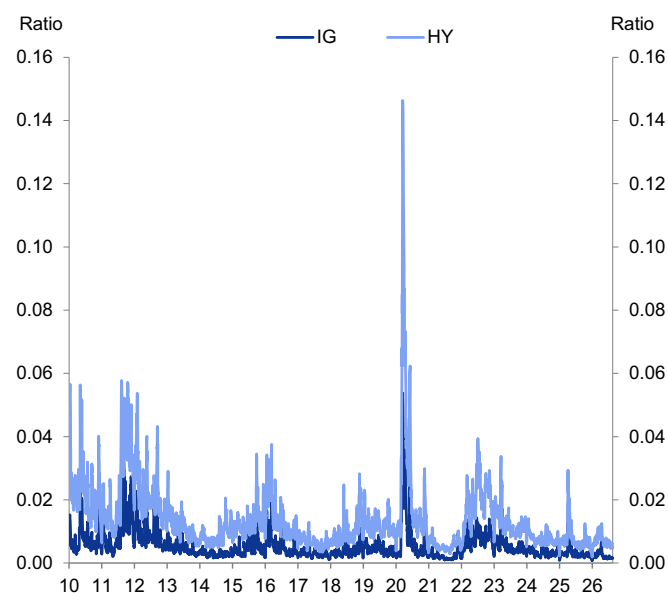
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 31: USD return dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 32: EUR return dispersion

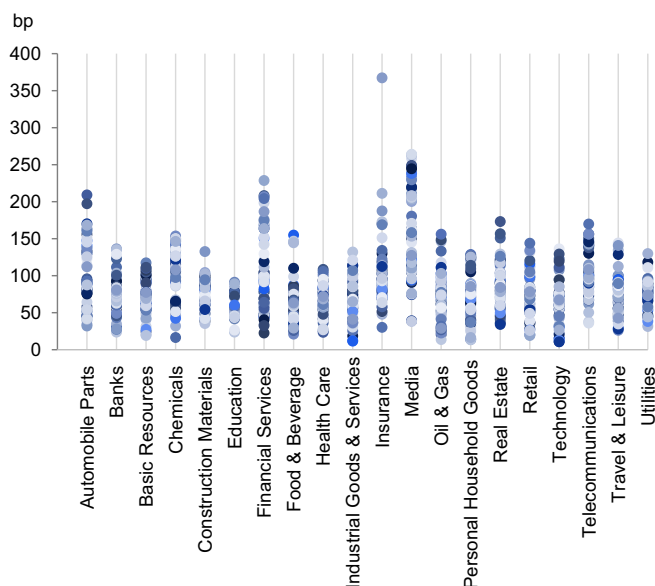


Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: Sector dispersion

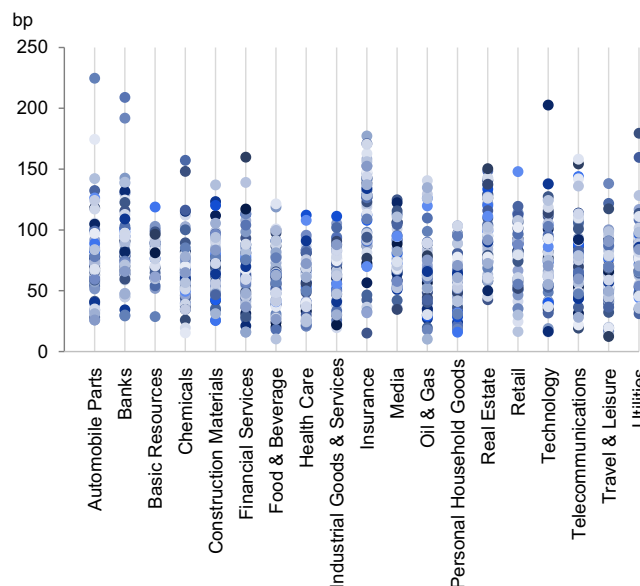
Issuer-level spreads are calculated using a market-value weighted average across the capital structure.

Exhibit 33: Distribution of issuer-level spreads by sector in the USD IG market



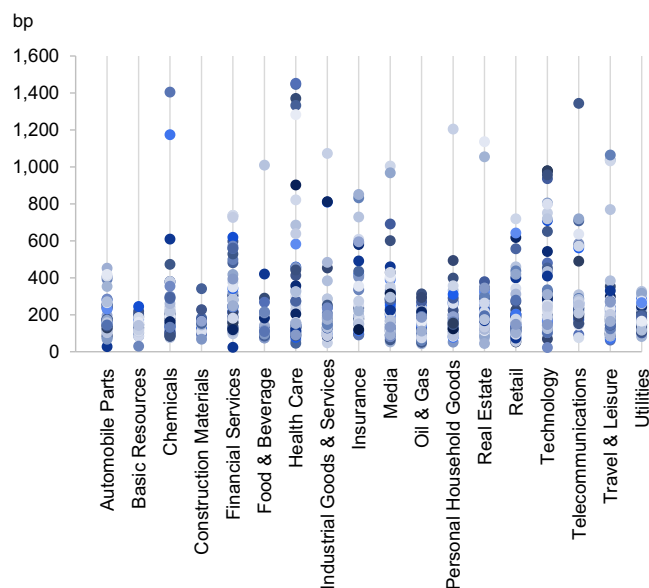
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 34: Distribution of issuer-level bond spreads by sector in the EUR IG market



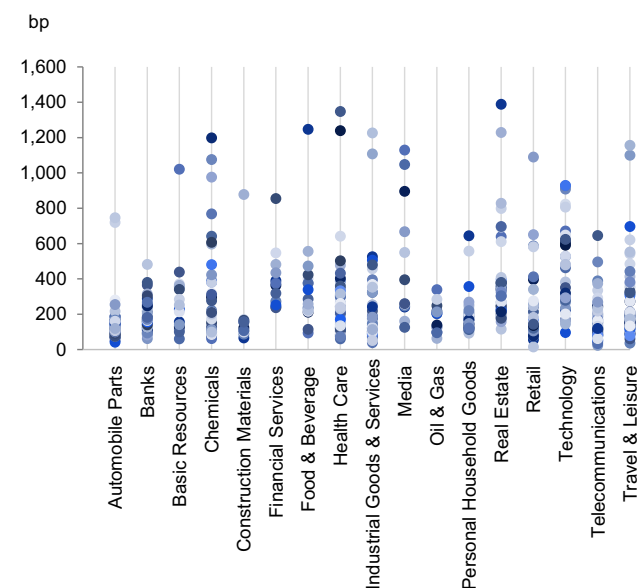
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 35: Distribution of issuer-level bond spreads by sector in the USD HY market



Source: Goldman Sachs Global Investment Research

Exhibit 36: Distribution of issuer-level bond spreads by sector in the EUR HY market



Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: Biggest movers in the USD bond market

Exhibit 37: Top and bottom 20 performers in the USD IG market

Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Comcast Corp	Media	CMCSA 5.35 05/15/53	1,600	A3/A-/A-	3.2%	2.6	84.53
Alphabet Inc	Internet	GOOGL 5 3/4 02/15/66	1,750	Aa2/AA+/-	3.0%	2.7	89.78
Salesforce Inc	Software	CRM 6.55 03/15/56	3,750	A2/A+/-	3.0%	2.8	95.66
Merck & Co Inc	Pharmaceuticals	MRK 4.9 05/17/44	750	Aa3/A+/-	2.8%	2.5	88.92
Paypal Holdings Inc	Commercial Services	PYPL 5.05 06/01/52	1,000	A3/A-/A-	2.7%	2.3	84.89
Meta Platforms Inc	Internet	META 6.45 05/15/66	4,000	Aa3/AA-/A-	2.7%	2.5	91.64
Conagra Brands Inc	Food	CAG 5.4 11/01/48	1,000	Baa3/BBB-/BBB-	2.2%	1.8	83.55
Oracle Corp	Software	ORCL 6 1/2 04/15/38	1,250	Baa2/BBB-/BBB	2.2%	2.1	96.59
Campbells Company/The	Food	CPB 5.4 03/21/34	1,000	Baa2/BBB-/BBB-	2.2%	2.1	96.53
Meta Platforms Inc	Internet	META 5 1/2 11/15/45	4,500	Aa3/AA-/A-	2.1%	1.8	87.03
Charter Comm Opt Llc/Cap	Media	CHTR 3.85 04/01/61	1,850	Ba1/BBB-/BBB-	2.1%	1.2	55.44
Oracle Corp	Software	ORCL 3.8 11/15/37	1,750	Baa2/BBB-/BBB	1.9%	1.5	76.51
Wells Fargo & Company	Banks	WFC 5.606 01/15/44	2,400	A3/BBB/A-	1.8%	1.7	93.71
Charter Comm Opt Llc/Cap	Media	CHTR 5 3/4 04/01/48	2,450	Ba1/BBB-/BBB-	1.8%	1.5	80.04
Meta Platforms Inc	Internet	META 5 5/8 11/15/55	6,500	Aa3/AA-/A-	1.8%	1.5	83.55
Merck & Co Inc	Pharmaceuticals	MRK 5.55 12/04/55	1,500	Aa3/A+/-	1.7%	1.6	92.79
Lyb Int Finance Iii	Chemicals	LYB 3 5/8 04/01/51	1,000	Baa3/BBB-/BBB	1.6%	1.1	63.82
Citigroup Inc	Banks	C 6 10/31/33	654	Baa2/BBB-/BBB+	1.6%	1.7	103.16
Charter Comm Opt Llc/Cap	Media	CHTR 6.484 10/23/45	3,500	Ba1/BBB-/BBB-	1.6%	1.4	87.71
Apple Inc	Computers	AAPL 2 3/8 02/08/41	1,500	Aaa/AA+/-	1.6%	1.1	68.31
Bottom 20 performers							
Assurant Inc	Insurance	AIZ 2.65 01/15/32	350	Baa2/BBB/-	-2.0%	-1.8	88.82
Sempra	Electric	SRE 6 7/8 10/01/54	1,100	Baa3/BBB-/BBB-	-2.0%	-2.1	104.05
Meta Platforms Inc	Internet	META 4.65 08/15/62	1,494	Aa3/AA-/A-	-1.9%	-1.4	71.63
Workday Inc	Software	WDAY 3.8 04/01/32	1,250	Baa1/BBB+/-	-1.9%	-1.8	95.06
Santander Holdings Usa	Banks	SANUSA 5.353 09/06/30	1,000	Baa2/BBB+/A-	-1.8%	-1.9	102.15
Concentrix Corp	Software	CNXC 6.85 08/02/33	550	Baa3/BBB-/BBB	-1.7%	-1.6	95.00
Freeport-Mcmoran Inc	Mining	FCX 5.45 03/15/43	1,706	Baa2/BBB-/BBB	-1.7%	-1.6	93.49
Brookfield Asset Managem	Diversified Finan Serv	BAMCN 4.653 11/15/30	600	-/A-/A-	-1.7%	-1.7	99.97
Unitedhealth Group Inc	Healthcare-Services	UNH 4 5/8 07/15/35	1,000	A2/A+/A	-1.6%	-1.6	96.94
Athene Holding Ltd	Insurance	ATH 6 7/8 06/28/55	600	Baa2/BBB-/BBB-	-1.5%	-1.5	94.46
Blackstone Private Cre	Investment Companies	BCRED 5 1/4 04/01/30	398	Baa2/BBB/-	-1.4%	-1.4	98.47
Comcast Corp	Media	CMCSA 4.6 10/15/38	969	A3/A-/A-	-1.4%	-1.2	89.02
Mastercard Inc	Diversified Finan Serv	MA 4.35 01/15/32	1,150	Aa3/A+/-	-1.4%	-1.4	99.31
Royal Caribbean Cruises	Leisure Time	RCL 5 3/8 01/15/36	1,500	Baa2/BBB/BBB	-1.4%	-1.4	98.48
Athene Holding Ltd	Insurance	ATH 6 5/8 10/15/54	600	Baa2/BBB-/BBB-	-1.3%	-1.3	93.15
General Motors Finl Co	Auto Manufacturers	GM 5.45 09/06/34	750	Baa2/BBB/BBB	-1.3%	-1.3	100.20
Blackstone Private Cre	Investment Companies	BCRED 6 01/29/32	1,000	Baa2/BBB/-	-1.3%	-1.3	99.56
Kyndryl Holdings Inc	Computers	KD 3.15 10/15/31	648	Baa3/BBB-/BBB	-1.3%	-1.1	84.08
Broadcom Inc	Semiconductors	AVGO 4.35 02/15/30	1,500	A3/A-/A-	-1.3%	-1.3	99.12
Ibm Corp	Computers	IBM 4.3 02/03/31	500	A3/A-/A-	-1.3%	-1.3	98.81

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

Exhibit 38: Top and bottom 20 performers in the USD HY market

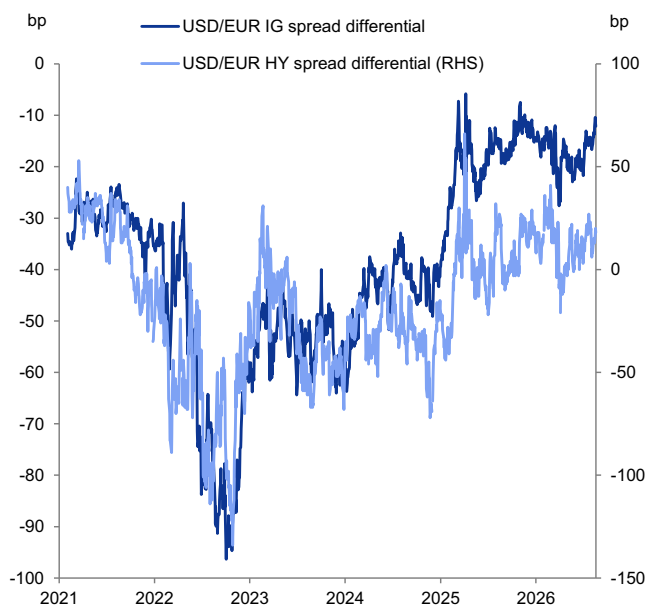
Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Chc Group Llc	Oil&Gas Services	CHHCF 11 3/4 09/01/30	550	--/--/--	5.1%	4.0	78.00
Gray Media Inc	Media	GTN 4 3/4 10/15/30	790	Caa1/CCC/--	1.9%	1.5	81.00
Virgin Medi O2 Vi Dac	Media	VMED 8 1/2 03/15/33	500	B3/B-/B-	1.8%	1.2	65.82
Merlin Entertainments Gr	Entertainment	MERLLN 7 3/8 02/15/31	500	B3/B-/B-	1.7%	1.3	76.91
Resideo Funding Ii Llc	Distribution/Wholesale	REZI 6 1/2 07/15/32	600	Ba3/BB/--	1.4%	1.4	98.81
Fmc Corp	Chemicals	FMC 8.45 11/01/55	750	Ba3/B+/BB-	1.3%	0.9	73.50
Domtar Corp	Forest Products&Paper	UFS 6 3/4 10/01/28	642	B2/CCC+/BB-	1.2%	0.9	77.75
Wr Grace Holding Llc	Chemicals	GRA 5 5/8 08/15/29	1,155	Caa2/CCC/B-	1.2%	1.1	93.86
Ineos Quattro Finance 2	Chemicals	STYRO 9 5/8 03/15/29	400	B3/B/B+	1.2%	1.0	84.50
Bcpe Flavor Debt/Issuer	Retail	SIZPLT 9 1/2 07/01/32	500	B3/B-/B-	0.9%	0.8	95.54
Mcafee Corp	Computers	MCFE 7 3/8 02/15/30	1,733	Caa1/CCC+/--	0.8%	0.7	88.75
Ellucian Holdings Inc	Software	DAEL 6 1/2 12/01/29	700	B2/B-/B-	0.8%	0.8	96.75
Rocket Software Inc	Software	ROCISO 9 11/28/28	800	B3/B-/BB-	0.7%	0.8	100.75
Paramount Global	Media	PARA 6 7/8 04/30/36	1,080	Baa3/BB+/BB+	0.7%	0.6	93.06
Gray Media Inc	Media	GTN 5 3/8 11/15/31	1,198	Caa1/CCC/--	0.6%	0.5	78.00
Vmed O2 Uk Finan	Telecommunications	VMED 4 3/4 07/15/31	1,400	Ba3/B+/BB	0.6%	0.5	80.38
Chs/Community Health Sys	Healthcare-Services	CYH 4 3/4 02/15/31	689	B3/B-/B-	0.6%	0.6	90.50
Hlf Fin Sarl Llc/Herbali	Pharmaceuticals	HLF 7 3/4 05/01/33	800	Ba2/BB/--	0.6%	0.6	101.36
Virgin Media Secured Fin	Media	VMED 4 1/2 08/15/30	915	Ba3/B+/BB	0.6%	0.5	82.50
California Resources Crp	Oil&Gas	CRC 7 1/4 01/15/35	550	B1/BB-/BB-	0.6%	0.6	99.98
Bottom 20 performers							
Mercer Intl Inc	Forest Products&Paper	MERC 5 1/8 02/01/29	872	Ca/CCC+/CCC	-41.8%	-15.1	36.20
Xerox Holdings Corp	Office/Business Equip	XRX 8 7/8 11/30/29	500	Caa3/CCC/--	-7.4%	-3.3	44.00
Iheartcommunications Inc	Media	IHRT 10 7/8 05/01/30	675	Caa3/CCC/--	-6.4%	-4.8	74.00
Jetblue Airways/Loyalty	Airlines	JBLU 9 7/8 09/20/31	2,000	Caa1/B/B	-4.8%	-4.3	89.75
Hertz Corp/The	Commercial Services	HTZ 12 5/8 07/15/29	1,250	Ba3/CCC+/B-	-4.5%	-3.2	72.03
Ion Plat Fin Us/Sarl	Internet	IONPLA 5 3/4 05/15/28	450	B2/B+/B-	-3.3%	-3.1	96.13
Csc Holdings Llc	Media	CSCHLD 11 3/4 01/31/29	2,050	Caa2/CCC/--	-3.1%	-1.8	57.00
Wilsonart Llc	Building Materials	WLSNRT 11 08/15/32	500	Caa2/CCC/--	-3.1%	-2.4	80.00
Stonebriar Abf Llc	Diversified Finan Serv	STONEB 8 1/8 12/15/30	750	Ba2/--/BB	-3.0%	-3.2	105.19
Lumen Technologies Inc	Telecommunications	LUMN 7.6 09/15/39	349	Caa1/B/BB	-3.0%	-2.8	91.68
Westjet Alb Partnership	Airlines	WJAPAR 8 02/14/31	500	B1/--/B	-3.0%	-3.0	100.00
Csc Holdings Llc	Media	CSCHLD 11 1/4 05/15/28	1,000	Caa2/CCC/--	-2.8%	-1.6	57.38
Ion Platform Finance Us	Commercial Services	IONPLA 7 7/8 09/30/32	1,500	B2/B+/B-	-2.8%	-2.2	78.75
Pagaya Us Holdings Co	Software	PGY 8 7/8 08/01/30	482	B3/B/B	-2.5%	-2.3	89.00
Iheartcommunications Inc	Media	IHRT 7 3/4 08/15/30	661	Caa1/CCC+/--	-2.4%	-2.0	85.00
Ion Plat Fin Us/Sarl	Internet	IONPLA 9 1/2 05/30/29	773	B2/B+/B-	-2.3%	-2.1	93.25
Stonebriar Abf Llc	Diversified Finan Serv	STONEB 7 08/15/31	500	Ba2/--/BB(EXP)	-2.2%	-2.3	100.75
Cordia Finanz Gmbh	Chemicals	PLATIN 9 3/8 10/03/31	900	B2/B-/B-	-2.2%	-2.1	92.25
Lumen Technologies Inc	Telecommunications	LUMN 7.65 03/15/42	285	Caa1/B/BB	-2.1%	-1.9	92.20
Coreweave Inc	Software	CRWV 9 5/8 07/15/32	1,250	B1/B/BB-	-2.1%	-1.9	92.19

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

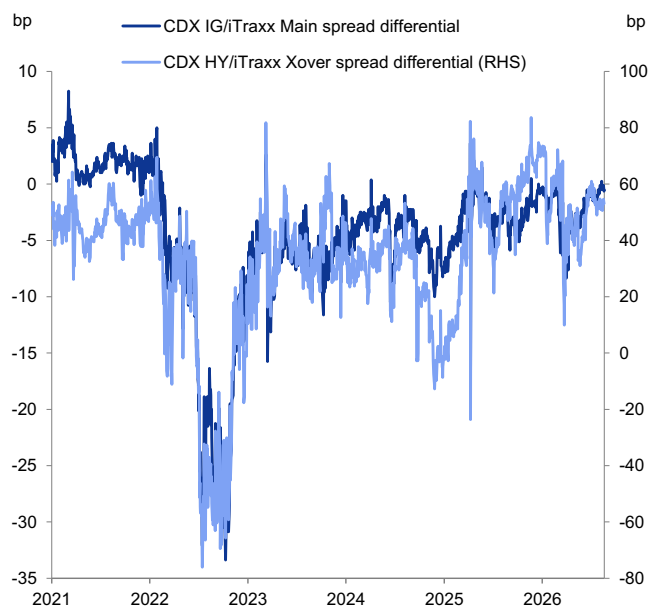
Relative value dashboard: Excess spreads in the USD and EUR markets

Exhibit 39: USD/EUR IG and HY spread differential



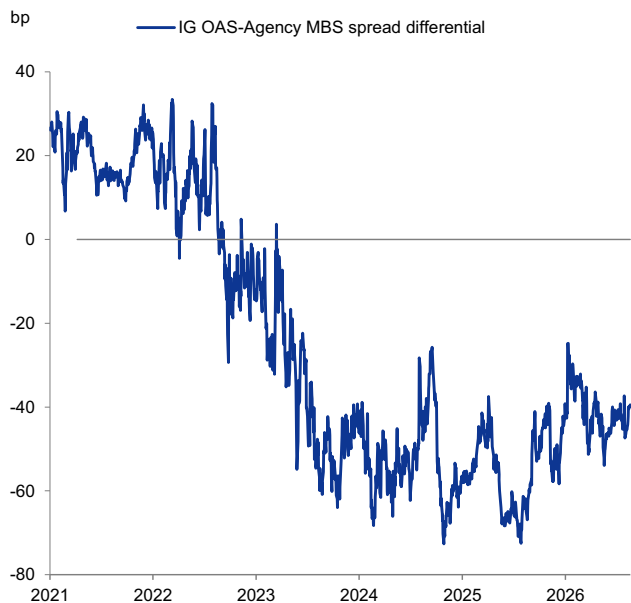
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 40: CDX IG/iTraxx Main and CDX HY/iTraxx Xover spread differential



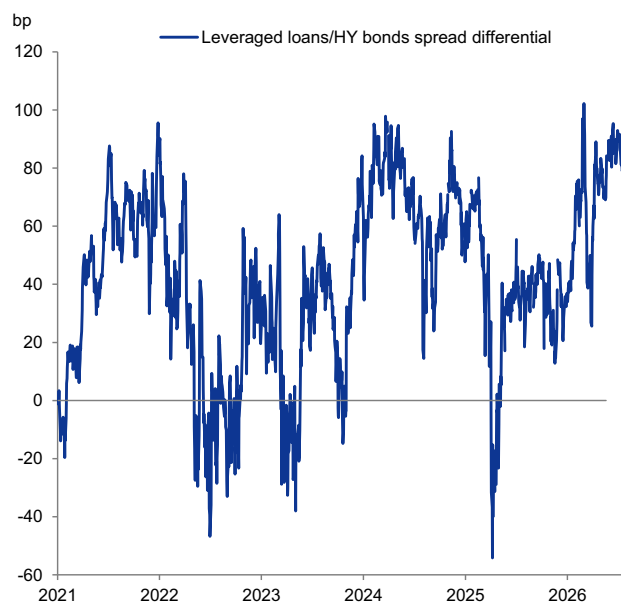
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 41: IG OAS/Agency MBS spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

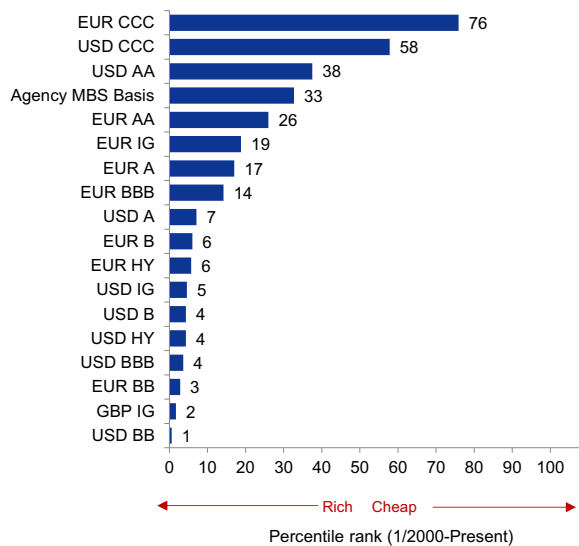
Exhibit 42: Leveraged loans/HY bonds spread differential



Source: Bloomberg, Morningstar LCD, Goldman Sachs Global Investment Research

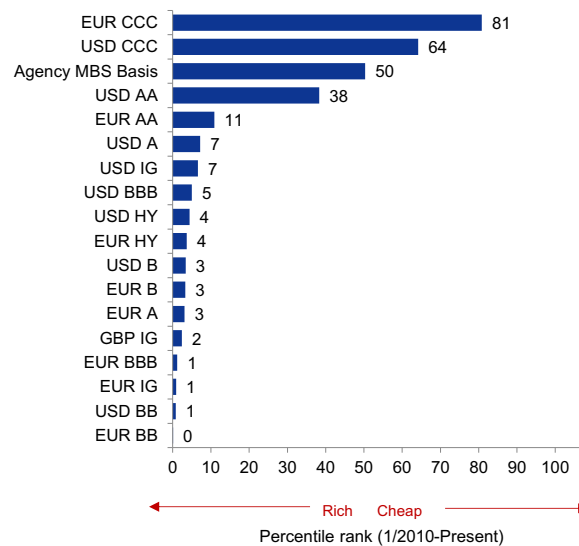
Valuation dashboard: Percentile ranks for spreads and yields

Exhibit 43: Percentile ranks for spreads across markets (2000-current)



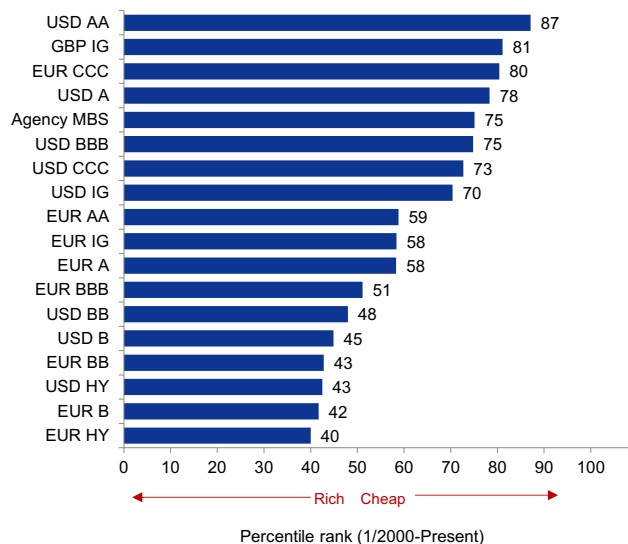
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 44: Percentile ranks for spreads across markets (2010-current)



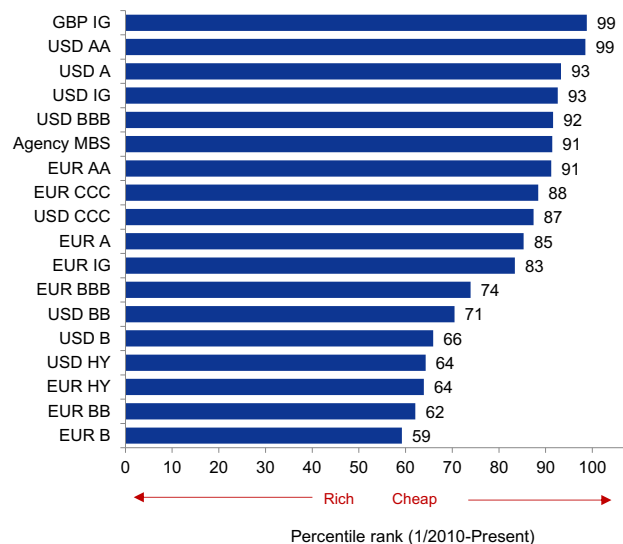
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 45: Percentile ranks for yields across markets (2000-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

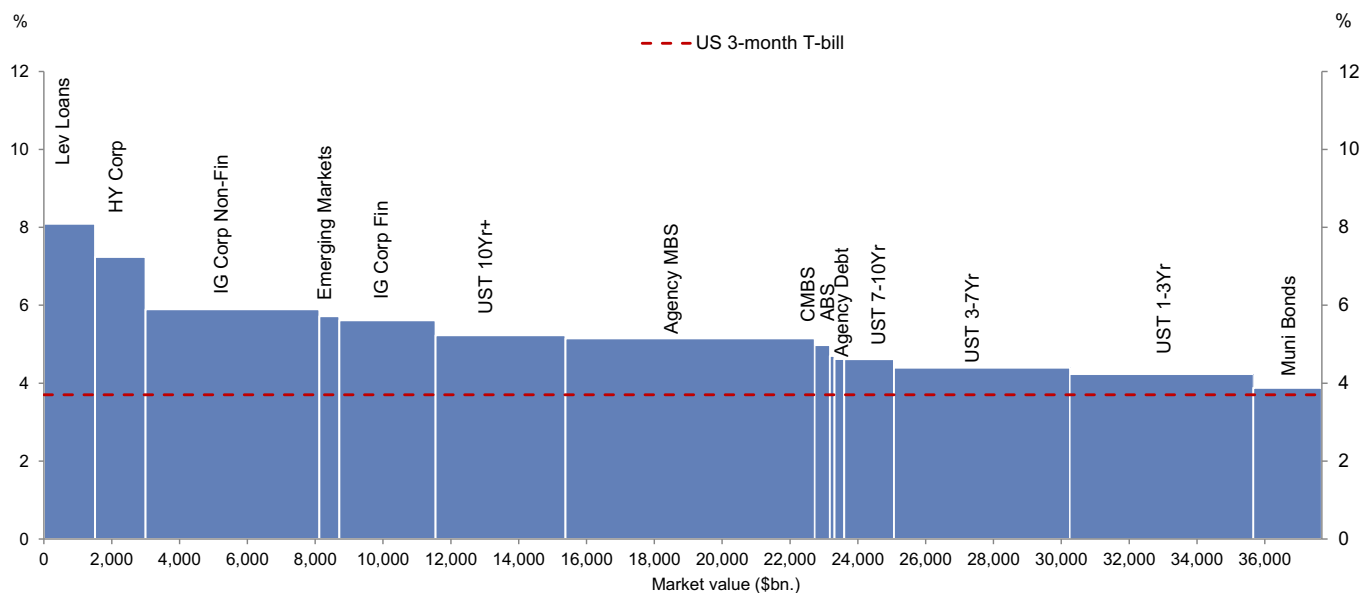
Exhibit 46: Percentile ranks for yields across markets (2010-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

Valuation dashboard: Sizing the carry in USD and EUR fixed income markets

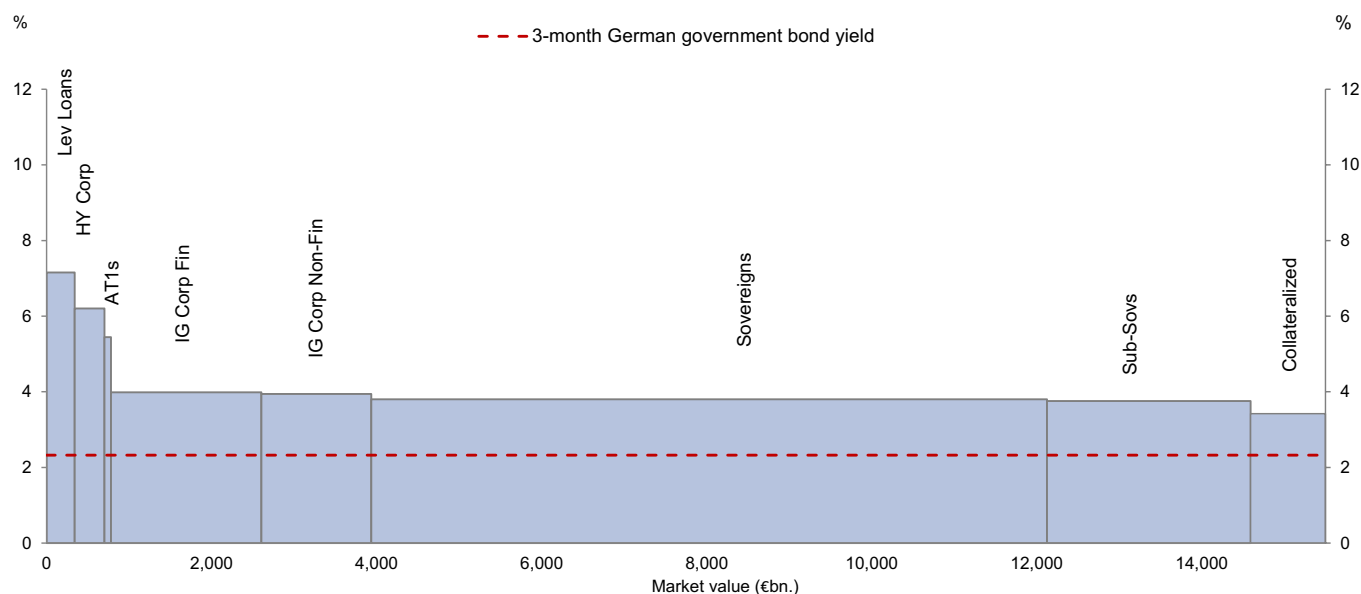
Exhibit 47: Average yield vs. total market value outstanding across the USD fixed income complex



Note: We use the iBoxx indices for the IG corporate credit and US Treasury markets, the Bloomberg indices for the HY, MBS, CMBS and ABS markets, and the Morningstar LSTA index for the leveraged loan market.

Source: iBoxx, Bloomberg, Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 48: Average yield vs. total market value outstanding for the EUR fixed income markets

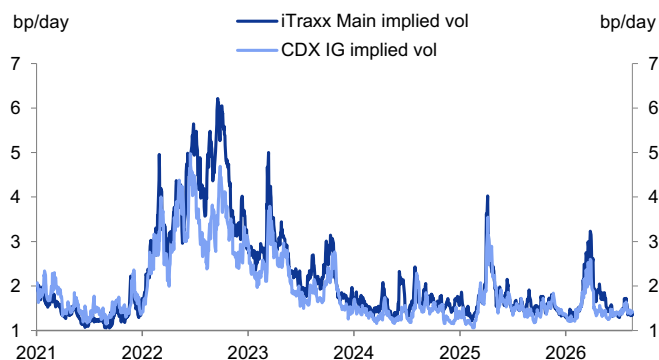


Source: iBoxx, Goldman Sachs Global Investment Research

Valuation dashboard: Macro index volatility

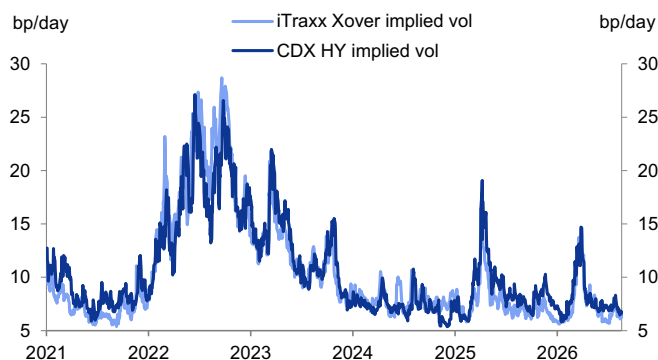
Note: Skew is calculated using difference in 3m implied volatility between 25-delta payers and receivers, divided by 3m ATM implied volatility.

Exhibit 49: CDX IG and iTraxx Main vol 3m ATM implied volatility



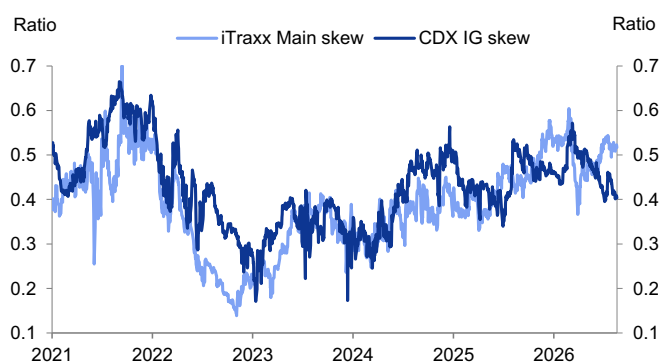
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 50: CDX HY and iTraxx Xover vol 3m ATM implied volatility



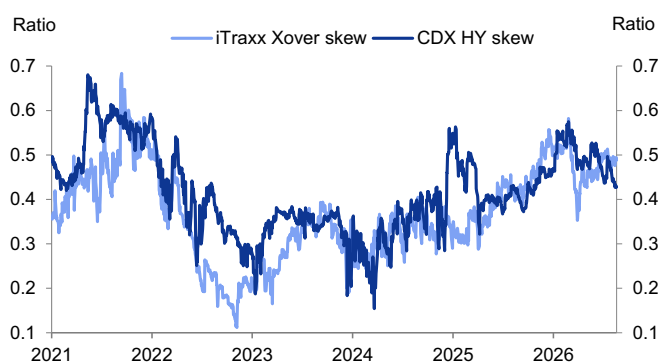
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 51: CDX IG and iTraxx Main skew



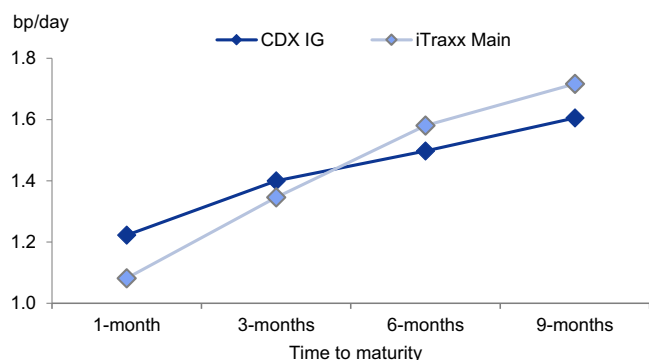
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 52: CDX HY and iTraxx Xover skew



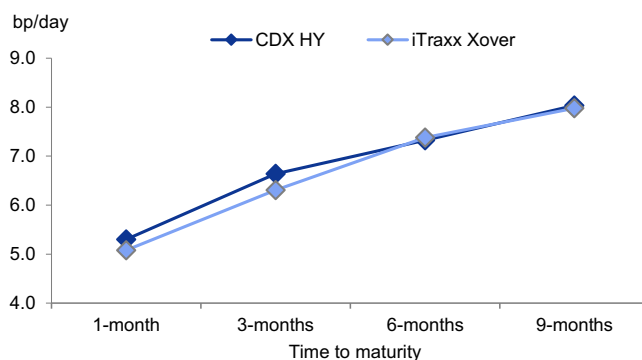
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 53: CDX IG and iTraxx Main ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 54: CDX HY and iTraxx Xover ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Valuation dashboard: CLOs

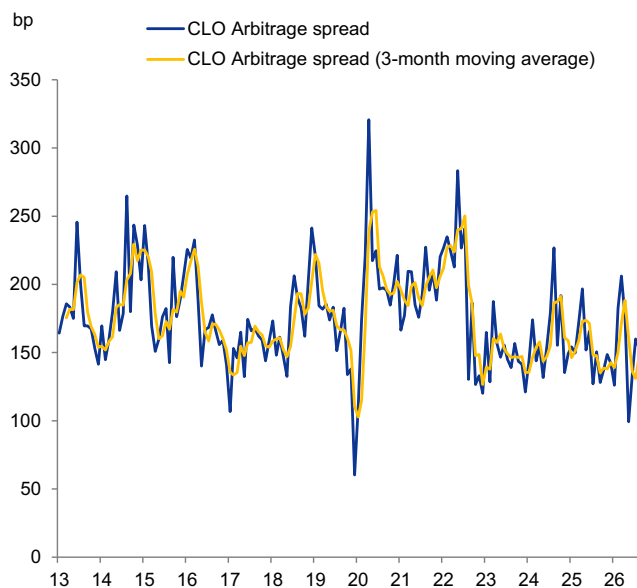
Exhibit 55: CLO spread percentile ranks

USD	Current spread (bp)	Percentile rank		
		3-year	5-year	10-year
AAA	112	22%	13%	9%
AA	143	8%	5%	2%
A	167	3%	2%	1%
BBB	290	38%	23%	17%
BB	772	70%	59%	69%

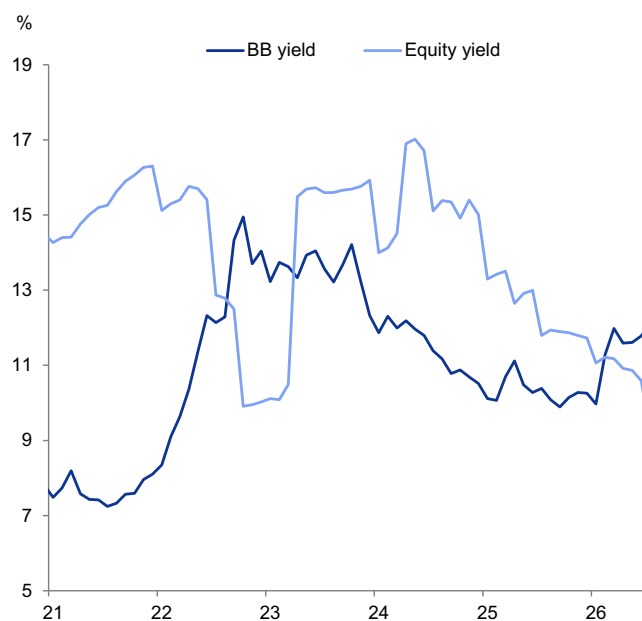
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 57: CLO new issue arbitrage spread

New issue CLO arbitrage spread, i.e., the difference between new issue loan spreads and new issue CLO spreads (across the capital stack)



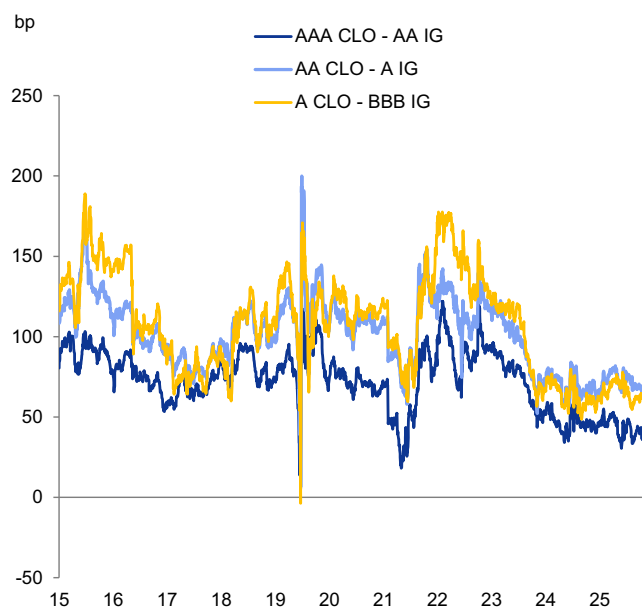
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 56: CLO equity vs. CLO BB yield

Source: Intex, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 58: CLO vs. USD IG spread differential

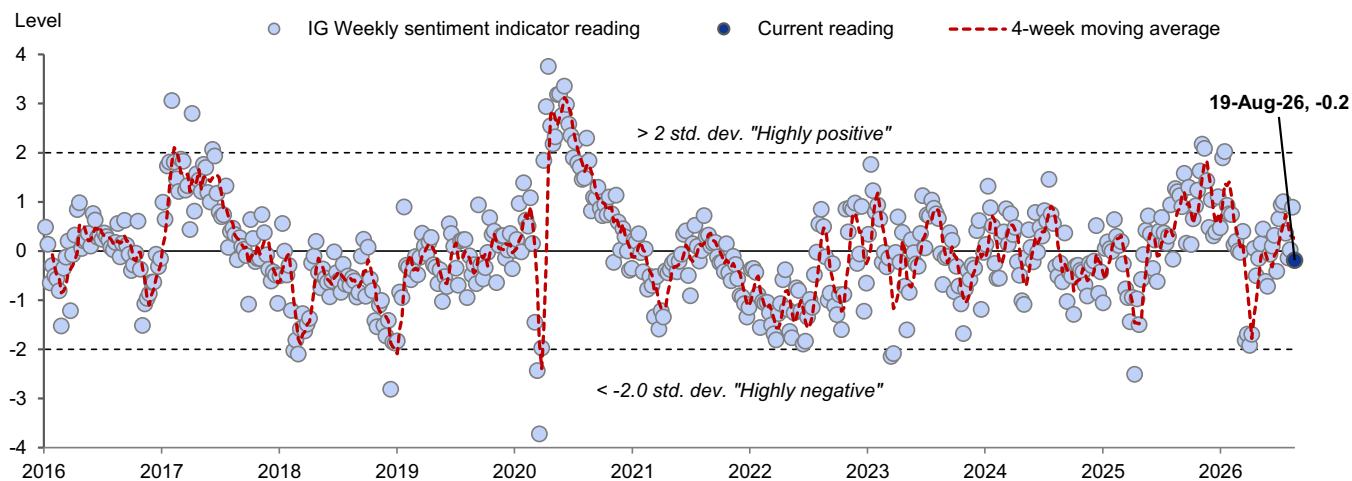
CLO by rating and USD IG bond by rating spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

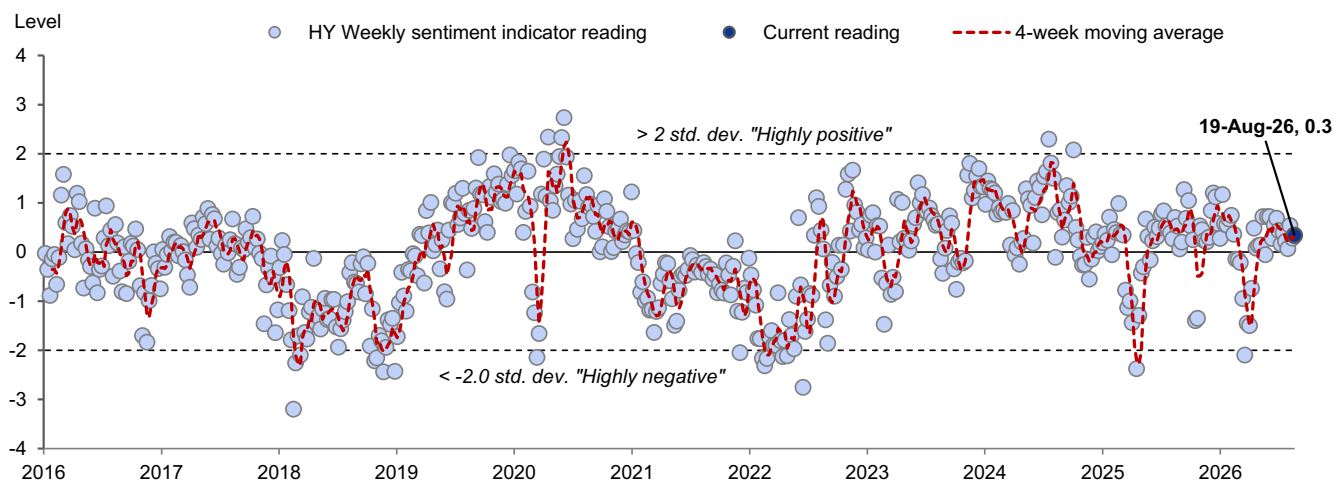
USD IG and HY Sentiment indicators

Exhibit 59: IG Weekly sentiment indicator



Source: See table below.

Exhibit 60: HY Weekly sentiment indicator



Source: See table below.

Exhibit 61: Variables included in the USD IG and HY weekly sentiment indicators

Measure	Frequency	Data Source
Synthetic index Positioning	Weekly	DTCC
Net Dealer Positioning	Weekly	FRBNY
Net Dealer Lift	Weekly	TRACE
Active US Credit Fund flows	Weekly	EPFR
Passive US Credit Fund flows	Weekly	EPFR
US Money Market Fund flows	Weekly	EPFR
Short interest on the largest credit ETFs	Semimonthly	GS FICC & Equities
Credit Mutual Fund cash positions	Monthly	ICI

Source: Goldman Sachs Global Investment Research

Microstructure dashboard: Top 20 most traded capital structures in the USD IG and HY markets

Exhibit 62: Top 20 most liquid issuers by weekly turnover in the USD IG market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
NVDA	NVIDIA Corp	32.5	1065.9	3.3	AA	1.5	7.0
META	Meta Platforms Inc	84.0	1653.8	2.0	AA-	1.3	9.4
AVGO	Broadcom Inc	54.1	969.1	1.8	A-	1.6	6.2
VALEBZ	Vale Overseas Ltd	6.5	114.3	1.8	BBB	7.6	7.9
INTU	Intuit Inc	5.5	92.6	1.7	A-	2.3	6.9
BG	Bunge Ltd Finance Corp	6.3	101.2	1.6	BBB+	2.5	4.8
BCRED	Blackstone Private Credit Fund	6.6	100.8	1.5	BBB-	1.5	3.8
ACN	Accenture Capital Inc	9.7	143.0	1.5	AA-	1.0	4.6
ATH	Athene Global Funding	16.3	209.9	1.3	A+	3.1	5.3
JBS	JBS NV/JBS USA Foods Group Holdings Inc/JBS USA Food C	15.8	201.2	1.3	BBB-	1.7	9.2
CAG	Conagra Brands Inc	5.8	73.6	1.3	BBB-	5.6	5.6
GOOGL	Alphabet Inc	51.0	638.1	1.3	AA	1.6	9.6
BBVASM	Banco Bilbao Vizcaya Argentaria SA	8.4	95.6	1.1	A-	1.5	4.1
SCCO	Southern Copper Corp	6.1	68.0	1.1	BBB+	11.9	9.3
PYPL	PayPal Holdings Inc	9.5	106.0	1.1	A-	3.8	7.1
ETN	Eaton Corp	13.9	155.0	1.1	A-	3.0	5.9
AMZN	Amazon.com Inc	120.5	1331.3	1.1	AA-	2.6	8.6
BAX	Baxter International Inc	6.4	70.7	1.1	BBB-	3.4	5.7
APH	Amphenol Corp	11.8	127.2	1.1	A-	2.0	6.3
DOW	The Dow Chemical Company	12.7	128.4	1.0	BBB-	7.8	9.2

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

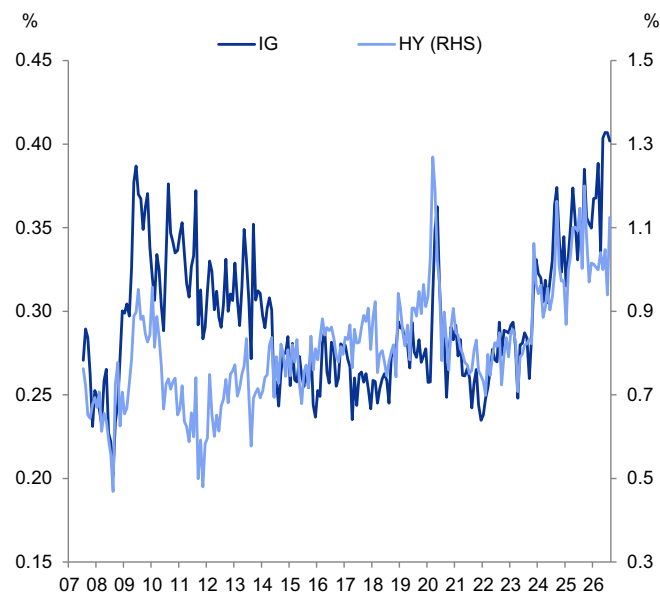
Exhibit 63: Top 20 most liquid issuers by weekly turnover in the USD HY market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
GALAXY	Galaxy Helios Data Centers II LLC	3.5	111.4	3.2	BB-	0.0	3.7
IHRT	iHeartCommunications Inc	2.1	52.7	2.6	CCC+	1.6	2.9
CRWV	CoreWeave Inc	7.8	198.8	2.6	B+	0.7	3.8
ACRISU	Acrisure LLC / Acrisure Finance Inc	4.3	106.1	2.5	B	3.0	3.0
IONPLA	ION Platform Finance US Inc / ION Platform Finance SARL	3.9	88.9	2.3	B	0.8	3.2
CORZ	Core Scientific Finance I LLC	3.3	75.2	2.3	BB-	0.2	2.9
MIK	Michaels Cos Inc/The	2.8	57.6	2.1	B-	0.4	5.0
GTN	Gray Media Inc	4.8	95.3	2.0	CCC	3.0	3.2
APLD	APLD ComputeCo LLC	2.4	42.6	1.8	BB-	0.7	2.8
PFORGE	APLD ComputeCo 2 LLC	2.2	38.7	1.8	BB-	0.4	3.1
GSYCN	goeasy Ltd	2.4	40.3	1.7	B-	1.9	2.8
CLGX	CoreLogic Inc	2.5	40.4	1.6	B-	0.0	4.1
JBLU	JetBlue Airways Corp / JetBlue Loyalty LP	2.0	32.7	1.6	B	1.9	3.9
ARGID	Ardagh Group SA	2.7	43.3	1.6	B-	0.7	3.1
GT	The Goodyear Tire & Rubber Company	4.0	62.9	1.6	B+	3.1	3.9
NOVHOL	Clydesdale Acquisition Holdings Inc	3.5	55.0	1.6	B+	2.8	3.2
AXL	American Axle & Manufacturing Inc	2.7	41.8	1.5	B+	1.7	3.9
RRD	RR Donnelley & Sons Co	2.4	35.9	1.5	B-	1.3	2.5
WHR	Whirlpool Corp	3.9	57.7	1.5	BB-	1.7	4.2
EQMSRM	EquipmentShare.com Inc	3.5	47.4	1.4	B	1.7	3.4

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

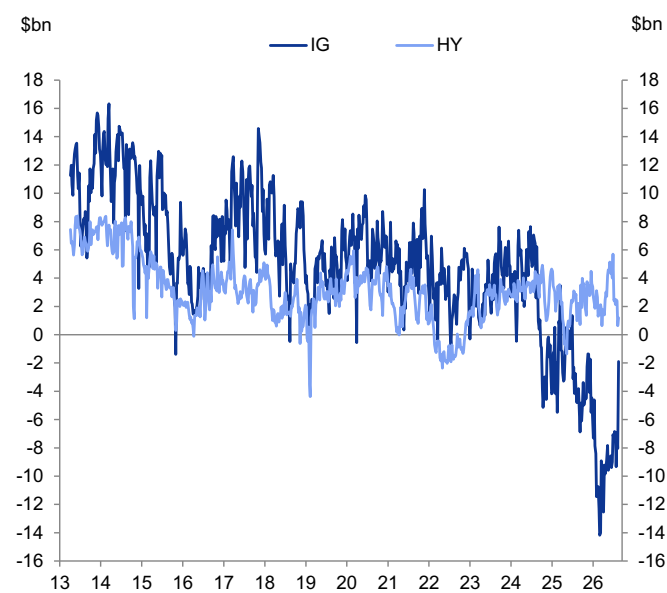
Microstructure dashboard: Liquidity conditions in the USD secondary bond market

Exhibit 64: Seasonally adjusted daily average turnover (trading volume/amount outstanding)



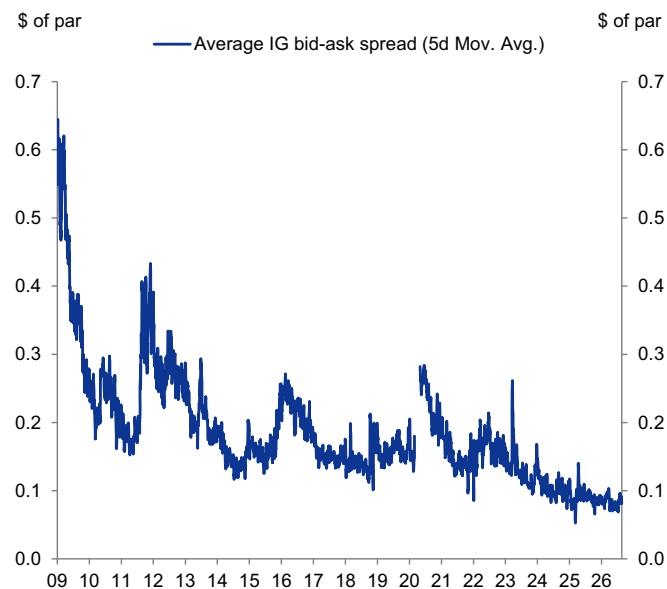
Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 65: Primary dealer net positioning in USD corporate bonds



Source: FRBNY, Goldman Sachs Global Investment Research

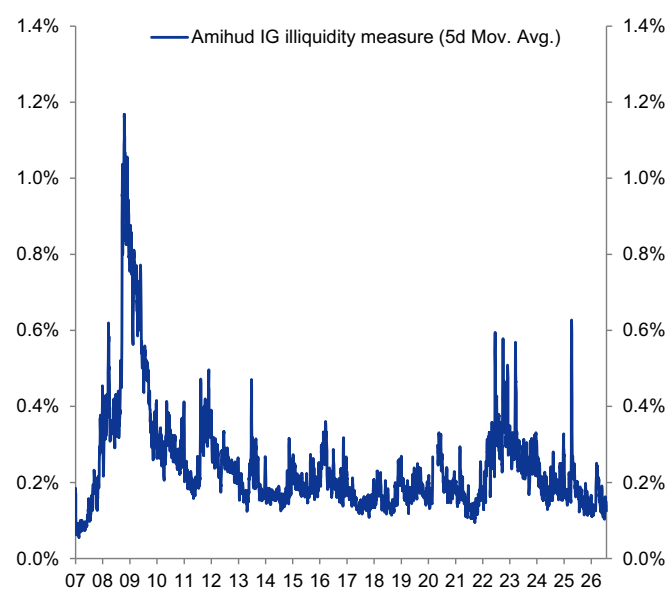
Exhibit 66: Average IG bid-ask spread 5-day moving average



Note: March and April 2020 are omitted.

Source: Trace, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 67: Amihud IG illiquidity measure 5-day moving average

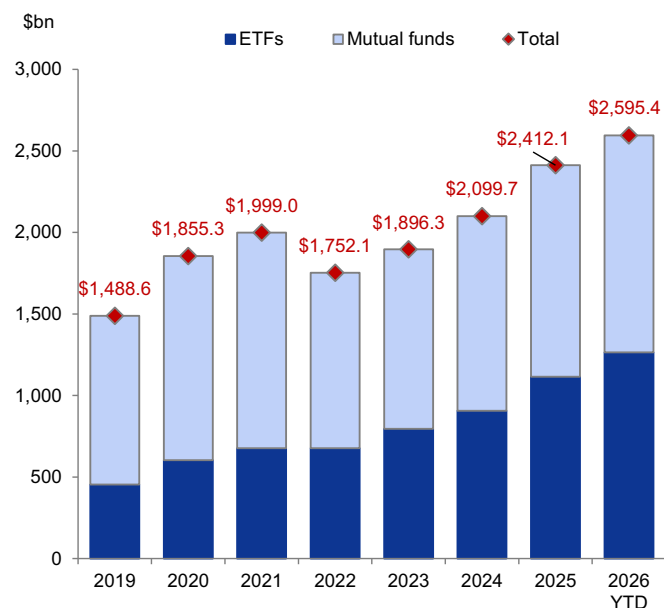


Note: March and April 2020 are omitted.

Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

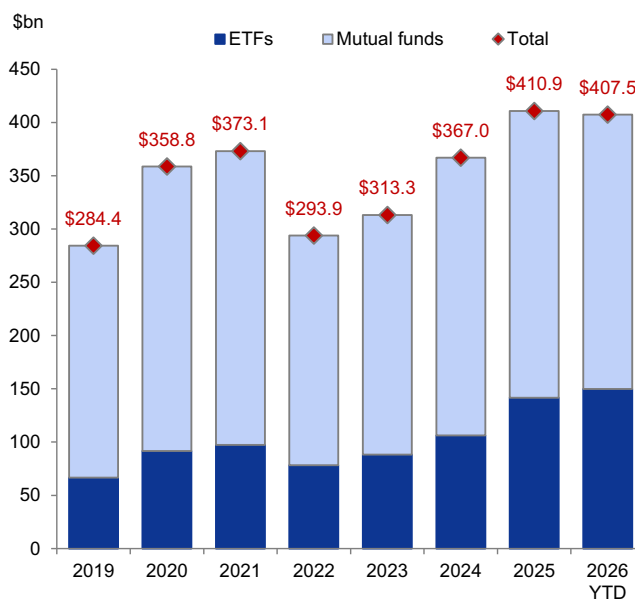
Flow dashboard: Mutual fund and ETF AUM

Exhibit 68: AUM in USD IG-focused Mutual Funds and ETFs



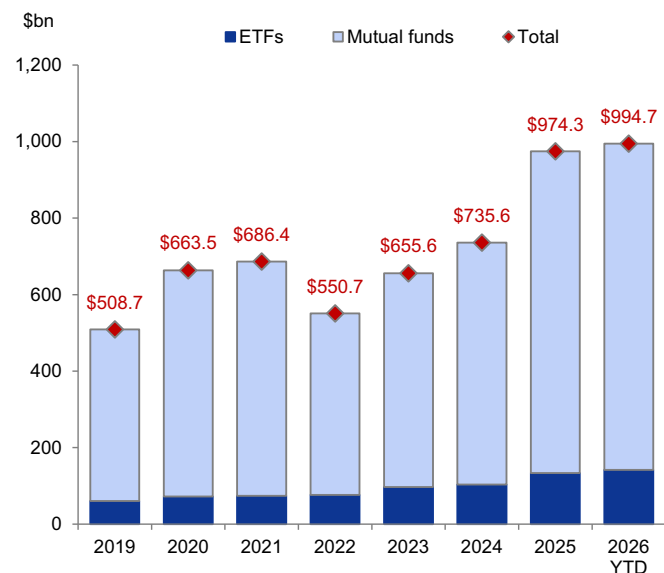
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 69: AUM in USD HY-focused Mutual Funds and ETFs



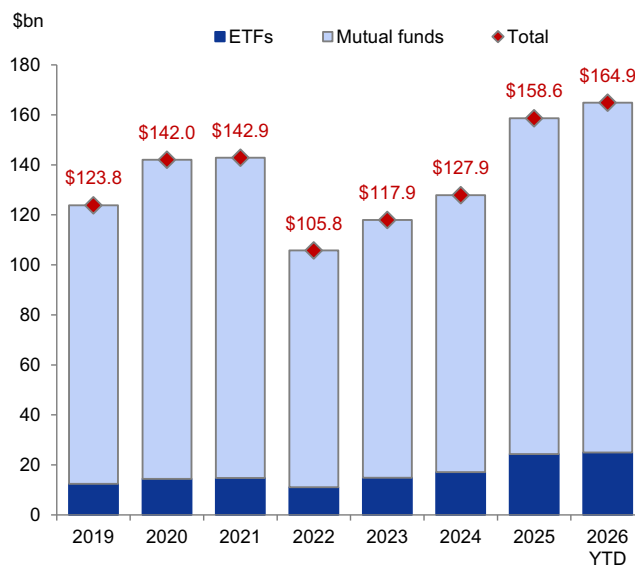
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 70: AUM in EUR IG-focused Mutual Funds and ETFs



Source: EPFR, Goldman Sachs Global Investment Research

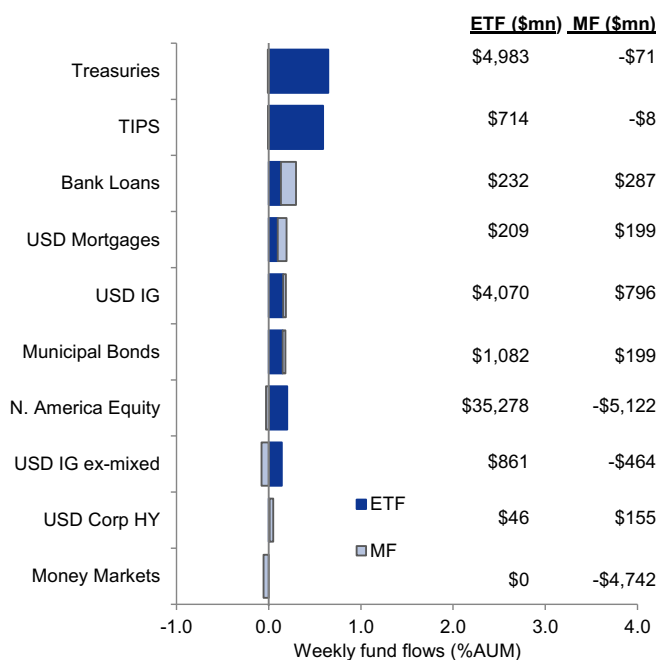
Exhibit 71: AUM in EUR HY-focused Mutual Funds and ETFs



Source: EPFR, Goldman Sachs Global Investment Research

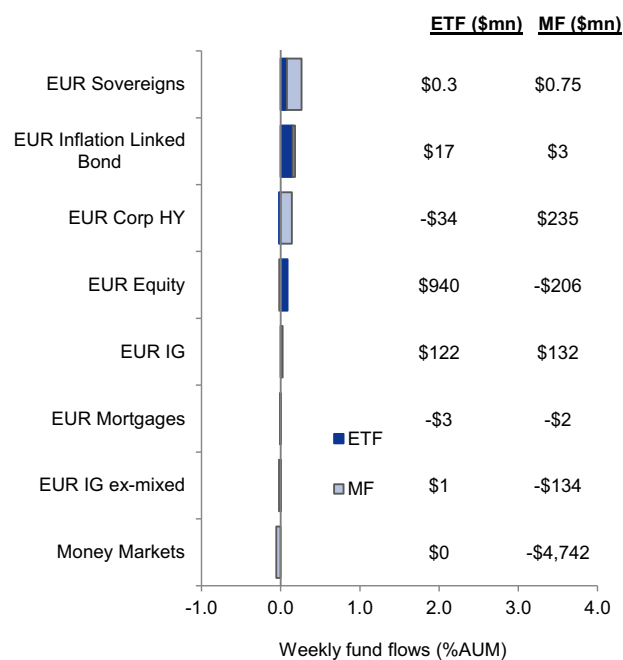
Flow dashboard: Mutual fund and ETF flows

Exhibit 72: Weekly mutual fund flows into US fixed income and equity markets



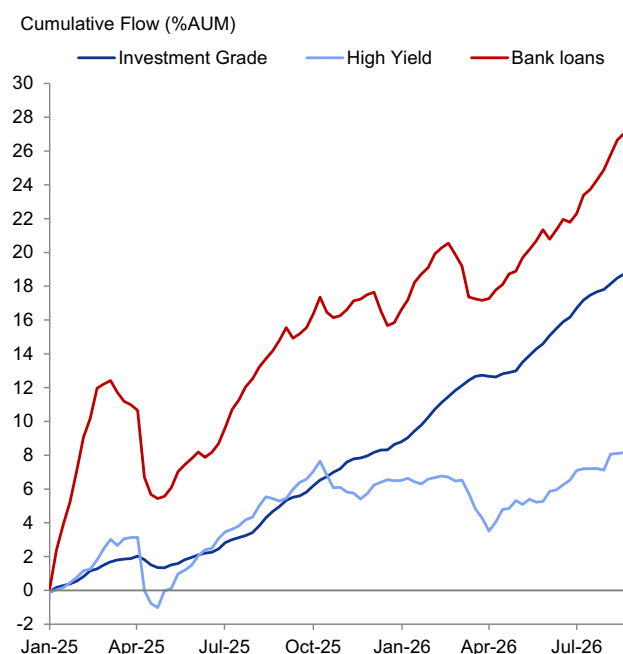
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 73: Weekly mutual fund flows into European fixed income and equity markets



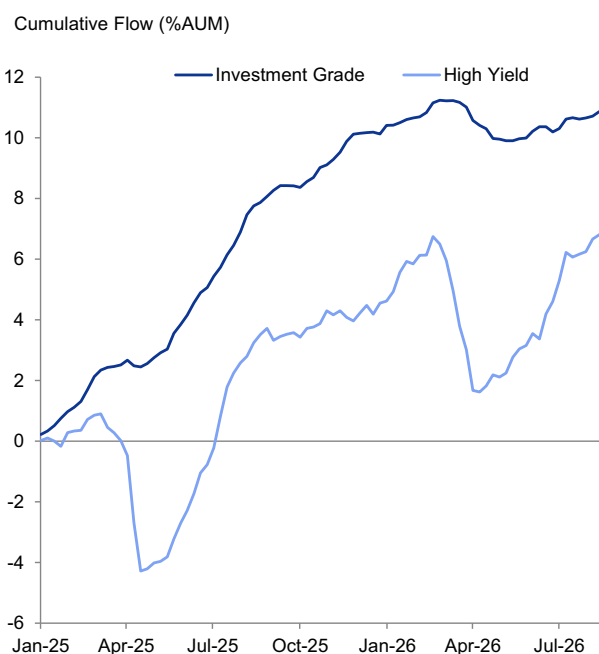
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 74: Cumulative flows as a percentage of beginning of the year AUM for US credit funds



Source: EPFR, Goldman Sachs Global Investment Research

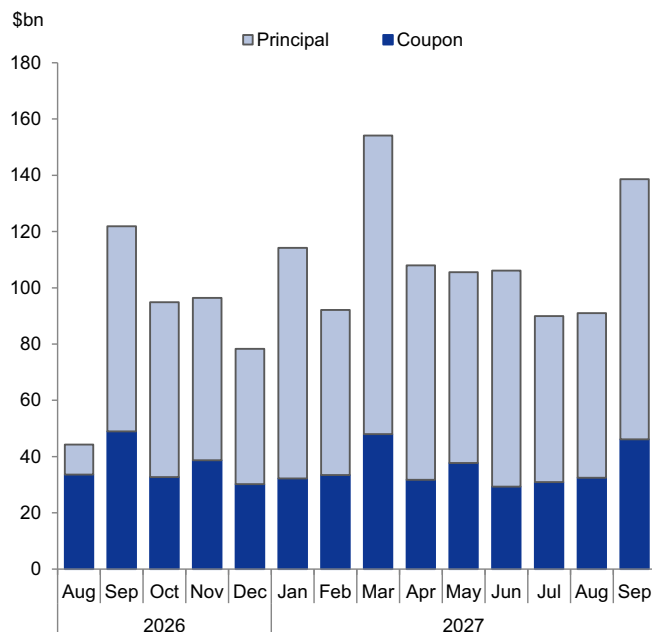
Exhibit 75: Cumulative flows as a percentage of beginning of the year AUM for European credit funds



Source: EPFR, Goldman Sachs Global Investment Research

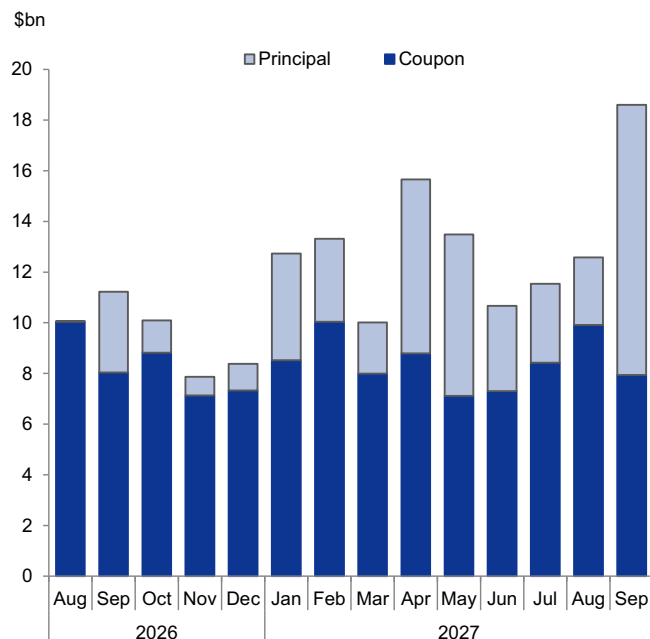
Flow dashboard: Upcoming coupon and principal payments

Exhibit 76: Our estimates of upcoming coupon and principal payments in the USD IG market



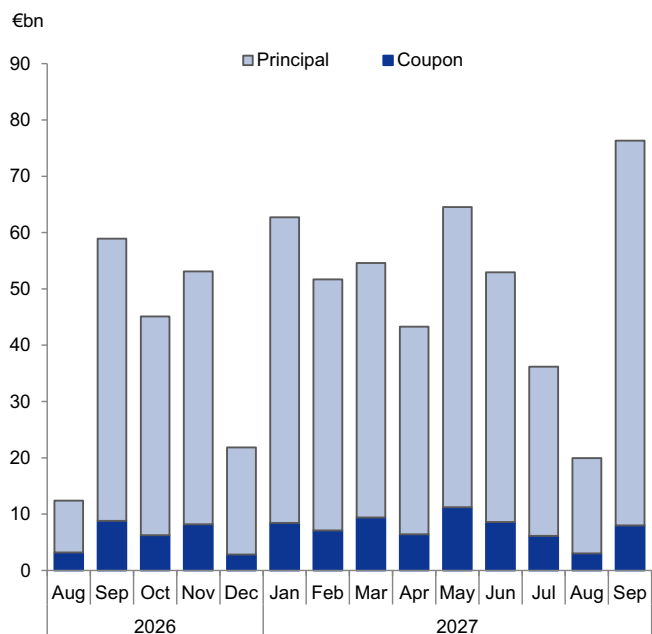
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 77: Our estimates of upcoming coupon and principal payments in the USD HY market



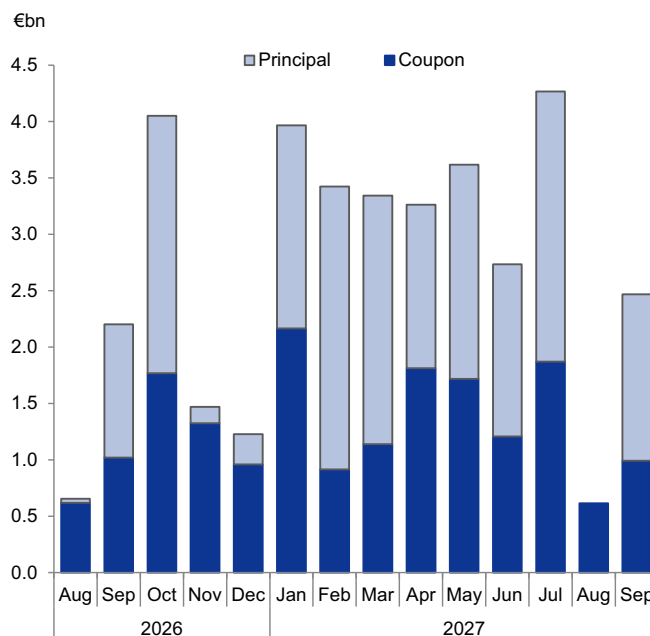
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 78: Our estimates of upcoming coupon and principal payments in the EUR IG market



Source: Bloomberg, Goldman Sachs Global Investment Research

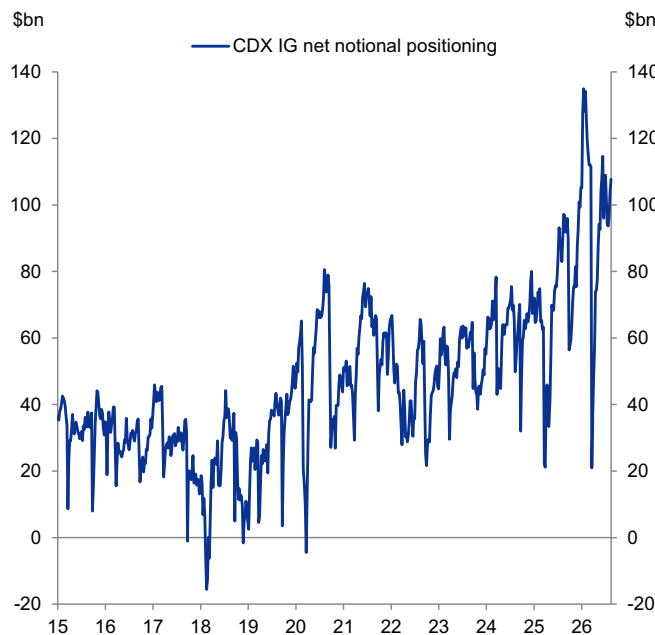
Exhibit 79: Our estimates of upcoming coupon and principal payments in the EUR HY market



Source: Bloomberg, Goldman Sachs Global Investment Research

Flow dashboard: Positioning in macro indices

Exhibit 80: Net notional positioning in the CDX IG 5-year on-the-run contract



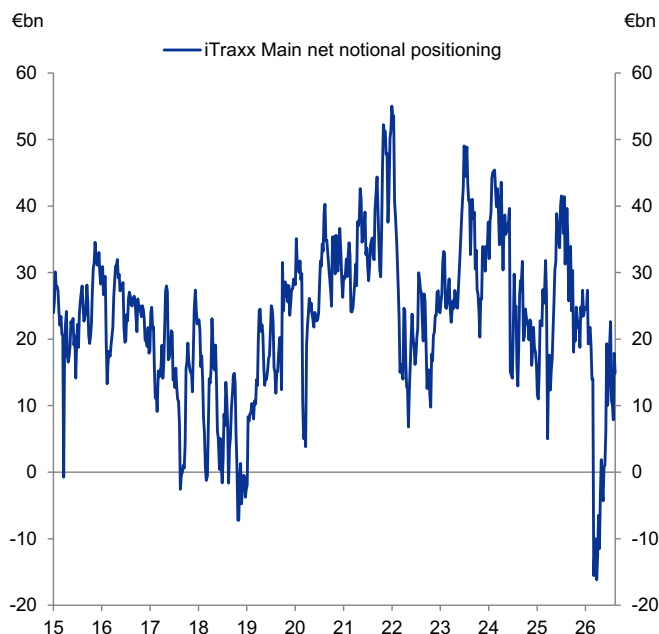
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 81: Net notional positioning in the CDX HY 5-year on-the-run contract



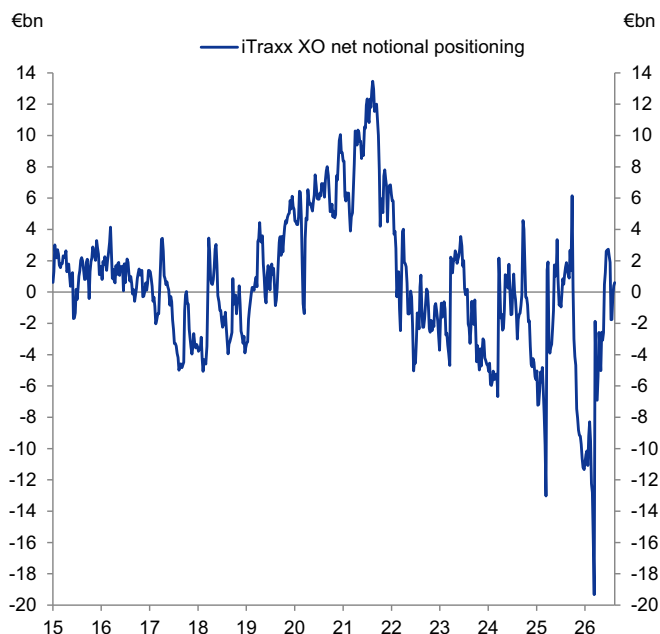
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 82: Net notional positioning in the iTraxx Main 5-year on-the-run contract



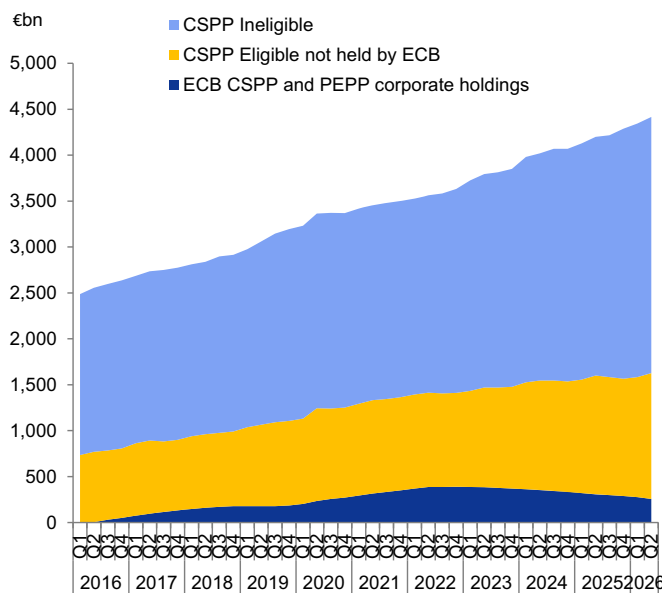
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 83: Net notional positioning in iTraxx over 5-year on-the-run contract

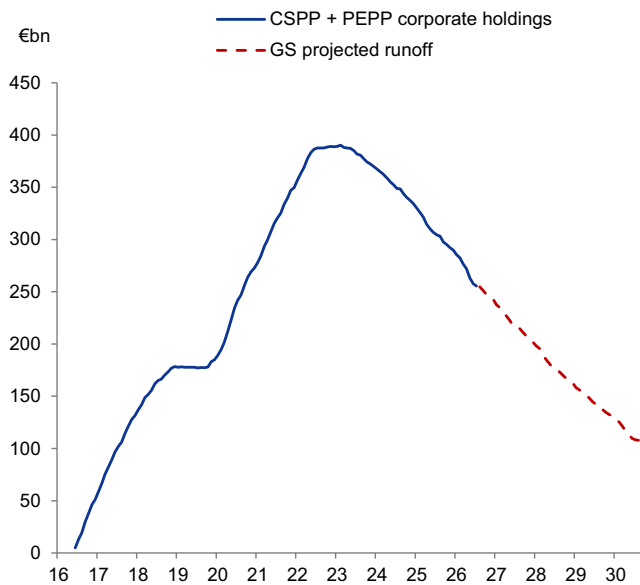


Source: DTCC, Goldman Sachs Global Investment Research

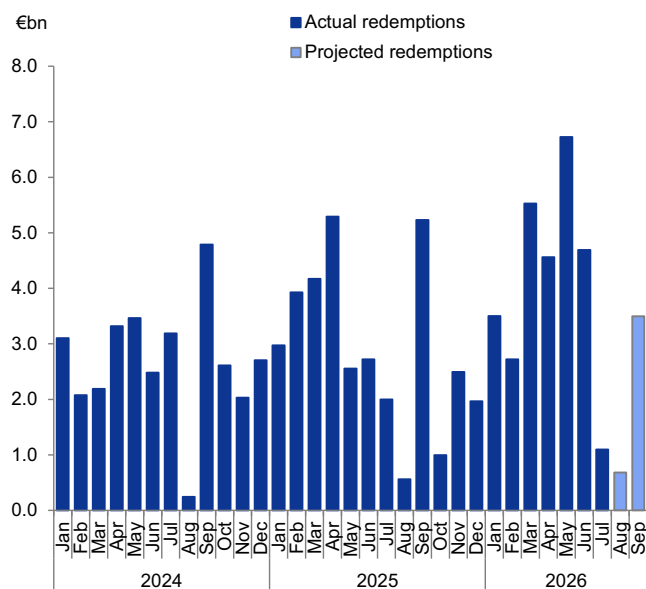
Flow dashboard: ECB QT

Exhibit 84: Ownership share of the ECB in the EUR IG market

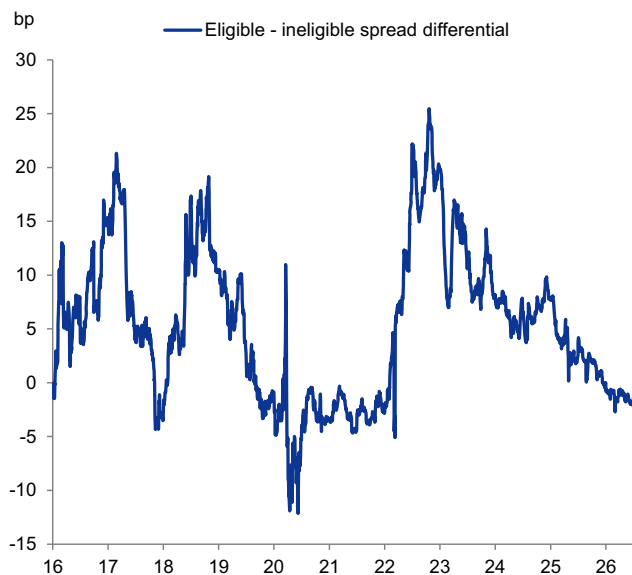
Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 85: Run-off projections for the ECB corporate bond holdings

Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 86: CSPP monthly run-off pace

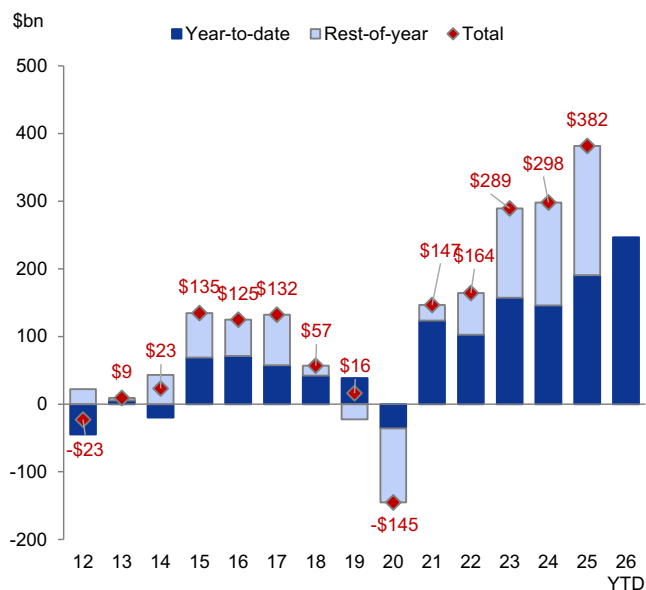
Source: ECB, Goldman Sachs Global Investment Research

Exhibit 87: Performance of CSPP-eligible bonds vs. non-eligible bonds

Source: iBoxx, Goldman Sachs Global Investment Research

Foreign demand and hedging cost

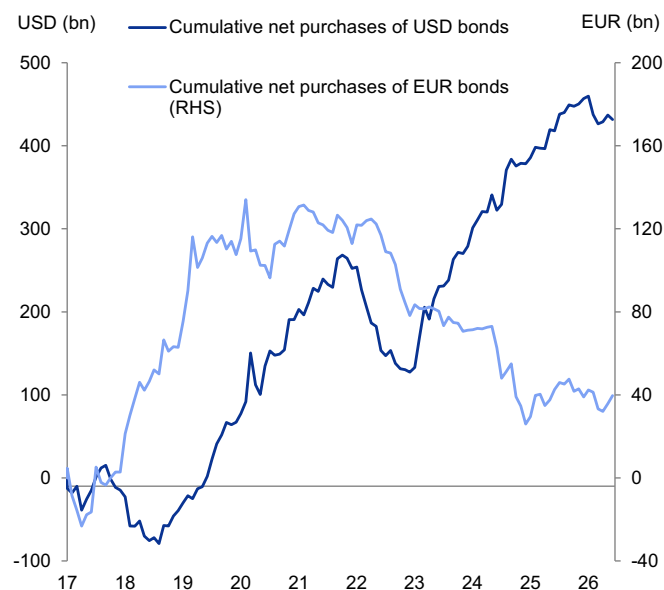
Exhibit 88: Net purchases of USD corporate bonds by foreign investors



Note: As of June 2026.

Source: Department of Treasury, Goldman Sachs Global Investment Research

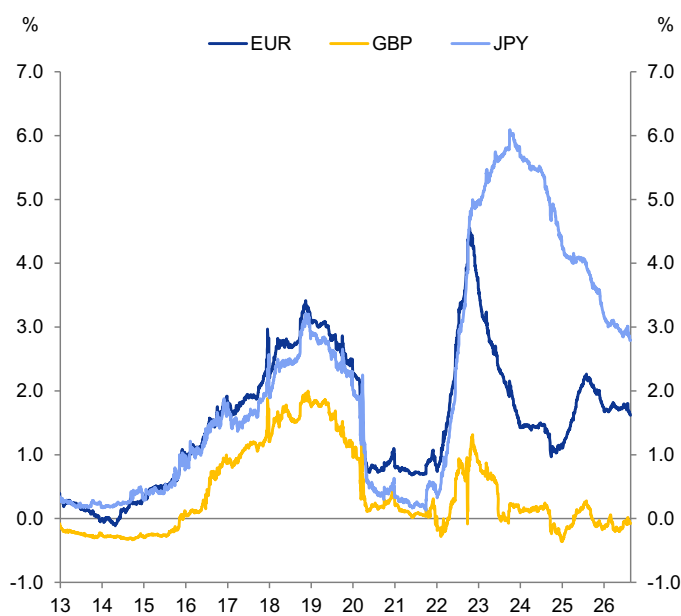
Exhibit 89: Cumulative net foreign purchases of USD and EUR bonds by Japanese investors



Note: As of June 2026.

Source: Bank of Japan, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 90: Annualized 3-month rolling cost of USD hedging against EUR, GBP and JPY

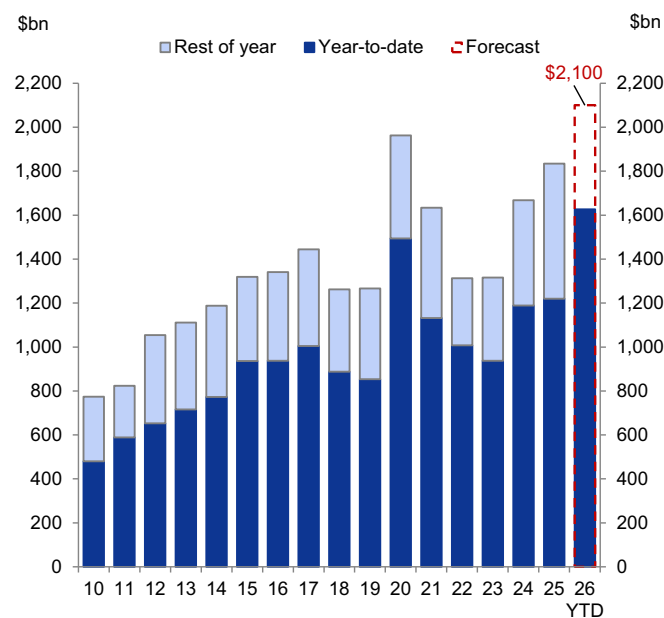


Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Primary market dashboard: USD gross issuance

Exhibit 91: USD IG gross issuance

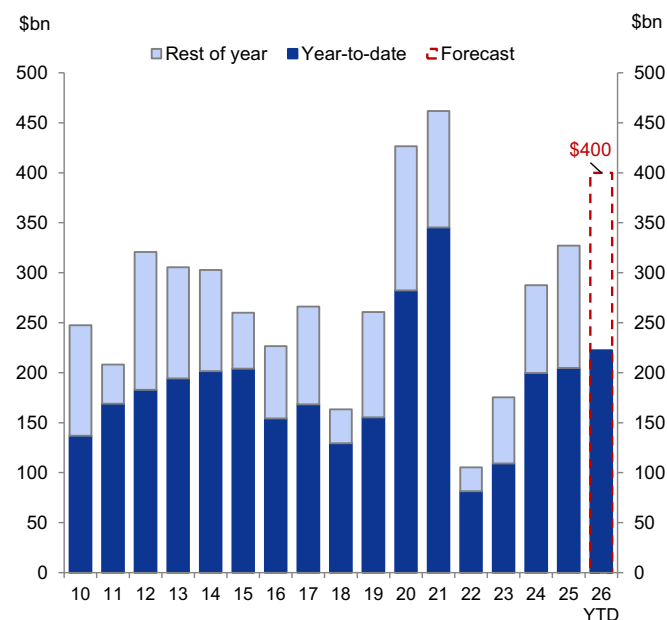
Historical investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 92: USD HY gross issuance

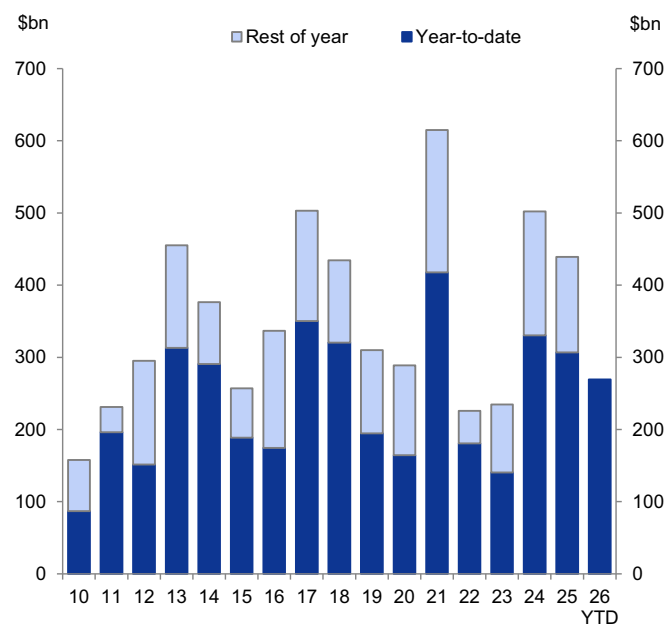
Historical USD high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 93: USD institutional leveraged loan gross issuance

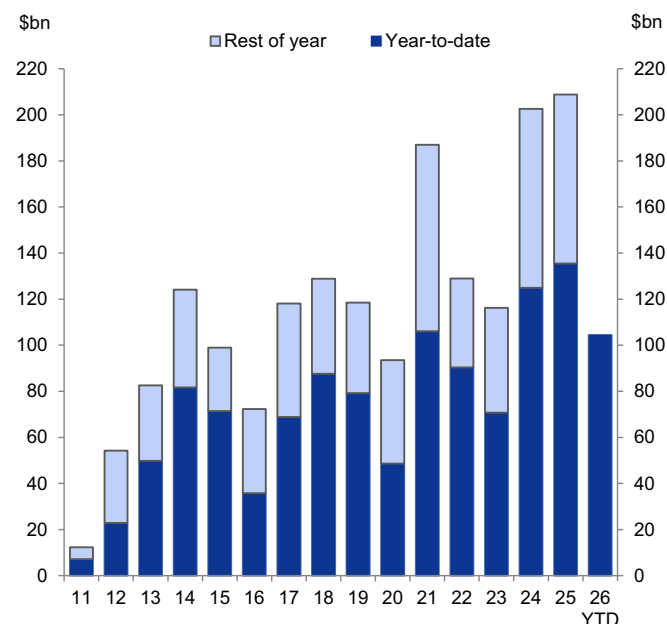
Historical leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 94: USD CLO creation

Historical USD CLO creation: Same period year to date vs. rest of the year

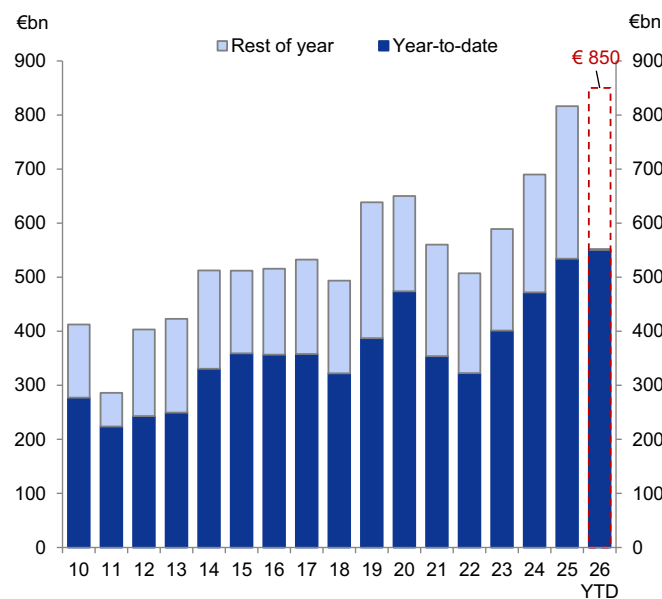


Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Primary market dashboard: EUR gross issuance

Exhibit 95: EUR IG gross issuance

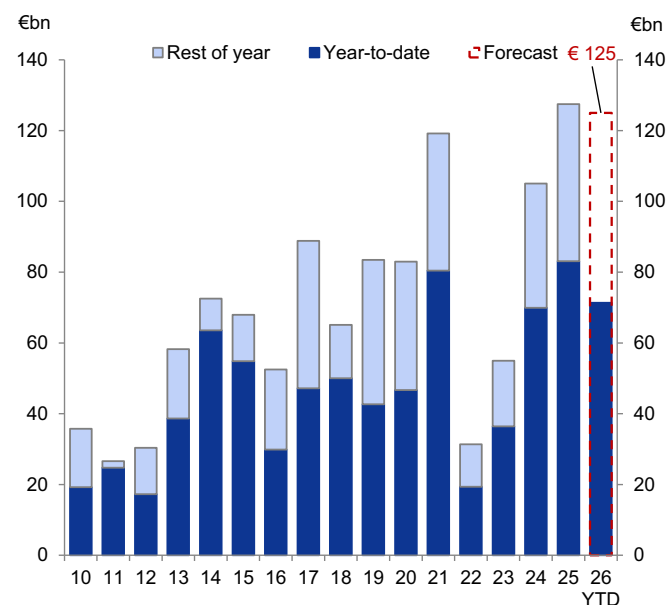
Historical EUR investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 96: EUR HY gross issuance

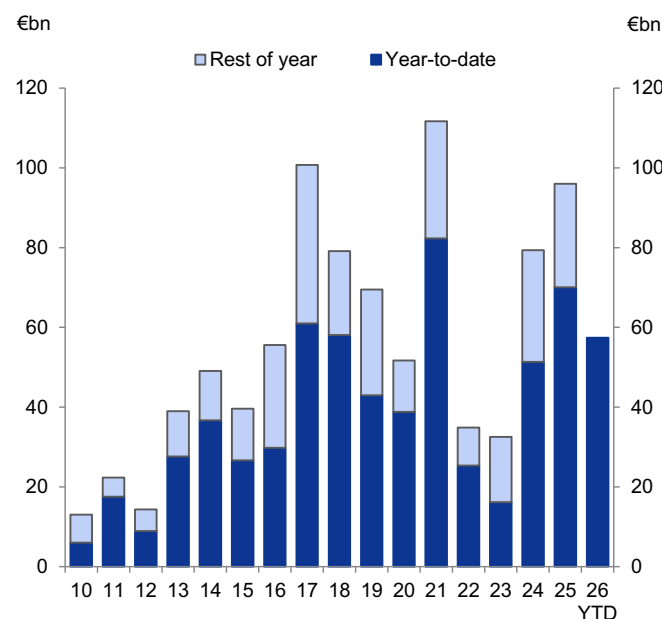
Historical EUR high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 97: EUR Leveraged loan gross issuance

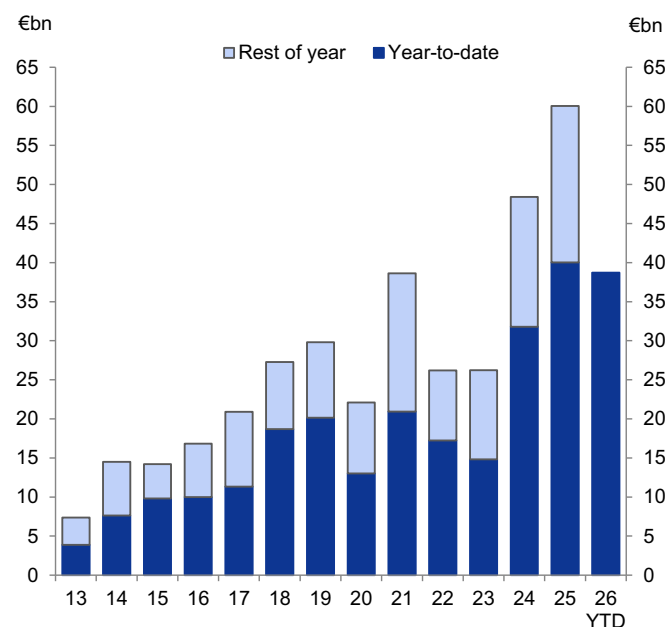
Historical EUR leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 98: EUR CLO creation

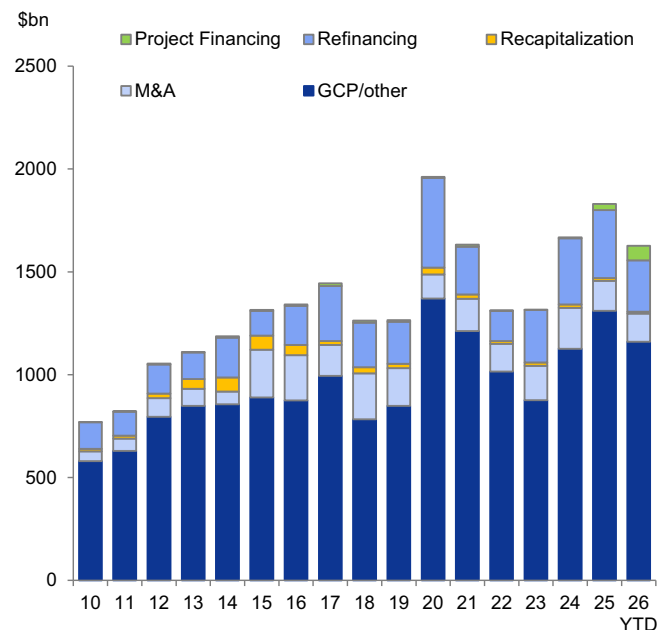
Historical EUR CLO creation broken: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

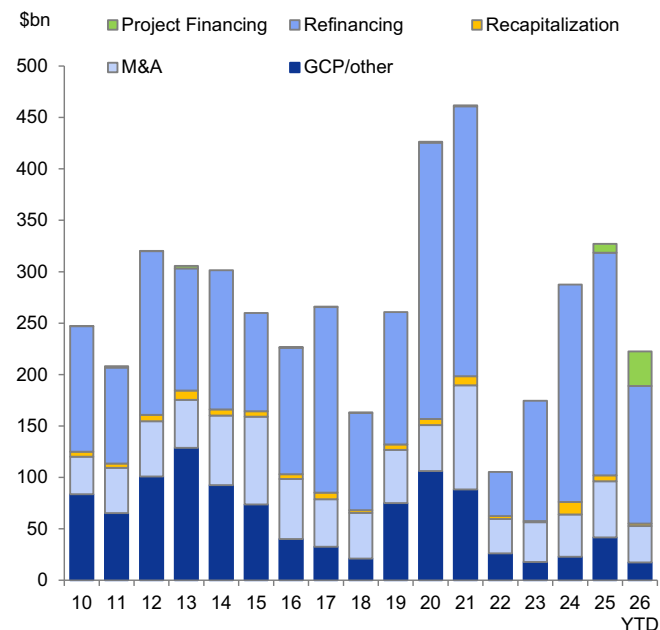
Primary market dashboard: Use of proceeds

Exhibit 99: Annual USD IG issuance volumes by use of proceeds



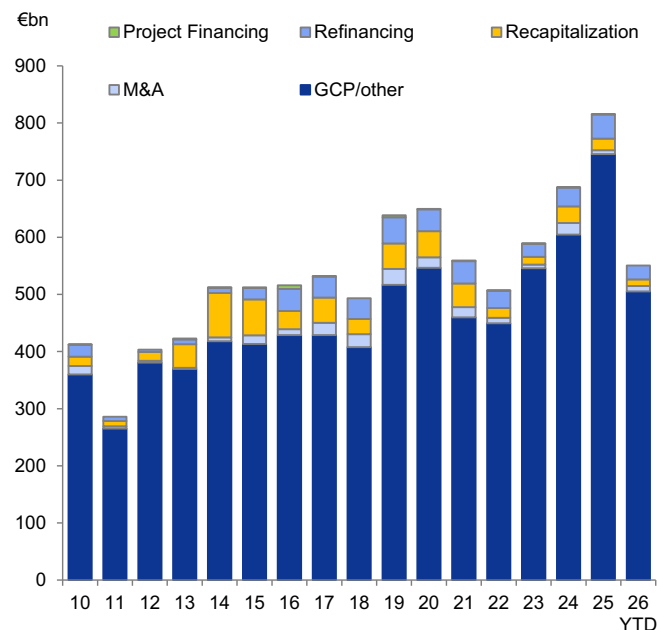
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 100: USD HY issuance volumes by use of proceeds



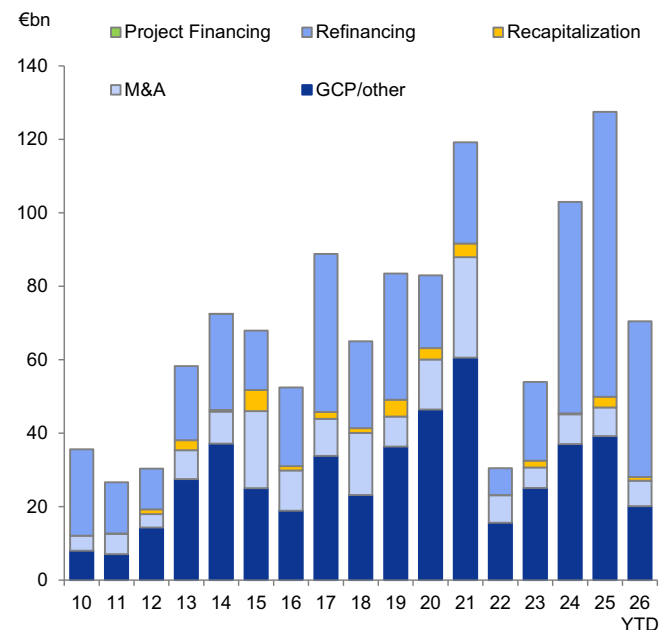
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 101: EUR IG issuance volumes by use of proceeds



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 102: EUR HY issuance volumes by use of proceeds

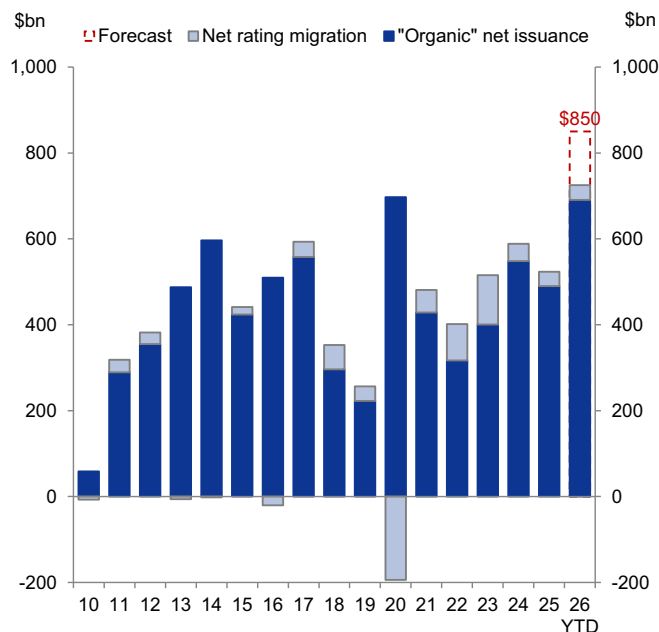


Source: Dealogic, Goldman Sachs Global Investment Research

Primary market dashboard: Net issuance

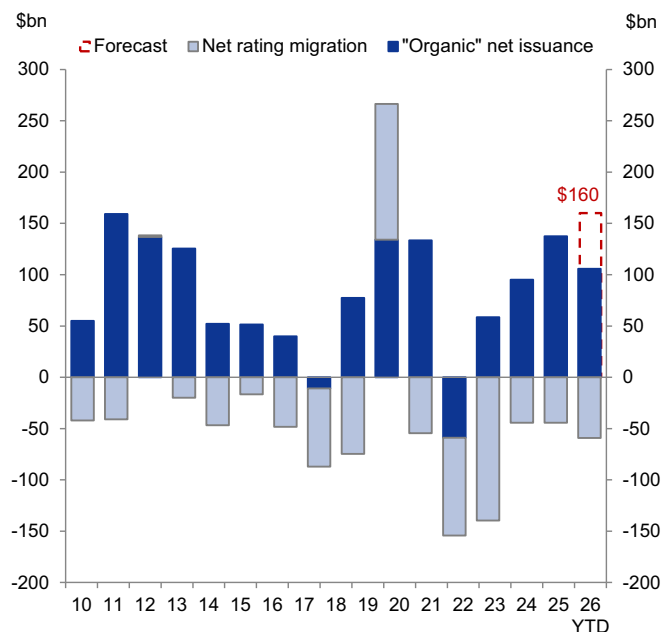
Note: Decomposition of the change in par value outstanding into net ratings migration vs. "organic" net issuance (gross issuance minus redemptions/matured bonds)

Exhibit 103: Annual USD IG net issuance



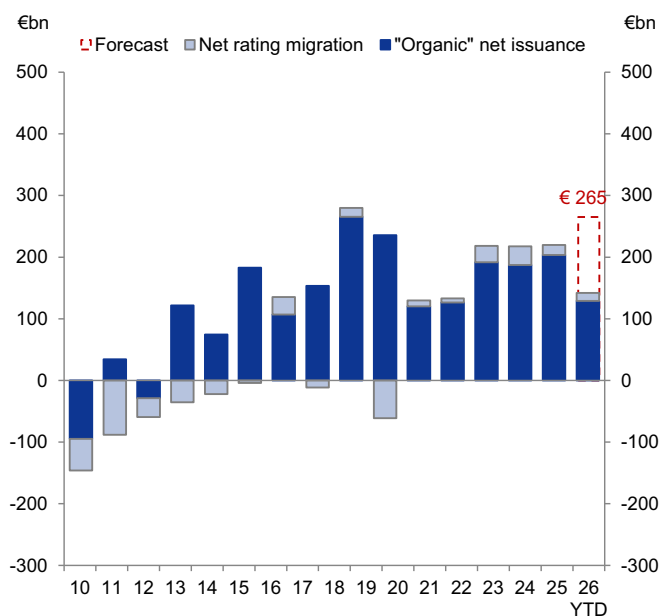
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 104: Annual USD HY net issuance



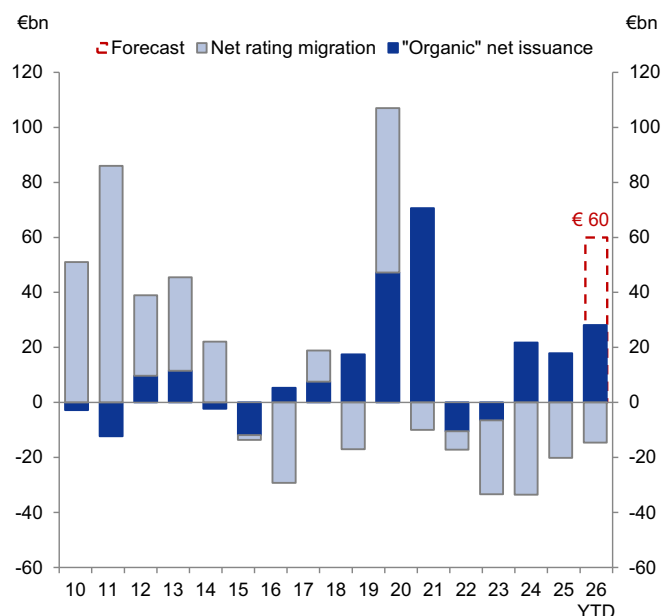
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 105: Annual EUR IG net issuance



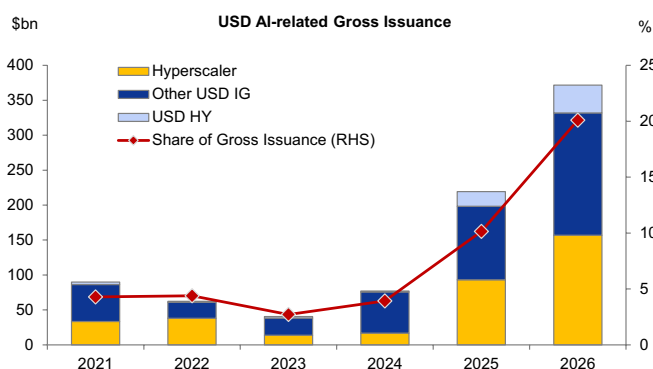
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 106: Annual EUR HY net issuance

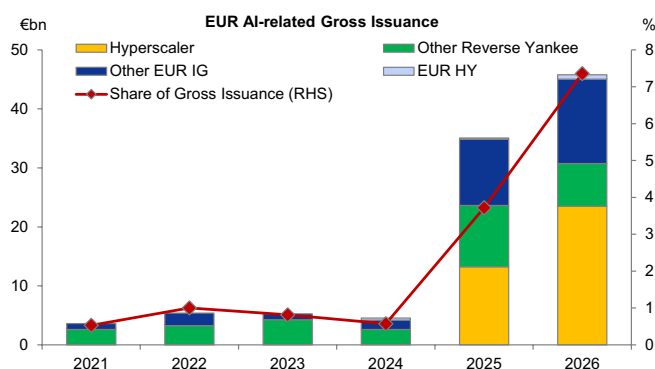


Source: Bloomberg, Goldman Sachs Global Investment Research

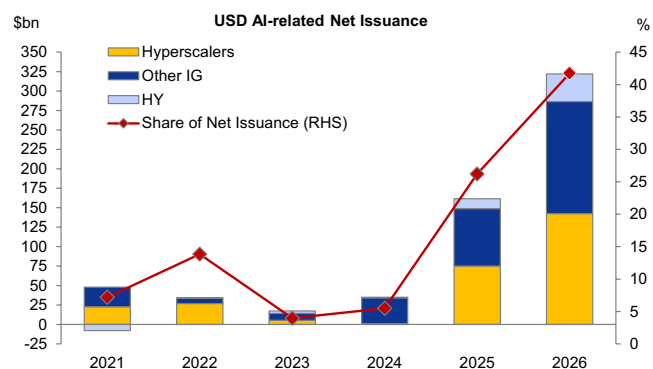
Primary market dashboard: AI-related issuance for the USD and EUR markets

Exhibit 107: USD AI-related Gross Issuance


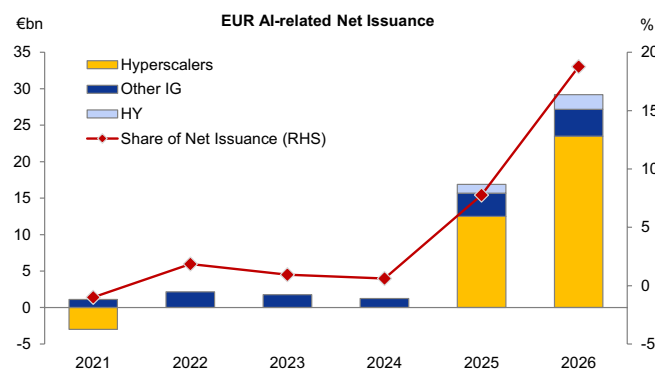
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 108: EUR AI-related Gross Issuance


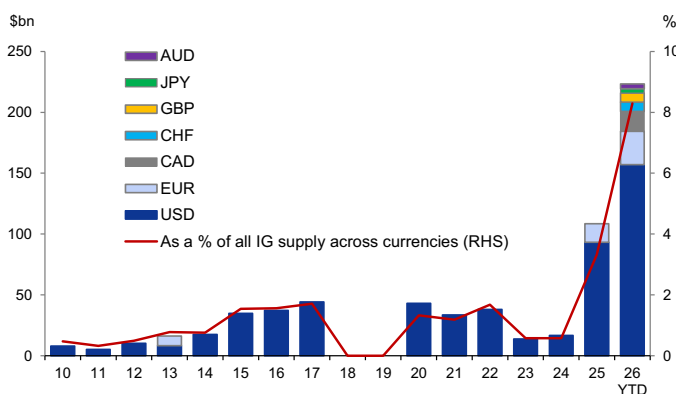
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 109: USD AI-related net issuance


Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 110: EUR AI-related net issuance


Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 111: Hyperscaler debt issuance across currencies (in USD-equivalent)


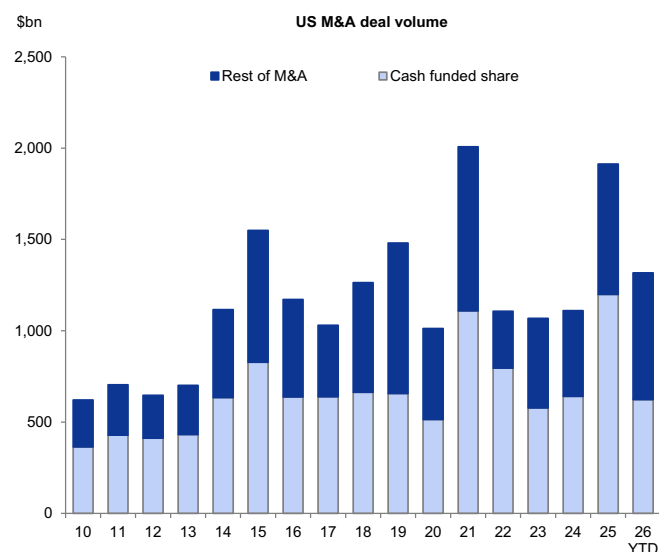
Note: Captures debt issued by the following companies: Alphabet (Google), Amazon, Microsoft, Meta, and Oracle.

Source: Dealogic, Goldman Sachs Global Investment Research

M&A volume

Exhibit 112: Annual US M&A deal volume

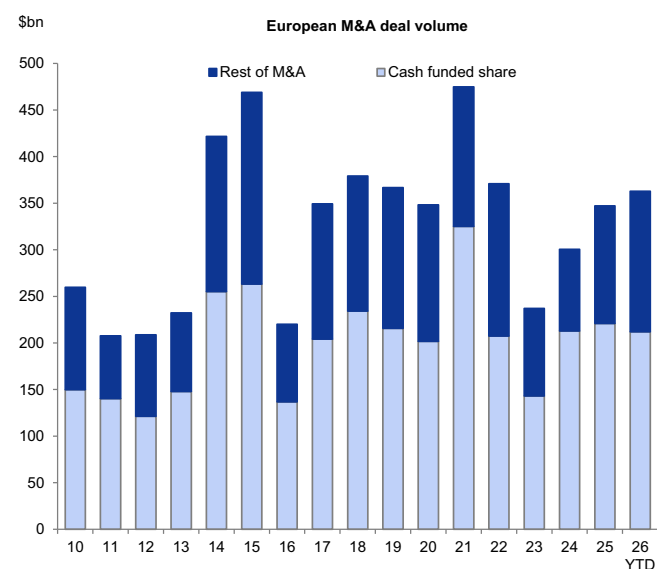
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 113: Annual European M&A deal volume

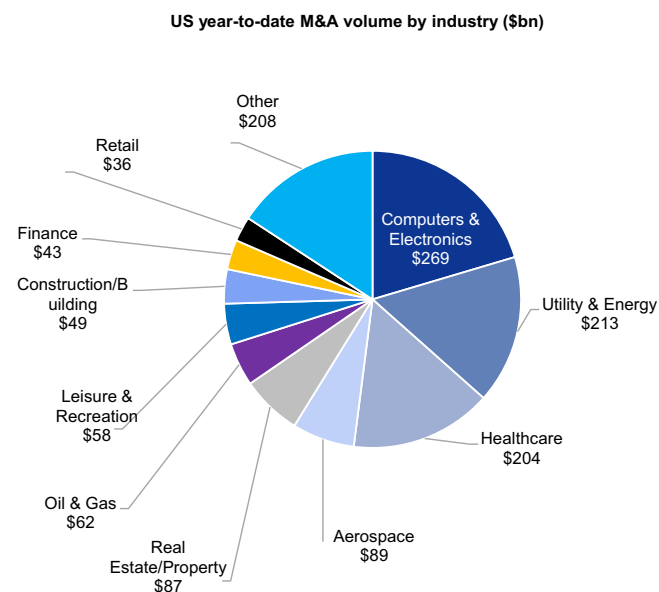
Includes strategic and sponsored deals



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 114: Year-to-date US M&A volume by sector

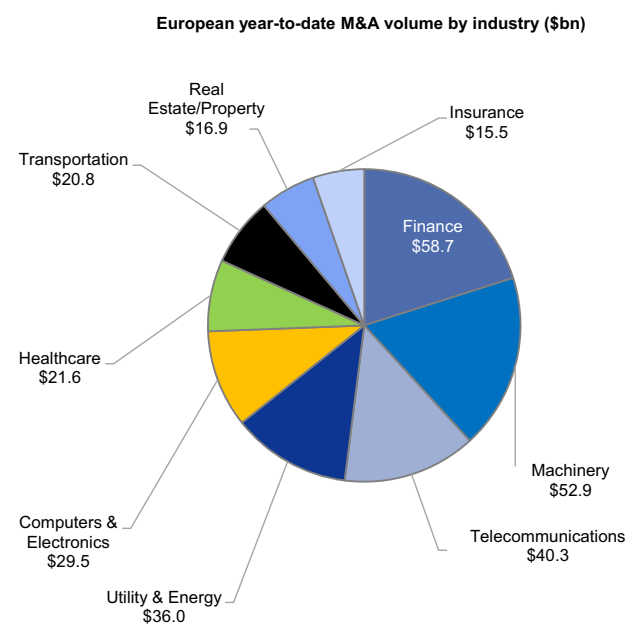
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

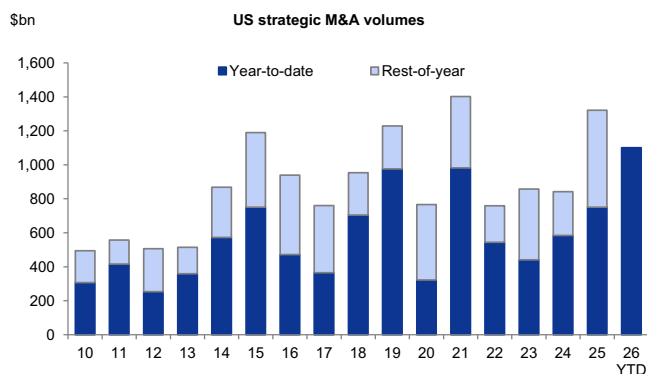
Exhibit 115: Year-to-date European M&A volume by sector

Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

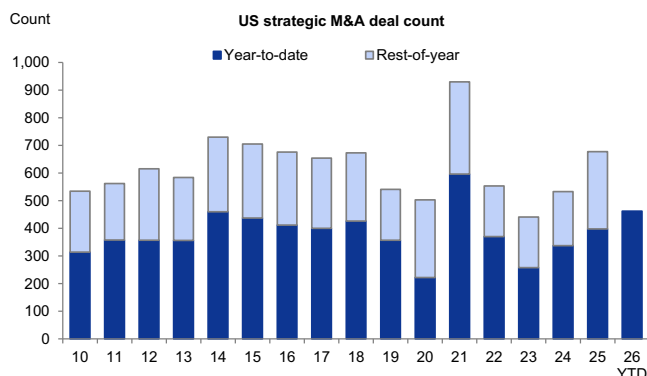
M&A volume (Strategic deals only)

Exhibit 116: Annual US strategic M&A volumes


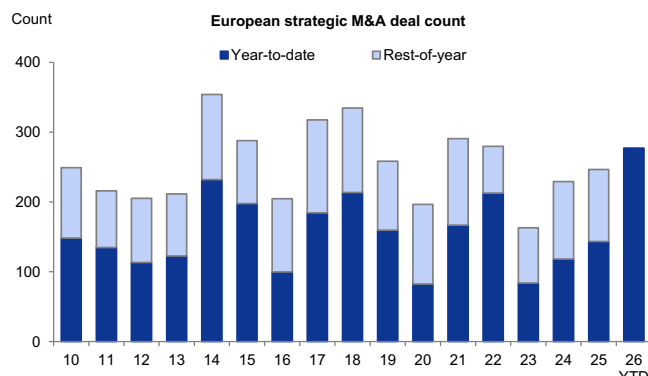
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 117: Annual European strategic M&A volumes

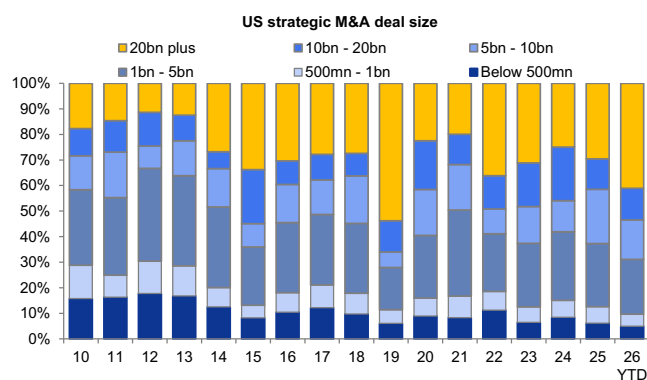

Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 118: Annual US strategic M&A deal count


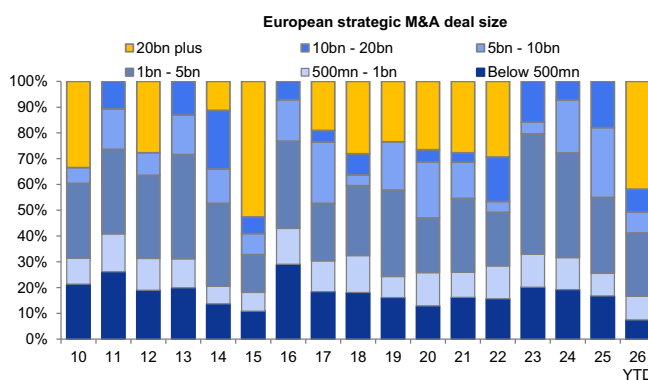
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 119: Annual European strategic M&A deal count


Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 120: Annual US strategic M&A deal size
 Percentages based on deal value, not count


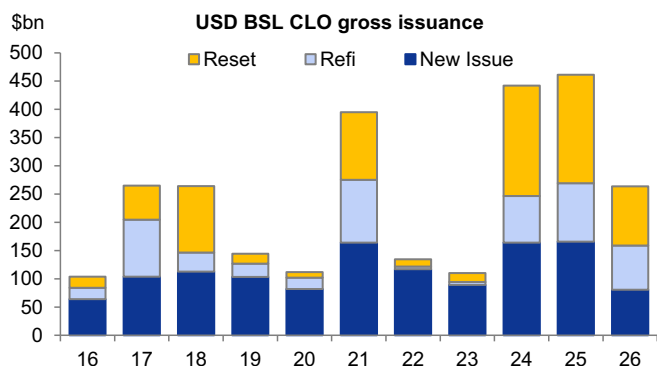
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 121: Annual European strategic M&A deal size
 Percentages based on deal value, not count


Source: Dealogic, Goldman Sachs Global Investment Research

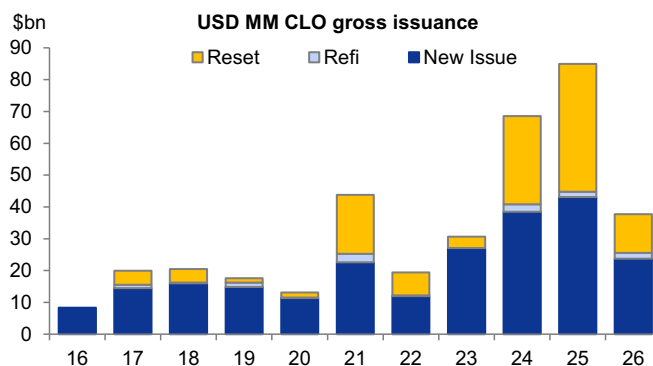
Primary market dashboard: CLO creation and refinancing

Exhibit 122: Annual USD broadly syndicated loan CLO creation



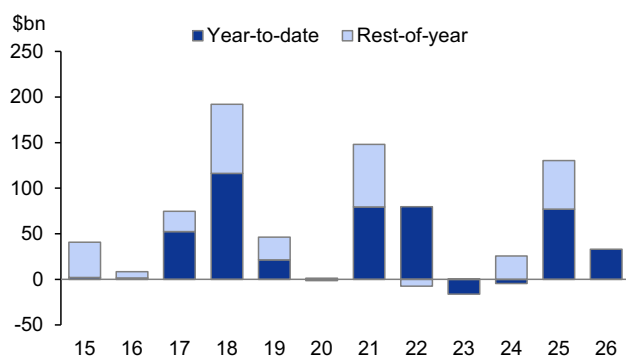
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 123: Annual USD middle market CLO creation



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

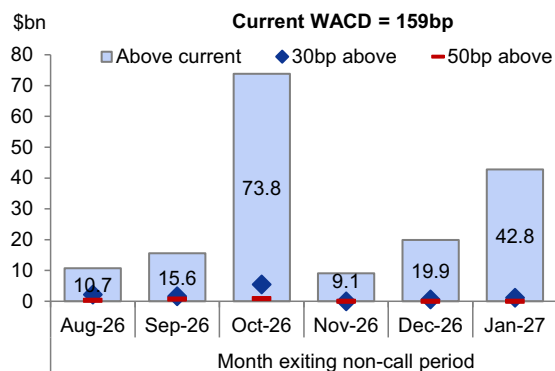
Exhibit 124: Annual USD leveraged loan net issuance



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 125: USD CLO refinance monitor

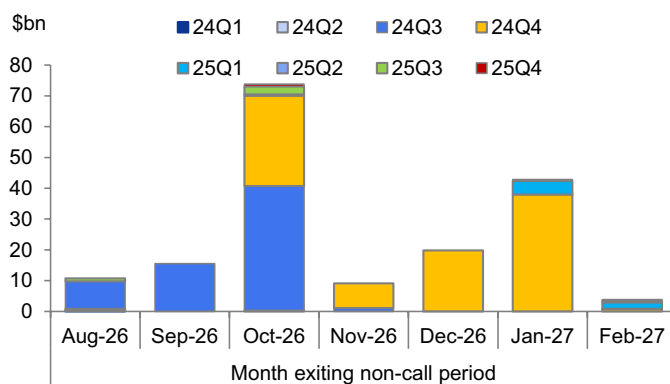
Upcoming notional above prevailing WACD



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 126: USD CLO refinance monitor

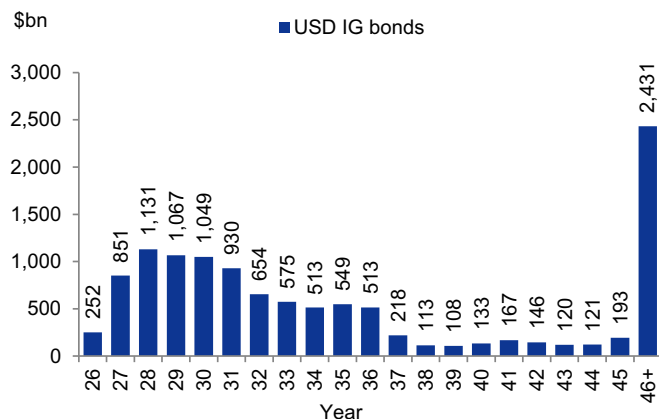
Notional above prevailing WACD by origination quarter



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

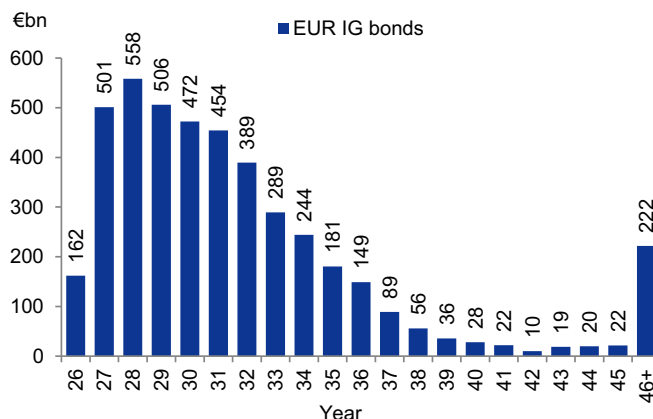
Primary market dashboard: Maturity walls

Exhibit 127: USD IG bond market maturity wall



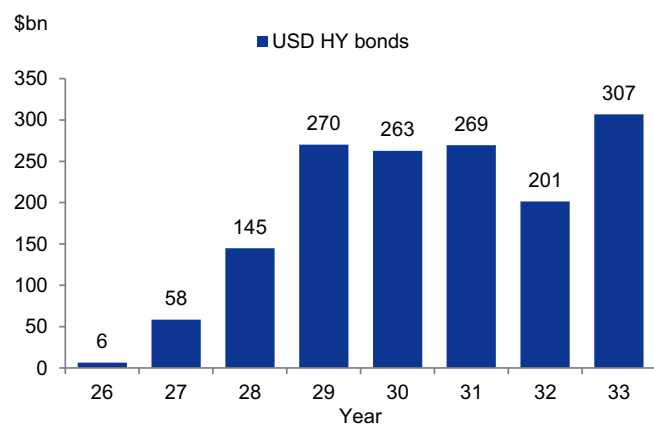
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 128: EUR IG bond market maturity wall



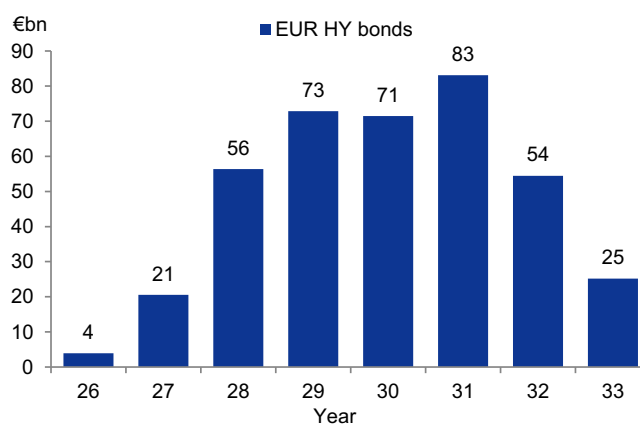
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 129: USD HY bond market maturity wall



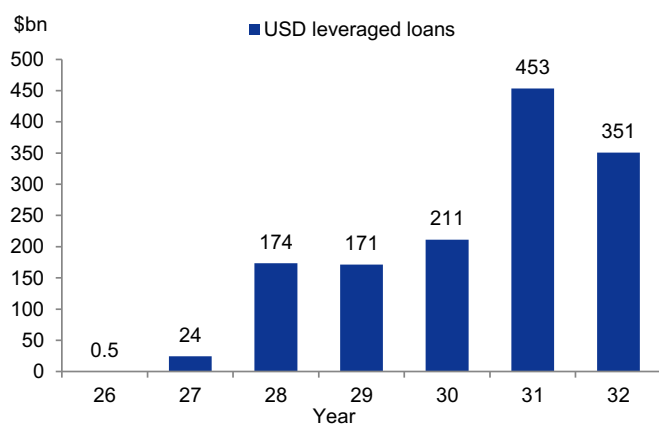
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 130: EUR HY bond market maturity wall



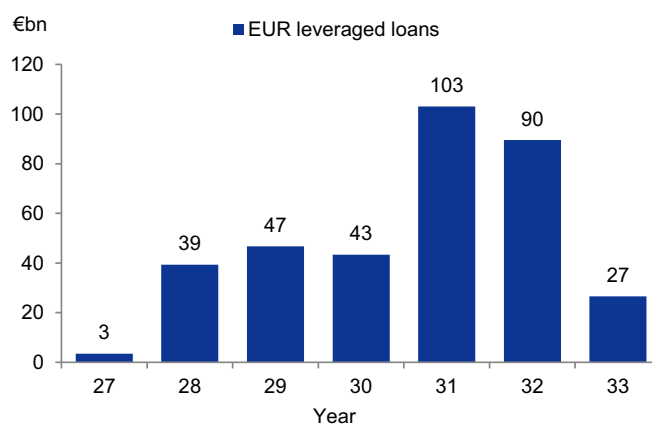
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 131: USD leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

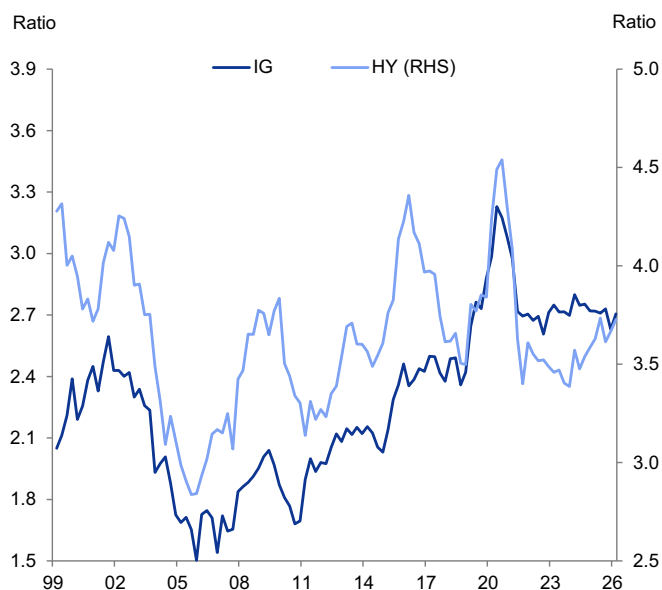
Exhibit 132: EUR leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

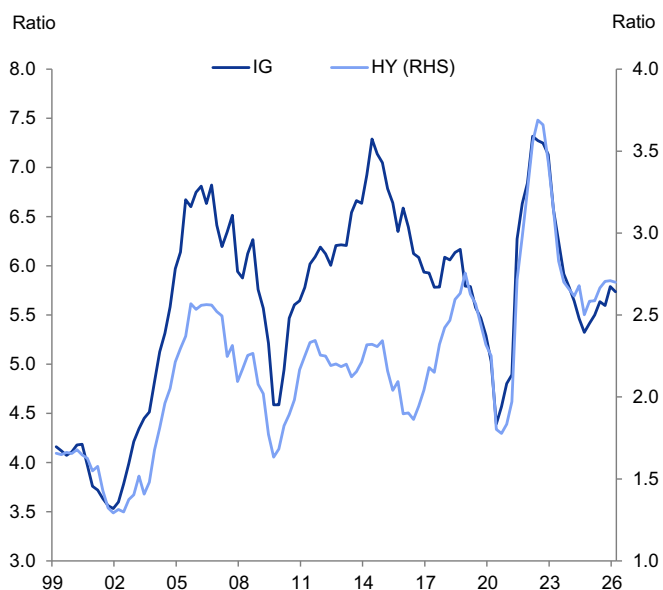
Fundamental dashboard: Credit metrics for US non-financial corporations

Exhibit 133: Net debt to EBITDA ratios for the median US IG and HY-rated non-financial corporations



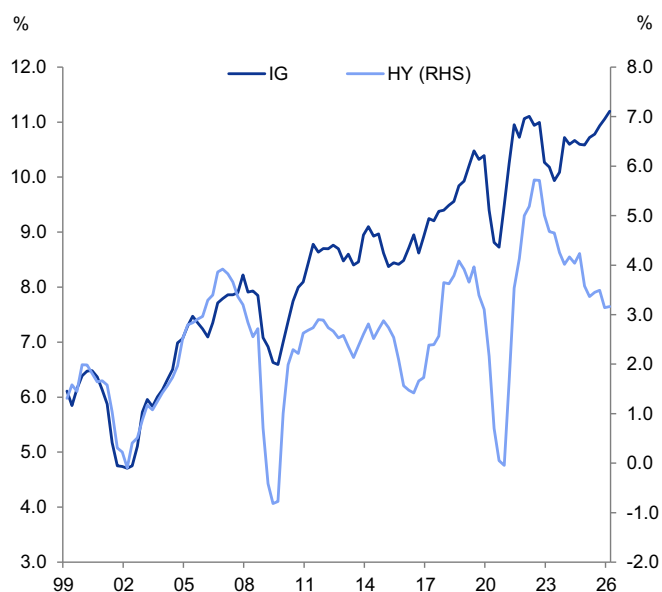
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 134: Interest coverage ratios for the median US IG and HY-rated non-financial corporations



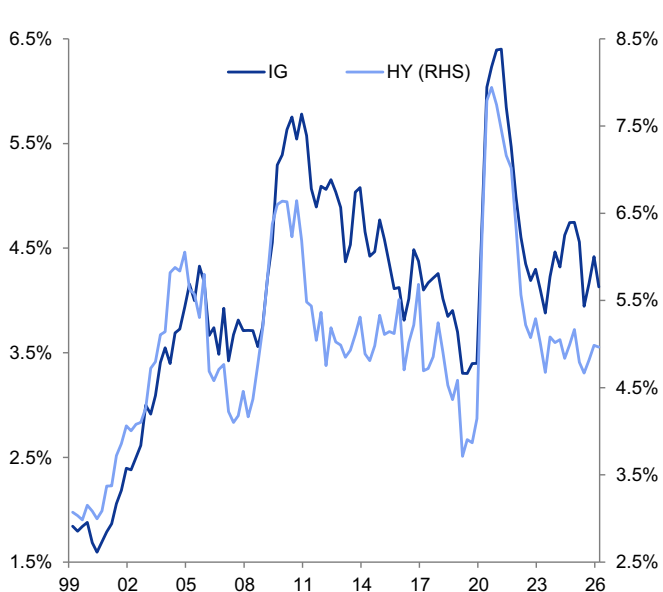
Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 135: Net margins for the median US IG and HY-rated non-financial corporations



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 136: Cash to total assets ratios for the median US IG and HY-rated corporations

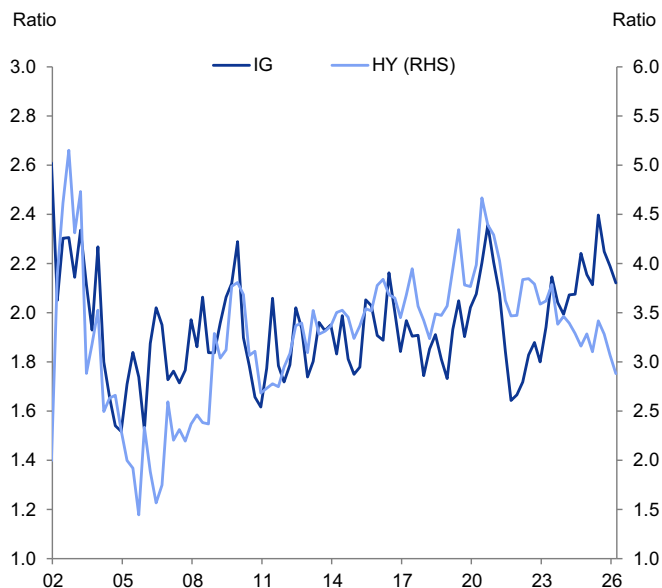


Source: FactSet, Goldman Sachs Global Investment Research

Fundamental dashboard: Credit metrics for European non-financial corporations

Exhibit 137: Net debt to EBITDA ratios for the median European IG and HY-rated non-financial corporations

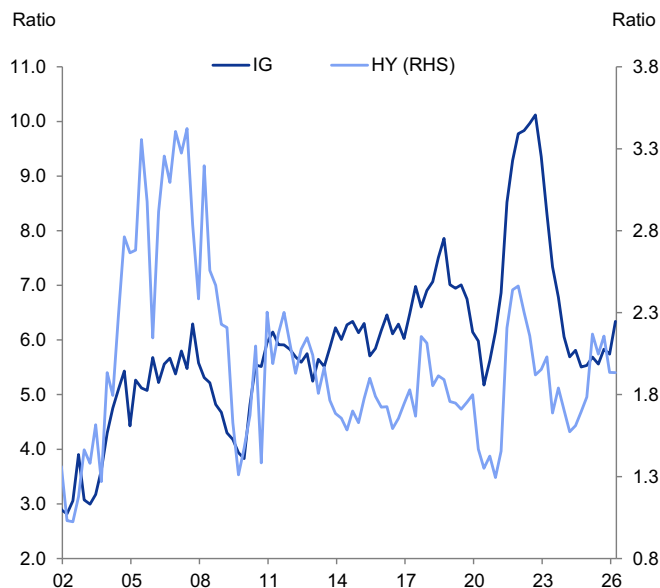
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 138: Interest coverage ratios for the median European IG and HY-rated non-financial corporations

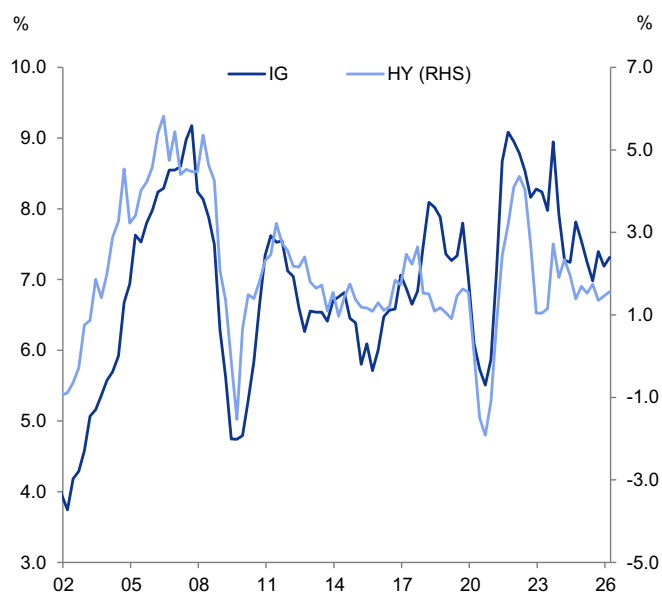
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 139: Net margins for the median European IG and HY-rated non-financial corporations

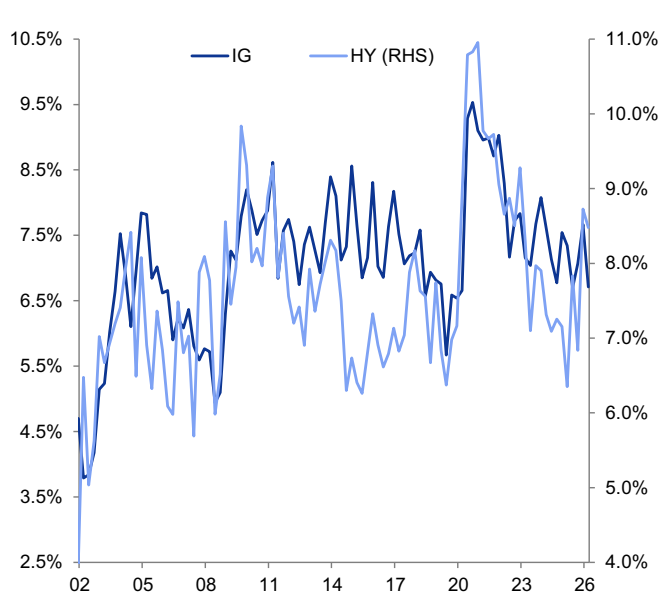
We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway



Source: FactSet, Goldman Sachs Global Investment Research

Exhibit 140: Cash to total assets ratios for the median European IG and HY-rated non-financial corporations

We include firms domiciled in the Euro area, UK, Switzerland, Sweden, Denmark and Norway

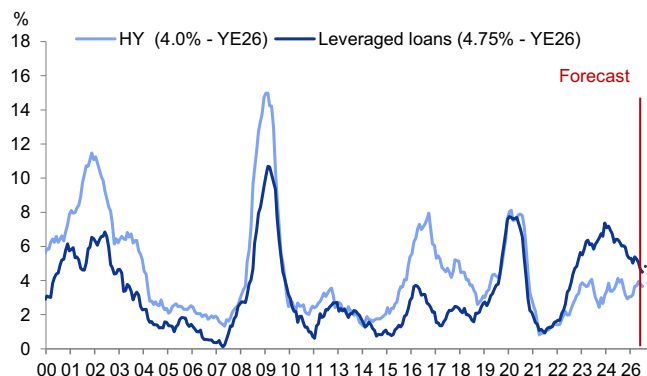


Source: FactSet, Goldman Sachs Global Investment Research

Default and ratings migration dashboard: Default trends

Exhibit 141: Default rates for US HY bond and leveraged loan issuers

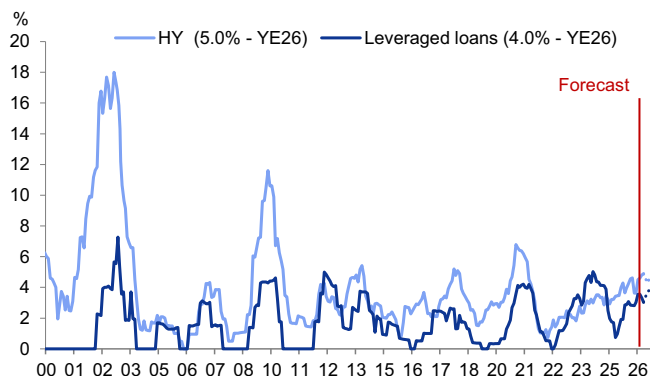
12-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 142: Default rates for European bond and leveraged loan issuers

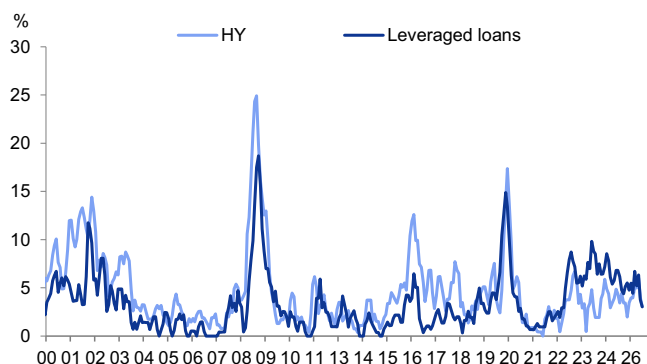
12-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 143: High frequency default rates for US HY bond and leveraged loan issuers

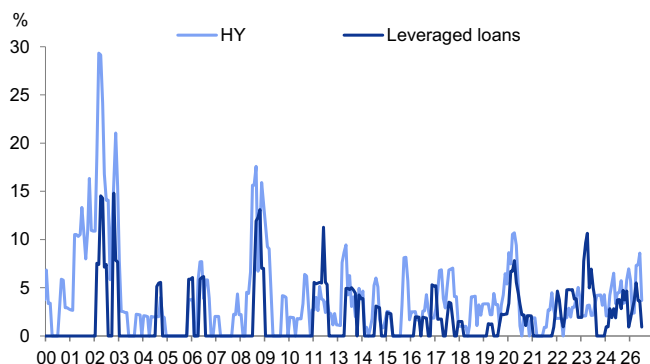
3-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

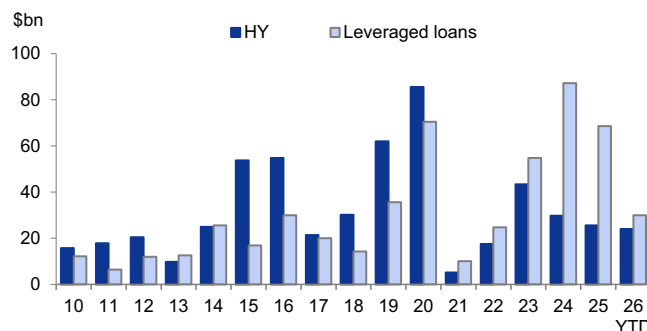
Exhibit 144: High frequency default rates for European HY bond and leveraged loan issuers

3-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



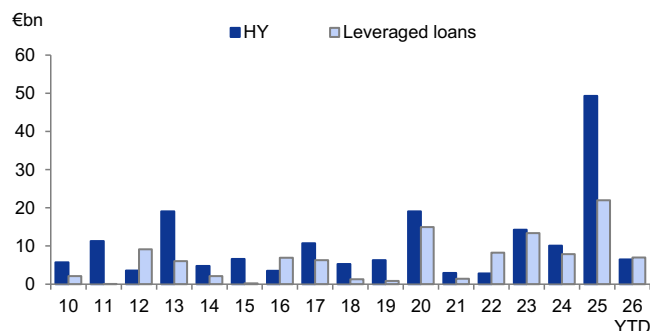
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 145: Annual par value amounts of defaulted USD HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 146: Annual par value amounts of defaulted EUR HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 147: Top 5 largest defaults in the USD HY bond and leveraged loan markets in the past 12 months

	Default date	Issuer	Industry	Default type	Notional amount (\$bn)
HY	6/30/2026	DISH DBS Corporation	Media: Broadcasting & Subscription	Prepackaged Chapter 11	9.8
	1/29/2026	LABL, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	2.9
	11/20/2025	New Fortress Energy Inc.	Energy: Oil & Gas	Missed interest payment	2.7
	1/14/2026	Saks Global Enterprises LLC	Retail	Chapter 11	2.7
	8/8/2025	Saks Global Enterprises LLC	Retail	Distressed exchange	2.1
Loans	9/28/2025	First Brands Group, LLC	Automotive	Chapter 11	4.7
	4/16/2026	QVC, Inc.	Retail	Prepackaged Chapter 11	2.9
	2/17/2026	Vibrantz Technologies Inc.	Chemicals, Plastics, & Rubber	Distressed exchange	2.6
	12/8/2025	Newfold Digital Holdings Group, Inc	High Tech Industries	Distressed exchange	2.5
	1/29/2026	LABL, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	2.3

Source: Moody's, data compiled by Goldman Sachs Global Investment Research

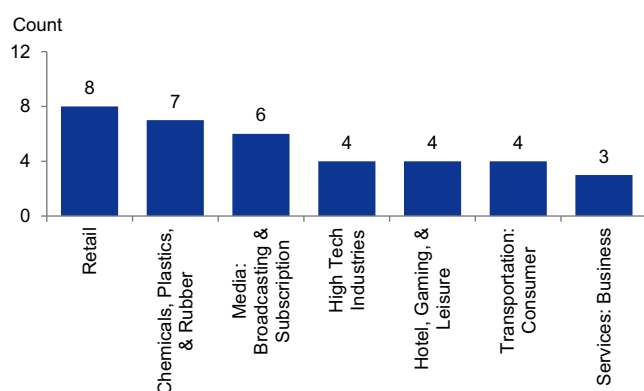
Exhibit 148: Top 5 largest defaults in the EUR HY bond and leveraged loan markets in the past 12 months

	Default date	Issuer	Industry	Default type	Notional amount (€bn)
HY	2/19/2026	Garfunkelux Holdco 3 S.A.	Finance	Distressed exchange	1.7
	11/12/2025	Ardagh Group S.A.	Containers, Packaging, & Glass	Distressed exchange	1.4
	4/29/2026	La Financiere ATALIAN S.A.S.	Services: Business	Missed interest payment	1.0
	11/12/2025	ARD Finance S.A.	Containers, Packaging, & Glass	Distressed exchange	0.9
	11/4/2025	Klockner Pentaplast of America, Inc	Containers, Packaging, & Glass	Prepackaged Chapter 11	0.8
Loans	4/30/2026	Poseidon BidCo S.A.S	High Tech Industries	Missed interest payment	1.6
	1/29/2026	Eos Finco S.a.r.l.	Wholesale	Distressed exchange	1.3
	4/7/2026	Biscuit Holding S.A.S.	Beverage, Food, & Tobacco	Missed interest payment	0.8
	11/4/2025	Klockner Pentaplast of America, Inc	Containers, Packaging, & Glass	Prepackaged Chapter 11	0.8
	2/18/2026	Accell Group Holding B.V.	Consumer goods: Durable	Distressed exchange	0.7

Source: Moody's, data compiled by Goldman Sachs Global Investment Research

Exhibit 149: Industry default count in the US

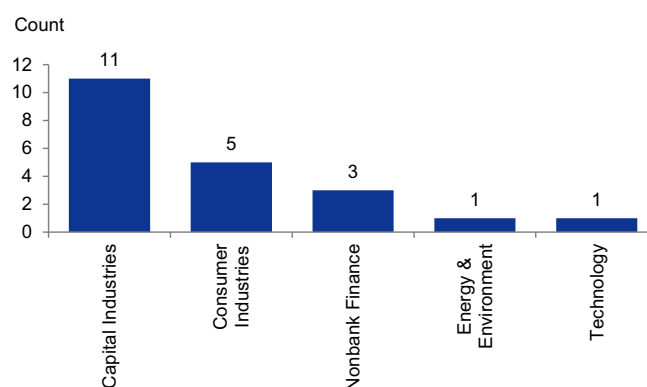
Trailing 12-month counts



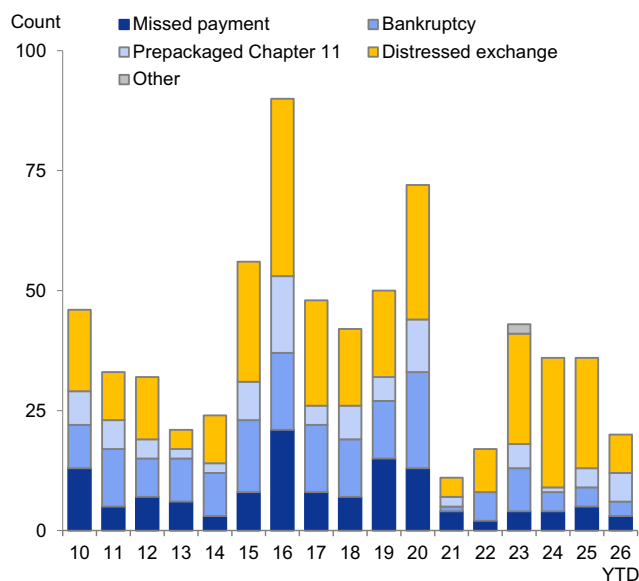
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 150: Industry default count in Europe

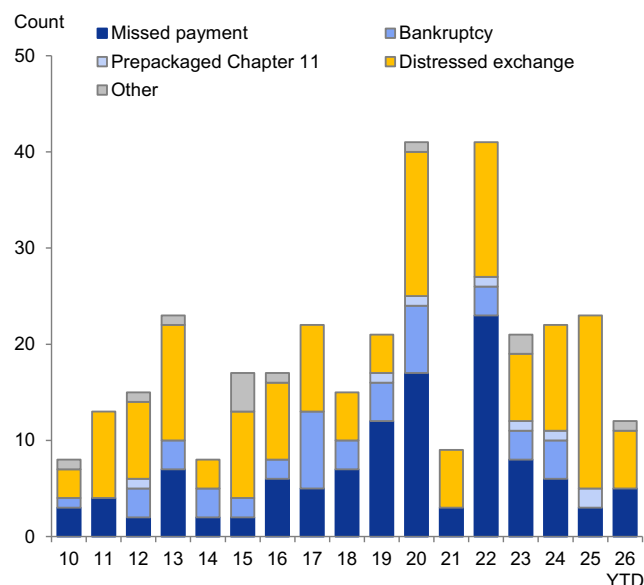
Trailing 12-month counts



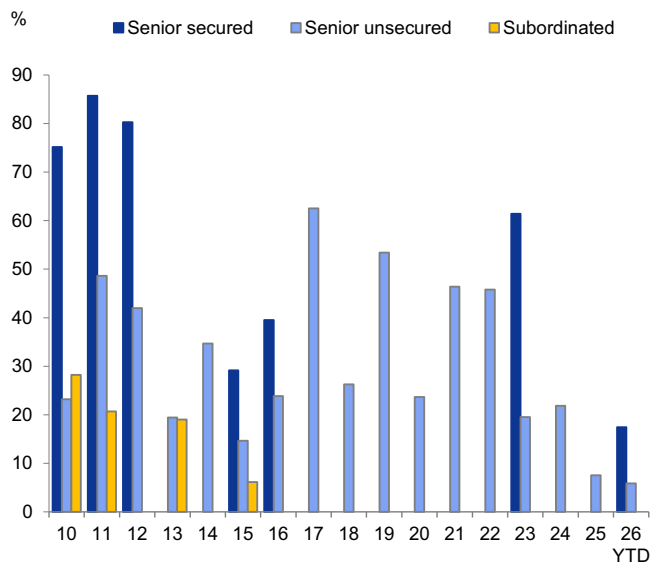
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 151: Composition of defaults in the USD HY bond market by type

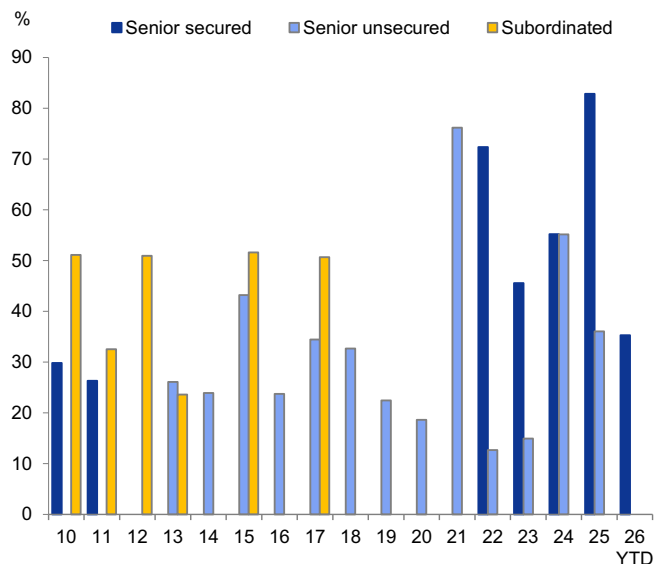
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 152: Composition of defaults in the EUR HY bond market by type

Source: Moody's, Goldman Sachs Global Investment Research

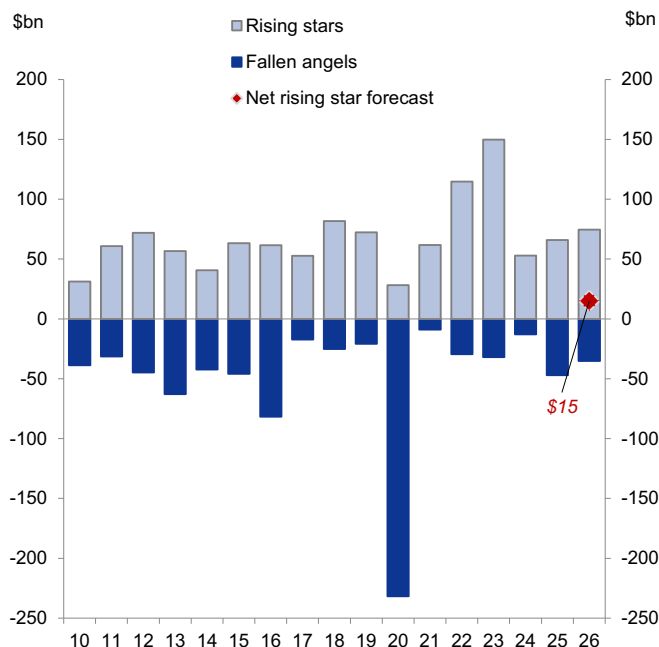
Exhibit 153: Annual average recovery rates by seniority for US issuers

Source: Moody's, Goldman Sachs Global Investment Research

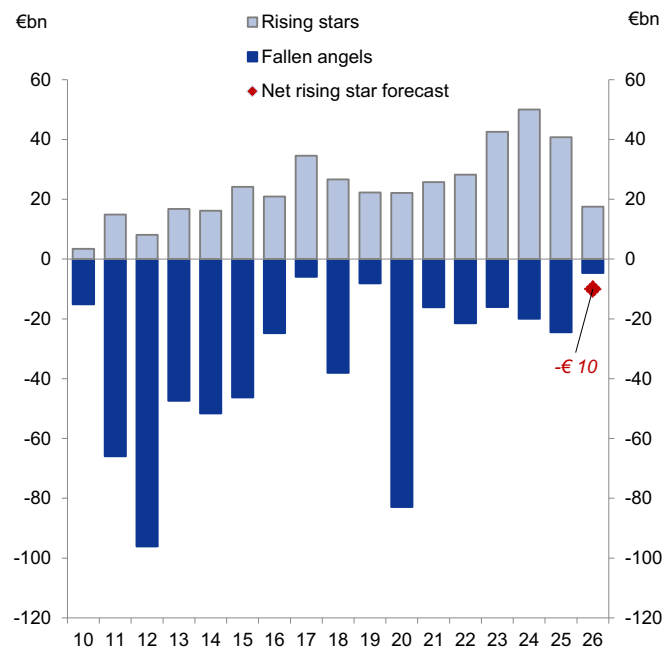
Exhibit 154: Annual average recovery rates by seniority for European issuers

Source: Moody's, Goldman Sachs Global Investment Research

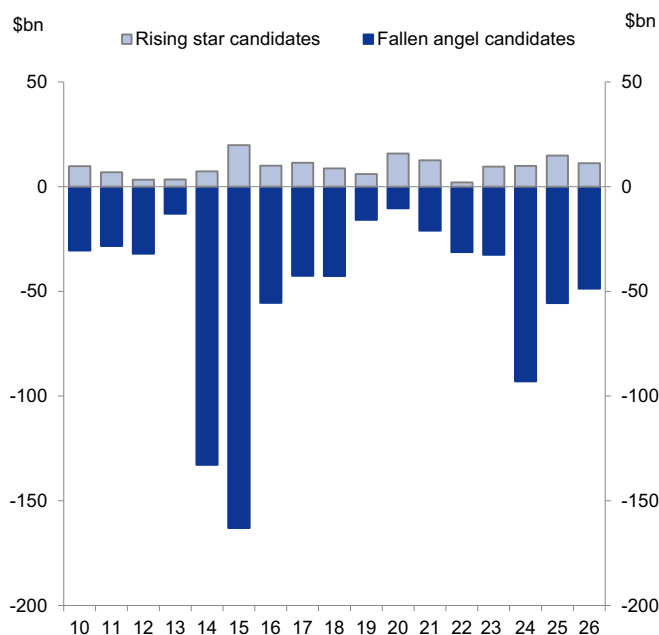
Default and ratings migration dashboard: Upgrades and downgrades

Exhibit 155: Fallen angels vs. rising stars in the USD market


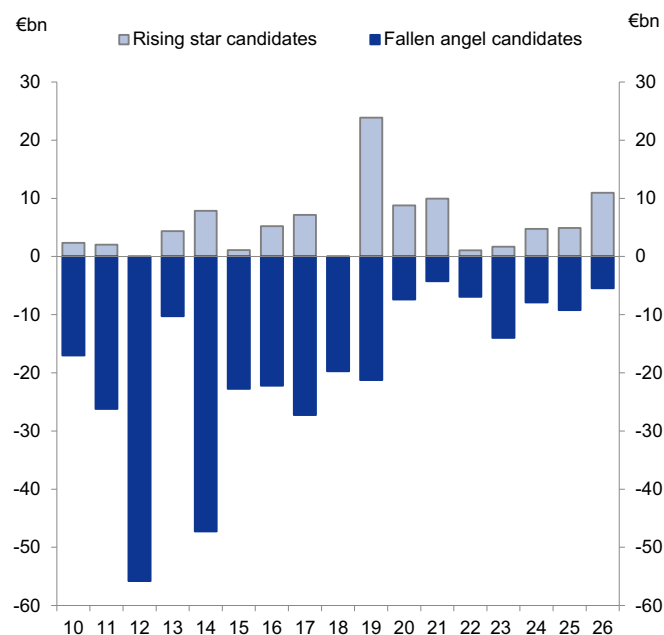
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 156: Fallen angels vs. rising stars in the EUR market


Source: Bloomberg, Goldman Sachs Global Investment Research

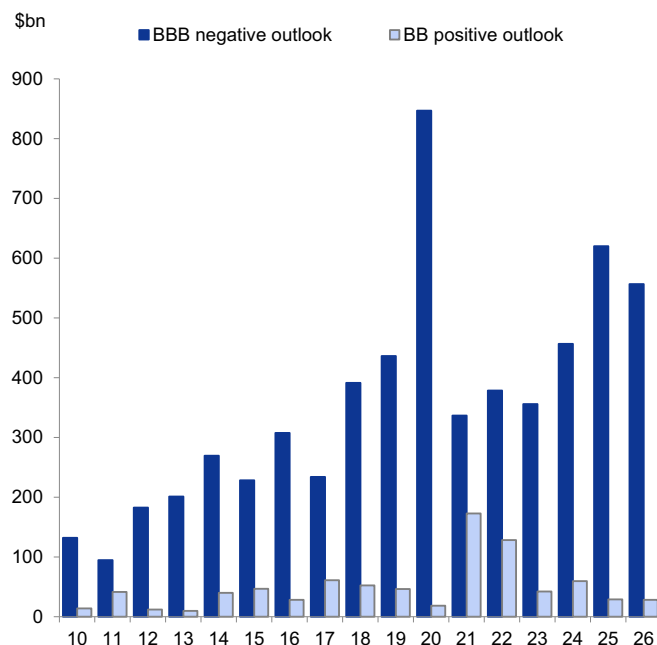
Exhibit 157: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the USD market


Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 158: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the EUR market


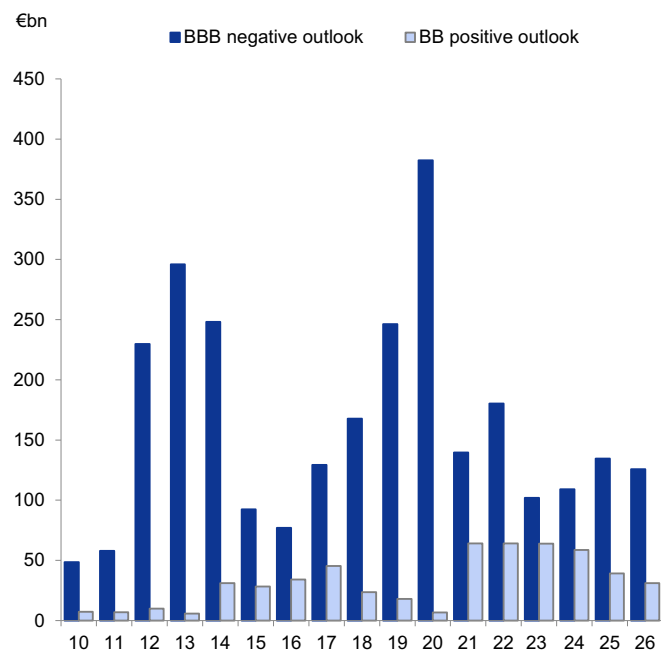
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 159: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the USD market



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 160: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the EUR market



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 161: Top 10 USD fallen angel issuers in the last 12 months

Company	Notional downgraded (\$bn)
Centene Corp	14.7
Paramount Global	13.0
Fs Kkr Capital Corp	3.8
Fmc Corp	2.5
Coty/Hfc Prestige/Int Us	1.7
Electronic Arts Inc	1.4
Banco Santander Mexico	1.3
Horizon Mutual Holdings	0.6
Coty/Hfc Prestige/Int Us	0.5
Ses Americom Inc	0.5

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 162: Top 10 USD rising star issuers in the last 12 months

Company	Notional upgraded (\$bn)
Carnival Corp Ltd	8.5
Teva Pharmaceuticals Ne	7.4
Vistra Operations Co Llc	6.3
Medline Borrower Lp	4.5
Office Cherifien Des Pho	4.1
1011778 Bc / New Red Fin	4.0
Permian Resourc Optg Llc	3.6
Unicredit Spa	3.5
Sba Communications Corp	3.0
Frontier Communications	2.8

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 163: Top EUR Fallen angel issues in the last 12 months

Company	Notional downgraded (€bn)
Ses Sa	3.4
Lanxess Ag	2.2
Verallia Sa	1.6
Mediobanca Di Cred Fin	1.5
Adecco Int Financial Svs	0.5
Illimity Bank Spa	0.3
Solaben Luxembourg	0.1

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 164: Top 10 EUR rising star issuers in the last 12 months

Company	Notional upgraded (€bn)
Teva Pharm Fnc Nl Ii	6.2
Banca Monte Dei Paschi S	3.0
Terna Rete Elettrica	1.9
Aib Group Plc	1.7
Carnival Uk Ltd	1.6
Unipol Assicurazioni Spa	1.3
Metro Ag	1.1
Bper Banca	1.0
Bank Millennium Sa	1.0
Piraeus Bank Sa	1.0

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Disclosure Appendix

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Contributing Authors: Amanda Lynam, CPA Goldman Sachs & Co. LLC, Spencer Rogers, CFA Goldman Sachs & Co. LLC, Sara Grut Goldman Sachs International, Shamshad Ali Goldman Sachs & Co. LLC.

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